Where and when to search for COTS on a reef

Fine-scale habitat suitability and metacommunity modelling for tactical and strategic decision making on the Great Barrier Reef · Technical report to the Great Barrier Reef Foundation, COTS Control Innovation Program, project CCIP-BR-13 · [TO CONFIRM: organisations as they should appear, e.g. AIMS, QUT, JCU, GBRMPA]

[TO CONFIRM: author list and order]

October 2026

Table of contents

Executive summary **2**

1 Background **4**
1.1 The management problem . . . 4
1.2 Earlier models this project builds on . . . 4
1.3 The decisions this work supports . . . 5
1.4 Project objectives . . . 5

2 Methods **6**
2.1 Observations and habitat information . . . 6
2.2 Models and their purposes . . . 7
2.3 Testing the search decisions . . . 8
2.4 Seasonal products and field assessment . . . 9
2.5 Modelling water movement and larval connections among reefs . . . 9
2.6 Simulating connected COTS and coral populations . . . 9

3 Results **10**
3.1 GBR fine-scale habitat predictions . . . 10
Habitat layers: a consistent first ordering on every reef . . . 10
3.1.1 Predicted habitat and recent evidence increase cull-site search efficiency 11
3.2 The 2027 products combine priorities with exploration . . . 14
3.3 Preliminary larval dispersal patterns vary across the reef cluster . . . 14
3.4 Historical COTS and coral dynamics provide a basis for testing control strategies 16

4 Implications for management **19**
4.1 Combining recent observations with habitat for daily searches . . . 19
4.2 Maintaining coverage during manta surveillance . . . 19
4.3 Assessing the culling program from 2012 to the present and beyond . . . 19
4.4 Projecting suppression of the fifth primary irruption wave . . . 19
4.5 Monitoring operational performance . . . 19

5 Limitations **20**

6 Recommendations and next steps **20**

Acknowledgements **21**

References **21**

Appendices **25**
Appendix A: Delivered layers and products 25
Appendix B: Predictors 26
Appendix C: HMSC joint model 27
Appendix D: Supplementary validation 29
Appendix E: Alignment with the CCIP-BR-13 Work Plan 33
Appendix F: Reproducibility 34
Appendix G: Detailed model specification and evaluation 35
Data and response definitions 35
Habitat models and mapped layers 36
Evaluation design and search-efficiency analyses 37
Seasonal products 39
Metacommunity model and historical culling 39
Software and reproducibility 41
Appendix H: Detailed tactical model comparisons 42
Recent evidence combined with habitat reduces the cull-site search 42
The gain holds season by season 43
Retaining dives above 0.04 CPUE 44
Each source covers the other's gaps 46
Manta surveillance: spread tows evenly, add where habitat ranks highest 50
The 2027 cull-site priority ranking and exploration areas 53
Appendix I: Updated metacommunity calibration and exploratory projections 55
Calibration updates and uncertainty 55
Recent densities and additional reefs 55
Initial control projections to 2050 57
Appendix J: Archived 15 October metacommunity comparisons 62
Metacommunity model: calibration and current limitations 62
Appendix K: Preliminary larval-concentration comparisons 66
Variation among releases 66
Cross-shelf patterns and the domain boundary 67
Appendix L: Management assessment and field-trial design 70
Combining recent observations with habitat for daily searches 70
Maintaining coverage during manta surveillance 70
Assessing the culling program from 2012 to the present and beyond 71
Projecting suppression of the fifth primary irruption wave 71
Monitoring operational performance 72
Developing an early-warning system 73

Suggested citation: [TO CONFIRM: Authors (2026). Where and when to search for COTS on a reef: fine-scale habitat suitability and metacommunity modelling for tactical and strategic decision making on the Great Barrier Reef. Report to the Great Barrier Reef Foundation, COTS Control Innovation Program (CCIP-BR-13). Australian Institute of Marine Science, Townsville. DOI/URL.]

Executive summary

The COTS Control Program uses vessel and diver time to search for crown-of-thorns starfish (COTS). Crews tow manta observers around reefs and dive designated cull sites, many holding few or no COTS. Project CCIP-BR-13 set out to make that search more efficient at the scale where crews decide, within a reef. It had two strands:

- fine-scale habitat suitability layers and rankings showing where on a reef to tow and dive first;
- a prototype metacommunity model for control strategies across reef clusters.

Every search-efficiency result below was tested on held-out data against a no-information baseline at the same effort.

Key findings

- **The ranking reduces searches, with larger reductions at lower retention.** Retaining 80%, 90%, 95% and 99% of dives above 0.04 CPUE required 67%, 80%, 90% and 98% of sites with habitat plus recent observations. At 95% retention, random selection required 96%. Annual full surveys and triggered checks could help recover missed populations, but their effort and detection success need field assessment. Lower-retention options allow more high-density dives to be missed initially.
- **The earlier ≥0.02 comparison improved on habitat alone season after season.** With the evidence model trained only on earlier seasons, the combined ranking avoided more low-density dives than habitat alone in every season from 2022, and searched fewer sites in every season from 2023.
- **Each source covers the other's gaps.**
  - Past cull records plus habitat forecast next season's problematic sites with an AUC of 0.68, against 0.57 for habitat alone.
  - History lies within reach of only 9% of reef cells, and 67% of designated sites are on reefs with no recent visits. There, habitat supplies the first search order.
  - Habitat also points to ground outside the current search footprint, where strong manta evidence was more common than on repeat ground (13.6% vs 5.8% of tows).
- **For manta surveillance, use habitat to guide added searches.** Habitat-only tow selection did not outperform evenly spaced thinning in the retrospective comparison. Thinning is an analytical benchmark, not a practical field route. The maps are most useful alongside recent observations and crew knowledge when identifying habitats needing additional coverage.
- **The metacommunity model reproduces historical waves and now supports exploratory control projections.** Updated simulations compare COTS and coral through 2026, with and without recorded culling. Initial projections to 2050 link diver effort to removals, but peak sizes, recent densities and outside larval supply remain uncertain. Program-impact and fifth-wave suppression estimates still require independent validation.

What was delivered

- A 2027 priority ranking of all 15,884 designated cull sites on 576 reefs (classes A–D; CSV and KMZ).
- 3,807 exploration areas on 488 reefs: high-ranked habitat outside the cull sites and away from recent tows.
- 30 m habitat layers for the Great Barrier Reef, a known-aggregation-sites layer, and a pipeline re-run each season.
- The COTSMod.jl model, integrated with ADRIA, with a reproducible calibration workflow for the Lizard Island cluster.

What managers can do now

- Use the 2027 ranking as a search order. Agree and test the tolerance for missed >0.04 dives before adopting a fixed exclusion rule.
- Thin manta surveillance evenly if effort must be cut, and add tows where habitat ranks highest.
- Add one or two exploration areas per reef visit and log the outcome.

Main limitations. On reefs with no recent visits the ranking relies on habitat and is close to chance. Rankings learned from past effort partly encode crew behaviour, and densities at never-dived sites cannot be verified. The products have not yet been trialled in the field, and the metacommunity model has not passed its validation gates.

1 Background

1.1 The management problem

Irruptions of crown-of-thorns starfish (*Acanthaster* cf. *solaris*; COTS) are a major cause of coral loss on the Great Barrier Reef (De'ath et al., 2012; Pratchett et al., 2017). The COTS Control Program suppresses irruptions by manual culling, which can protect coral when sustained at the right reefs (Matthews et al., 2024). A substantial portion of operational effort goes into searching for COTS, creating opportunities to improve where surveillance and culling are directed.

Manta tow surveillance records COTS and feeding scars and assesses reef irruption status. Crews then choose cull sites using manta results and previous culling records. The current EoTR register contains 15,884 designated sites on 576 reefs, of which approximately 1,300 sites across 200 actioned reefs are typically culled each year (Great Barrier Reef Marine Park Authority, 2026). The dashboards bring together recent observations to support these decisions (Matthews et al., 2025). In the held-out cull records from 2018 to 2026, 18% of dives found no COTS (ghost dives), and a further 21% found COTS at CPUE below 0.02 COTS per minute of bottom time (low-density dives).

The operational strategy links decisions from individual sites to voyages and regional priorities (Fletcher et al., 2020). In practice, tow routes mostly follow established cull sites around reef perimeters. Many sites were initially delineated from satellite maps in 2018, leaving gaps in lagoon bommie fields, shallow crests and isolated patches. Search can therefore repeatedly trigger dives where few COTS are found while overlooking other potential habitat. Reducing repeated searches in low-priority areas could free effort to search these gaps, improving both efficiency and the chance of finding previously undetected populations.

Crews and GBRMPA analysts already hold useful information about recent culling, empty sites, and COTS or scars seen on manta tows. This knowledge is strongest on frequently visited reefs and within the established search footprint. Fine-scale habitat maps cover places with little recent evidence, including potential search areas outside that footprint, but say little about whether COTS are present this season. This project combines these sources to prioritise familiar sites and identify additional places to search.

1.2 Earlier models this project builds on

Most COTS modelling has treated whole reefs as its spatial units, either across the GBR for regional planning (Matthews et al., 2020b; Condie et al., 2021; Bozec et al., 2022) or on individual reefs to inform culling thresholds (Morello et al., 2014; Plagányi et al., 2020). Reef-level species distribution models (SDMs), which relate species observations to environmental conditions, showed that larval connectivity and water quality explain much of the spatial distribution of irruptions (Matthews et al., 2020a), and a hybrid machine-learning model estimates each reef's annual irruption probability (Stewart et al., 2026). Population and ecosystem models help compare regional control strategies and ecological targets, but their whole-reef units do not identify where within a reef to search or cull.

Recent developments support finer spatial resolution. Coral species distribution models use satellite-derived multiscale terrain and benthic variables to map habitat for key coral taxa at tens of metres (Guillaume et al., 2025). Their predictions and environmental predictors are inputs to the habitat modelling here, which maps COTS suitability at 30 m and combines it with recent program observations.

In parallel, advances in hydrodynamic and COTS larval-connectivity modelling provide a route to representing connections among smaller areas of reef. This project adapts COTSMod, a

spatially explicit model of COTS populations and coral recovery (Matthews et al., 2020b), to represent site-level dynamics in the northern GBR irruption initiation box. It uses AIMS's high-throughput ADRIA ecosystem modelling framework and CoralBlox coral dynamics (de Almeida et al., 2026; open-AIMS, 2026b). The prototype covers the Lizard Island cluster, with 2,895 sub-reef sites on 113 reefs. Its present transport inputs remain at reef scale; the fine-scale hydrodynamic network has not yet replaced them. This distinction matters when interpreting its capacity to represent within-reef differences (Section 2.6).

1.3 The decisions this work supports

The tactical work addresses two operational uses: reducing manta tow effort while retaining detection of COTS aggregations, and searching fewer cull sites while avoiding empty or low-density dives. For management, the priority is to retain dives above 0.04 COTS per minute of bottom time. Missing some dives between 0.02 and 0.04 is acceptable under the proposed operational decision criterion; these should not carry the same penalty as missed dives above 0.04. The 0.04 value is the management target used here, while ecological sustainability also depends on coral cover (Plagányi et al., 2020; Great Barrier Reef Marine Park Authority, 2026). The existing retrospective analyses used ≥0.02 to define a problematic dive and <0.02 for a low-density dive. Those results are retained under their original definitions, alongside a separate evaluation of the unchanged rankings against the >0.04 management criterion (Section 3.1.1).

Reducing search effort can miss COTS even when sites are dropped without a model. The evaluation therefore asks whether a ranking finds more aggregations or retains more high-density dives than random selection at the same effort. For manta surveillance, it also compares model selection with evenly spaced tows. These comparisons distinguish the value of the ranking from the saving achieved simply by searching less. Additional searches outside the familiar footprint address a separate question: whether effort released from repeat searches can improve coverage and detection elsewhere.

A strategic decision sits above these: how to allocate control among connected reefs in a cluster or region. When should culling stop on a heavily affected reef if its remaining coral cannot be protected? When should control continue on a reef whose COTS population cannot be brought below a local target, because reducing its larval export could protect other reefs? A site-level model of population growth, collapse, dispersal and coral consumption is needed to compare those consequences. Evidence of renewed population growth at Lizard Island (Chandler et al., 2023; Uthicke et al., 2024) also motivates assessing whether the emerging fifth primary irruption wave can be suppressed, and what resources this would require.

1.4 Project objectives

The project addressed three objectives:

1. Develop GBR-wide predictive fine-scale habitat suitability models by connecting fine-scale COTS data to new high-resolution benthic and coral habitat data.
2. Optimise and standardise surveillance paths to improve irruption detection and operational efficiency, including a comparison against traditional perimeter-based surveillance.
3. Refine site-level management strategies for individual reefs through predicted suitable habitat and metapopulation modelling.

For Objectives 1 and 2, the report evaluates habitat layers, manta thinning and cull-site rankings combining habitat with recent evidence (Section 3.1). The 2027 product also identifies exploration areas outside designated sites and recent tow paths, so that search planning can extend beyond places already visited. Retrospective thinning tests the retention of observed

aggregations; it does not establish the performance of new surveillance routes or exploration areas.

For Objective 3, the metacommunity prototype provides a platform to assess the effects of the culling program from 2012 to the present and compare future control strategies. The intended comparisons address stopping rules where local coral protection is no longer achievable, continued control of larval-source reefs despite limited local protection, and the allocation and additional resources needed to suppress the fifth primary irruption wave. Outcomes include COTS populations, larval export, and local and downstream coral loss and recovery. Historical culling is implemented as prescribed removals, and an initial effort-based model now supports exploratory projections. Independent validation and agreement on operational scenarios remain necessary before these comparisons can guide management (Section 3.4). Methods describe the prototype and its calibration (Section 2.6); Appendix E maps the Work Plan outputs to the report.

2 Methods

2.1 Observations and habitat information

The analyses combined records of where COTS had been found with maps describing reef habitat. The full cull archive covered 2002–2026. After excluding records with missing counts or invalid bottom time, records matched to designated cull sites covered 4,299 sites on 339 reefs. Catch per unit effort (CPUE) was the number of COTS culled per minute of bottom time. The habitat models summarised cull outcomes for each site and year; search performance was tested against individual dives.

Two thresholds answered different questions. The original habitat models predicted cull CPUE ≥0.02. Empty or "ghost" dives found no COTS; low-density dives found COTS below 0.02. The management assessment focused on retaining dives above 0.04, the Control Program's adopted ecological target on reefs with less than 40% coral cover (Plagányi et al., 2020; Great Barrier Reef Marine Park Authority, 2026). Dives from 0.02 to 0.04 inclusive were treated as intermediate: some could be omitted without the penalty assigned to missed dives above 0.04. The original model response was retained when assessing this management criterion.

Manta records supplied observations along tow paths. Strong evidence meant a COTS sighting, at least two counted feeding scars, or scars recorded as common in older surveys; records with neither COTS nor scars were negatives. The manta-based habitat models used 198,852 labelled tows in 1,015 combinations of reef and survey year (reef-years). Contributing monitoring streams included Control Program crews from 2016, the Reef Joint Field Management Program from 2012, and the AIMS Long-Term Monitoring Program from 1992. These are source-program periods; each analysis used its specified data subset.

Scooter-assisted large area diver-based (SALAD) surveys (Pratchett et al., 2022) and Reef Health and Impact Surveys (RHIS) (Beeden et al., 2014) supplied additional COTS and coral observations for the multi-observation habitat model. ReefScan imagery (Bainbridge and Coleman, 2024) supplied a separate check at recorded sightings, rather than training data for that model. Survey-effort tracks were unavailable for that check, so it could not estimate detection rates or absence. Appendix G describes the role of each source.

Habitat predictors described depth, seabed shape, wave exposure, turbidity and predicted habitat for three *Acropora* taxa (Guillaume et al., 2025). They were assembled on a 30 m grid. ReefGuide supplied additional environmental layers (open-AIMS, n.d.); their shallower coverage left gaps in some deeper reef areas, particularly beyond about 15–20 m. GBR10 bathymetry and geomorphic maps supplied further reef information (Great Barrier Reef Marine Park

Authority, 2021a, 2021b; Roelfsema et al., 2021). Appendix B lists predictors and sources; Appendix G gives the fitting and mapping details.

2.2 Models and their purposes

The habitat models asked whether environmental conditions could rank places within a reef. The search-ranking model then added recent observations to identify where crews should search first. These tasks differ from simulating how culling affects connected COTS populations and coral, which was the role of the metacommunity model. Table 1 defines each model before its results are discussed.

Table 1: Models and comparisons. Predictor counts for the two annual models include survey year. Original code labels and alternative combinations are mapped to these descriptions in Appendix G.

Model or comparison	Purpose	Information used
Annual full-extent habitat model, 13 predictors	Predict cull outcomes across the full habitat-map extent	Twelve terrain and coral-habitat predictors plus survey year; fitted to cull site-years
Annual limited-extent habitat model, 18 predictors	Test whether added environmental information improves the annual prediction	The same predictors and year, plus five ReefGuide layers; fitted to cull site-years
Manta habitat models	Rank tow locations with strong COTS evidence	The environmental predictors above, fitted to tow observations; versions with year and with differences among reef-years accounted for were compared
Recent-evidence site ranking	Set a search order for designated cull sites	Habitat rank plus the preceding season's cull dives and nearby manta tows
Multi-observation habitat model	Explain habitat associations while allowing for different observations	Cull, SALAD and RHIS evidence, terrain, geomorphic classes and coral habitat; implemented with a hierarchical model of species communities
Habitat updated by longer cull history	Test how much past finds add to a habitat forecast	Multi-observation habitat estimates and spatially weighted earlier cull records
COTS–coral metacommunity model	Simulate connected populations, coral consumption and responses to removals	Population and coral processes, larval supply, disturbance and historical culling in the Lizard Island cluster

The search-planning map used the annual model with additional environmental information wherever its layers were available, and the full-extent model elsewhere. Scores were placed on a common scale before cells were ranked within each reef. The manta habitat models were

tested as alternatives. The multi-observation model supplied ecological interpretation and uncertainty; it was not the operational search map.

The recent-evidence ranking combined habitat with last season's observations at or near each designated site, including whether a site had been searched without finding problematic densities. It therefore formalised information already used by crews as well as adding habitat information. Its performance must be interpreted in that operational context.

2.3 Testing the search decisions

Predictions were compared among locations on the same reef in the same year, because those are the choices crews make. We withheld reefs or used earlier seasons to predict later ones, as set out in Table 2. Comparisons kept effort or the share of high-density dives retained equal. Random selection provided a reference for the gains expected simply from searching fewer sites. Manta thinning was also compared with evenly spaced tows.

Table 2: Evaluation of the tactical decisions. The season-by-season evidence model used earlier outcomes, but its habitat models were withheld by reef and were not also restricted to earlier years.

Decision	How the prediction was tested	Comparison and outcome
Where habitat predicts COTS	Each reef was scored by models fitted without its group of reefs	Ability to rank observations on the same reef and in the same year
Which cull sites to search	All designated sites were ranked, including never-dived sites; recent evidence came only from earlier seasons	Habitat alone, recent evidence and their combination; sites searched and dive outcomes retained
Whether the ranking works in later seasons	The evidence model was fitted on seasons before each test season, 2019–2026	The same search outcomes, evaluated season by season
Whether >0.04 dives are retained	Saved rankings were assessed at the higher threshold without refitting	Missed high-density dives, low-density and ghost dives avoided, and intermediate dives retained
Which manta tows to omit	Recorded surveys were thinned while keeping the same number of tows for each strategy	Habitat-ranked, random and evenly spaced tows; observed aggregations still detected

An observed manta aggregation comprised at least two strong-evidence tows linked within 500 m in a survey. It was retained if any of its tows remained. Tow order was approximated from position around the reef because recorded sequence was unavailable. The test therefore assessed thinning of existing surveys, not new route design.

For the >0.04 comparison, "sites searched" counted the selected designated sites across the evaluated reef-years, including sites that had never been dived. The assessment asked what proportion of observed dives fell at those selected sites; it could not determine COTS densities at unvisited sites. We compared search effort at 80%, 90%, 95% and 99% retention of high-density dives and assessed intermediate dives separately. These retention levels explored the trade-off; none was an agreed definition of negligible omissions. Confidence intervals resampled

whole reef-years, or whole surveys for aggregation detection, and repeated each comparison. Appendix H provides the full comparisons and uncertainty calculations.

2.4 Seasonal products and field assessment

The 2027 product ranked every designated site using habitat and the latest 12 months of cull and manta observations, from June 2025 to May 2026. Sites were grouped into four priority classes. Additional exploration areas identified high-ranked habitat outside designated sites and recent tow paths. These were proposed places to search, not confirmed COTS aggregations.

A field trial will record sites searched, tows, dives, effort and COTS found, comparing the ranking with crews' normal planning. It will also count visits to exploration areas and later checks of deferred sites. Annual or pre-closure checks remain necessary: postponing a search does not necessarily reduce annual effort. Appendix G gives the product settings and proposed assessment design.

2.5 Modelling water movement and larval connections among reefs

Hydrodynamic model and particle tracking. [TO WRITE: Sev and Owen — describe development of the fine-scale Lizard Island hydrodynamic model, its spatial domain and resolution, environmental forcing and checks against observations. Explain how particles representing COTS larvae were released and tracked, including release dates and source locations, larval behaviour, mortality, competency and the treatment of particles crossing the model boundary. Identify the model versions, data sources and supporting references.]

Connectivity matrices and network analysis. [TO WRITE: Sev and Owen — explain how particle trajectories were consolidated into connections among sub-reef sites and reefs. Define the matrix entries, their direction and units, how release numbers and source reef area were accounted for, and the settlement and competency rules. Describe the network analyses used to assess exchange within reef groups, larval sources and the relative isolation of Ribbon Reefs, including comparisons with distance alone, variation among releases and checks for model-boundary effects. Distinguish completed analyses from planned tests and state which connections were supplied to the population model.]

The initial dispersal assessment used modelled daily mean larval concentrations at ages 5, 9 and 10 days for 10 releases from December 2022 to January 2023. Concentrations were averaged by polygon area to sub-reef sites and then reefs, and ranked within each release before comparisons were combined. This assessed relative spatial patterns without letting differences in total larval load dominate. We examined how often each site ranked in the highest fifth at day 10, variation among releases, and cross-shelf patterns within latitude bands. These descriptive comparisons do not identify larval origins or settlement probabilities. Appendix K provides the supporting comparisons and their limits.

2.6 Simulating connected COTS and coral populations

The metacommunity prototype represents 2,895 sub-reef sites on 113 reefs in the Lizard Island cluster. COTSMod supplies population growth, coral consumption, reproduction and larval dispersal; ADRIA supplies the domain, disturbance and scenario handling, coupled to CoralBlox coral dynamics (Matthews et al., 2020b; de Almeida et al., 2026; open-AIMS, 2026c, 2026b). Larval connections currently operate between whole reefs. Habitat and recorded removals vary among sites, but fine-scale water-flow connections have not yet replaced those inputs.

Historical simulations were compared with COTS and coral records from Lizard Island, Eyrie, MacGillivray and North Direction. The tests asked whether the model reproduced the timing

and size of irruptions, low densities between waves, and coral change. Recorded culling was applied as removals, limited by the simulated animals available. A mismatch between available animals and recorded catches was treated as a model diagnostic.

The intended strategic assessments compare matched histories with and without culling from 2012 onward, then examine future stopping rules, control of larval-source reefs and resources for fifth-wave suppression. They must track local and downstream coral outcomes. The latest projections translate diver-hours into removals using a density-dependent catch rate, capped by the available adult population. They compare no further culling, recent effort, increased effort and reallocation among reefs under two climate pathways through 2050. These exploratory scenarios require independent validation and agreement on operational constraints. Appendix I documents the current calibration and projections; Appendix G retains detailed model methods. [TO CONFIRM: connectivity attribution, delivery of the fine-scale network, final model release and operational scenario settings.]

3 Results

3.1 GBR fine-scale habitat predictions

Habitat layers: a consistent first ordering on every reef

The habitat layers provide a first search order across the GBR, including reefs with little recent evidence. Their absolute values vary with survey year and region (Figure 15); within reefs, the cull-trained layers place their highest ranks along reef margins and outer slopes (Figure 1).

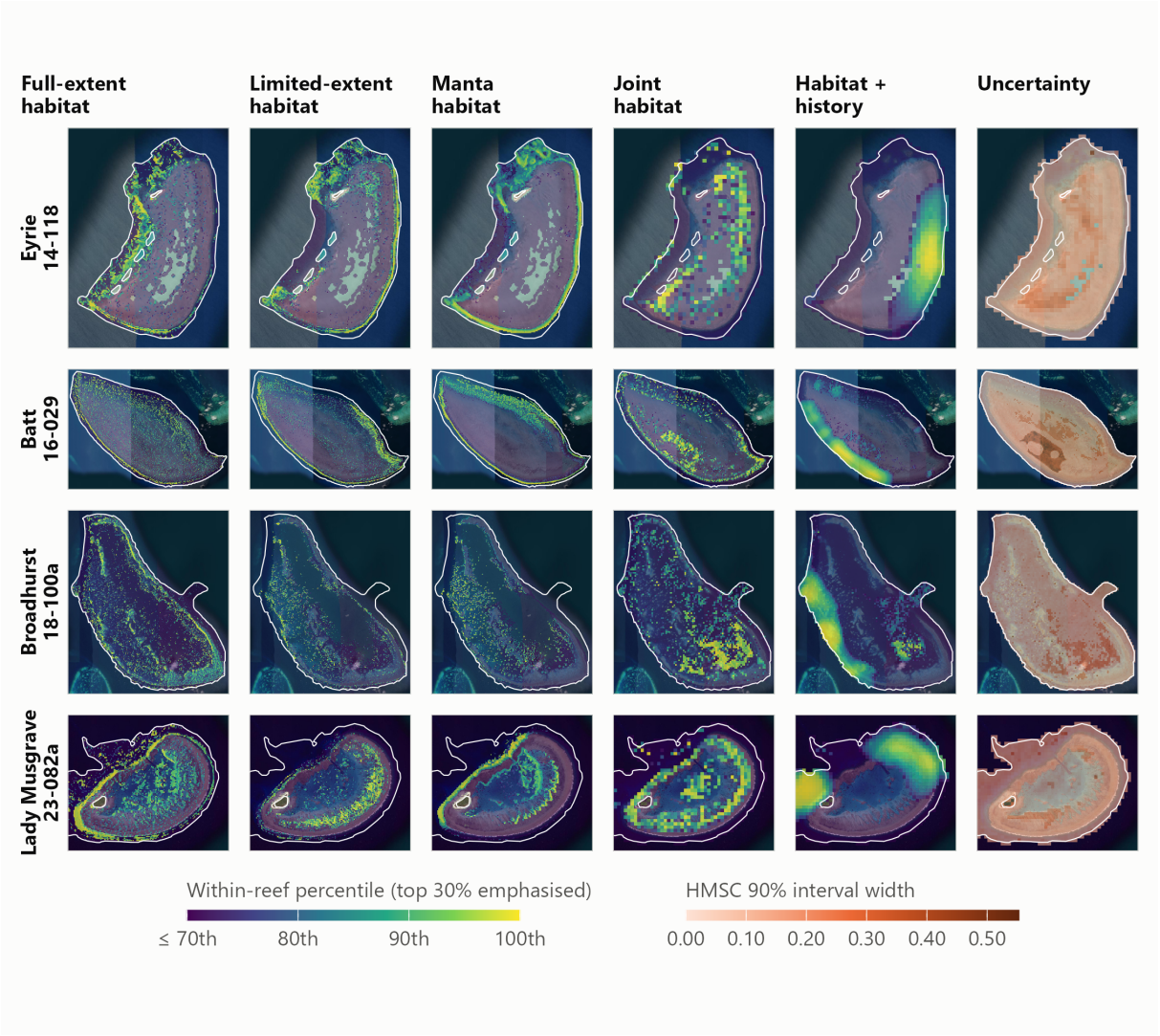


Figure 1: Within-reef habitat ranks on Eyrie, Batt, Broadhurst and Lady Musgrave reefs. Columns show the full-extent and limited-extent cull-trained models, the manta-trained model, the joint habitat model, habitat updated with cull history for 2027, and joint-model uncertainty (90% posterior interval width). The top 30% of each reef is emphasised. Satellite imagery: Esri World Imagery; full attribution is in Appendix H.

On their own, the layers discriminate modestly between places on the same reef. On independent manta tows, the limited-extent habitat model reached a within reef-year AUC of 0.576 (95% CI 0.548–0.602) (Table 13, Figure 16). It placed 23% more strong-evidence tows in each reef's top-ranked 20% than chance. The manta model, trained on tows but scored leave-reef-out, reached 0.612 (95% CI 0.578–0.636) (Appendix H).

3.1.1 Predicted habitat and recent evidence increase cull-site search efficiency

Combining recent cull and manta observations with habitat improved search priorities and efficiency compared with habitat alone. This pattern held across seasons (Appendix H). The assessment compared sites selected by the rankings and the share of high-density dives (>0.04 CPUE) they retained. In the test over 2019–2026 (28,431 dives in 861 reef-years), retaining 80% of those dives required searching 66.6% (95% CI 62.0%–71.1%) of designated sites with habitat estimates plus recent observations; retaining 90% required 80.1% (95% CI 73.9%–85.2%).

Both required fewer sites than habitat alone or random selection at the same retention (Table 3, Figure 2).

The combined ranking avoided 28.7% of low-density cull dives (positive CPUE <0.02) and 34.5% of ghost cull dives (0 CPUE) at 80% retention. At 90% retention it avoided 17.5% and 21.8%, respectively (Table 3).

Search reductions became smaller as retention increased. Retaining 95% required 90.2% (95% CI 85.4%–93.4%) of sites with the combined ranking, avoiding 8.9% of low-density dives and 11.2% of ghost dives. Retaining 99% required 98.2% (95% CI 97.2%–98.7%) of sites and avoided only 1.8% and 2.2% of those dive classes, respectively.

The evaluation did not test whether annual full surveys or later checks would recover missed populations; proposed follow-up measures are discussed in Section 4.5.

Habitat alone separated locations with and without COTS only weakly. Its practical contribution was to cover places where recent observations were absent: 67% of designated sites in the 2027 product were on reefs without dives or tows in the preceding 12 months. Those sites relied on habitat and should be treated as initial search priorities. Past observations carried most of the information on recently visited reefs, but also reflected where crews chose to work. Maps, habitat comparisons and tests beyond the familiar footprint are in Appendix H.

Table 3: Designated sites searched and low-density or ghost cull dives avoided at matched retention of dives above 0.04 CPUE, 2019–2026. All values are percentages; brackets give 95% reef-year bootstrap intervals. Low-density means positive CPUE below 0.02; ghost dives found no COTS. Intermediate dives (0.02–0.04 inclusive) are separate. Existing rankings were reassessed without refitting: habitat models were withheld by reef and the evidence model used earlier seasons. Follow-up surveys and annual effort are not included.

High-density retained (%)	Ranking	Sites searched (%)	Low-density avoided (%)	Ghost dives avoided (%)
80	Habitat + recent observations	66.6 [62.0–71.1]	28.7 [24.5–32.8]	34.5 [30.3–39.2]
80	Habitat alone	72.6 [67.1–77.2]	21.6 [17.7–26.5]	25.3 [20.9–30.6]
80	Random selection	80.4 [80.1–80.8]	20.0 [19.7–20.3]	20.0 [19.6–20.3]
90	Habitat + recent observations	80.1 [73.9–85.2]	17.5 [13.5–22.3]	21.8 [17.0–27.9]
90	Habitat alone	85.8 [81.4–88.7]	11.1 [9.0–14.4]	13.4 [10.9–17.2]
90	Random selection	90.5 [90.1–91.0]	10.0 [9.6–10.4]	10.5 [10.0–10.9]
95	Habitat + recent observations	90.2 [85.4–93.4]	8.9 [6.2–12.9]	11.2 [8.0–16.8]
95	Habitat alone	94.0 [89.7–95.5]	4.9 [3.7–8.2]	6.8 [4.9–10.5]
95	Random selection	95.6 [95.3–95.9]	4.9 [4.6–5.2]	5.3 [4.9–5.8]
99	Habitat + recent observations	98.2 [97.2–98.7]	1.8 [1.2–2.7]	2.2 [1.6–3.6]

Table 3: Designated sites searched and low-density or ghost cull dives avoided at matched retention of dives above 0.04 CPUE, 2019–2026. All values are percentages; brackets give 95% reef-year bootstrap intervals. Low-density means positive CPUE below 0.02; ghost dives found no COTS. Intermediate dives (0.02–0.04 inclusive) are separate. Existing rankings were reassessed without refitting: habitat models were withheld by reef and the evidence model used earlier seasons. Follow-up surveys and annual effort are not included.

High-density retained (%)	Ranking	Sites searched (%)	Low-density avoided (%)	Ghost dives avoided (%)
99	Habitat alone	98.8 [98.3–99.1]	1.0 [0.7–1.4]	1.4 [1.0–2.0]
99	Random selection	99.1 [99.1–99.2]	1.0 [0.9–1.0]	1.1 [1.0–1.2]

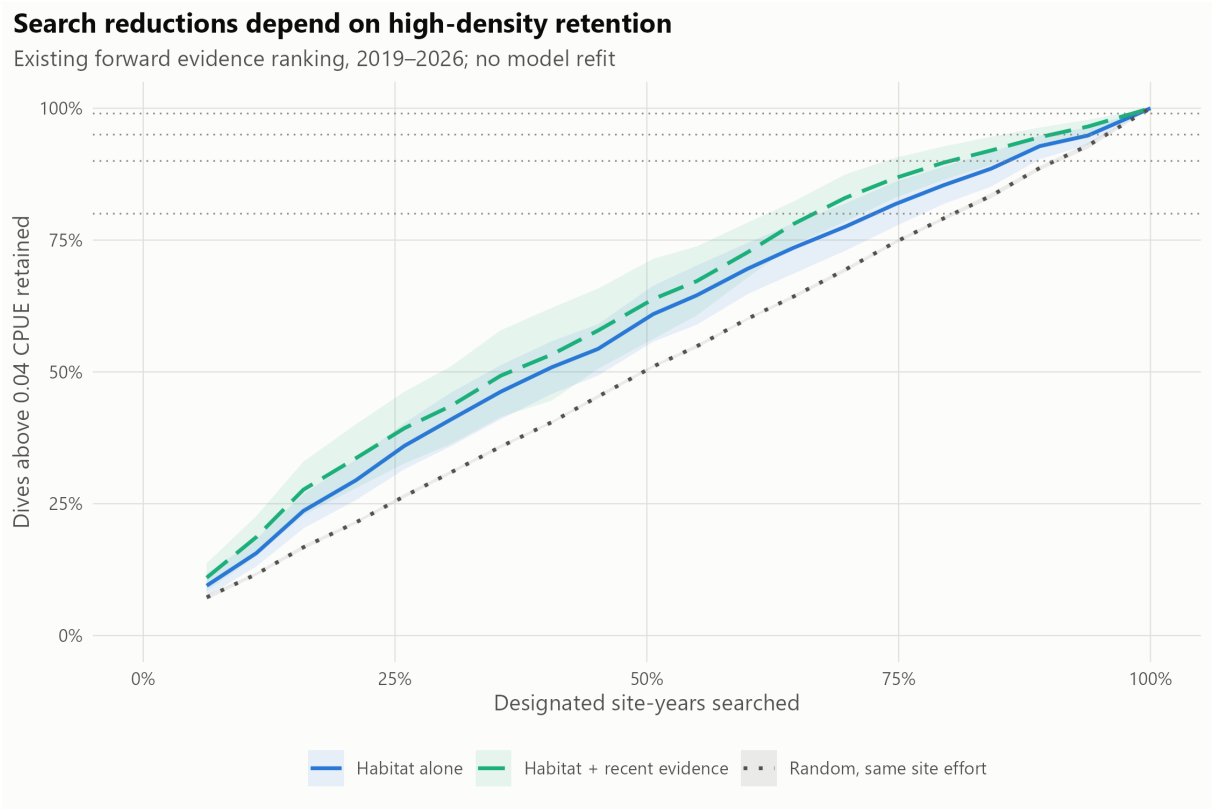


Figure 2: Retention of observed dives above 0.04 CPUE against actual designated site-years searched, using the existing forward evidence rankings. Random selection has the same site count on each reef. Shading shows 95% reef-year bootstrap intervals; dotted horizontal lines mark 80%, 90%, 95% and 99% retention. Habitat fits were held out by reef, while the evidence regression used earlier seasons. Follow-up surveys were not evaluated.

In the retrospective manta comparison, habitat-only selection did not outperform evenly spaced thinning. That comparison is an analytical benchmark; practical searches require crew judgement and recent observations alongside habitat. Appendix H retains the surveillance comparisons.

3.2 The 2027 products combine priorities with exploration

The delivered ranking covers 15,884 designated sites on 576 reefs. Its map shows search priority and the recent evidence behind each site.

The product also identifies 3,807 exploration areas on 488 reefs, outside designated sites and recent tow paths (Figure 3). These allow operators to direct some effort toward overlooked habitat. Historical observations on new ground supported investigating such gaps, but the proposed exploration areas themselves have not been trialled. Their detection yield and the annual effort required for added searches and deferred-site checks remain to be estimated or measured.

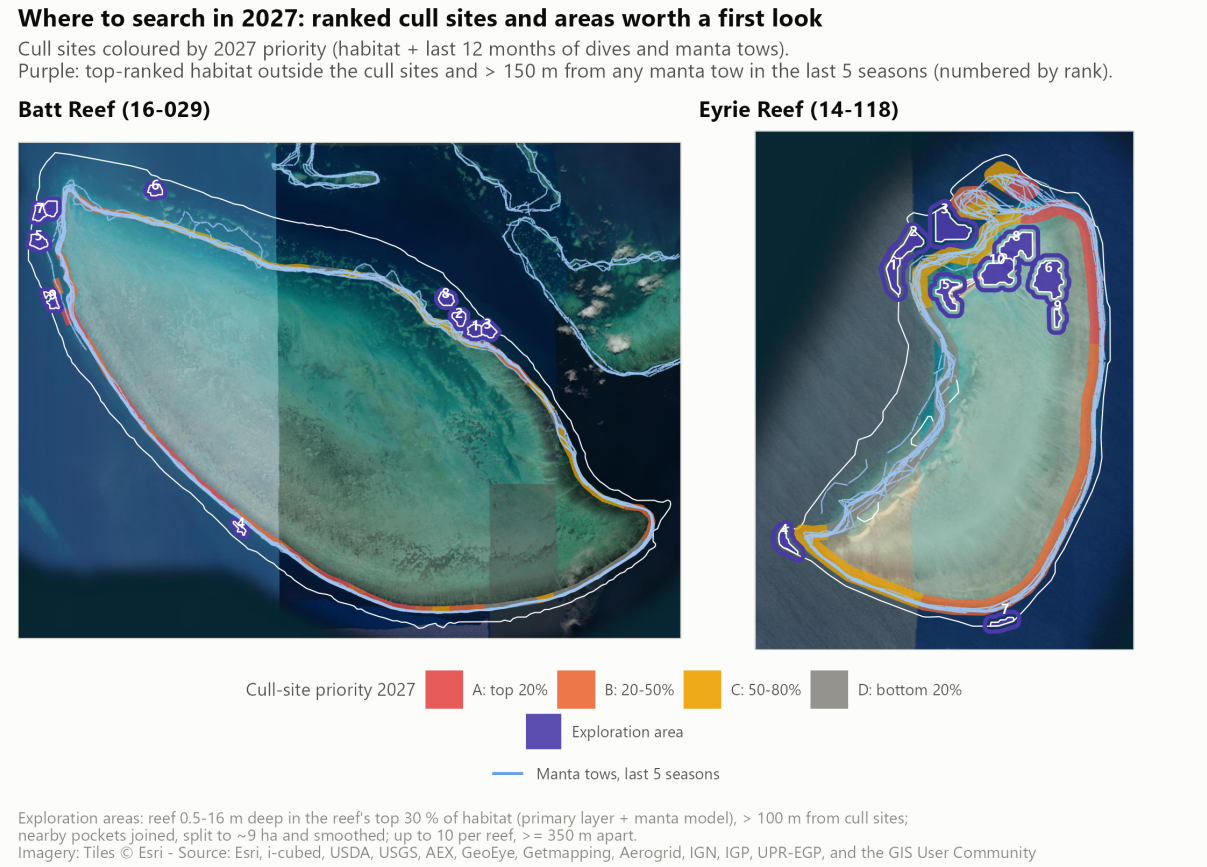


Figure 3: Batt Reef and Eyrie Reef: 2027 cull-site priority (A red to D grey), manta tows of the last five seasons (blue) and exploration areas (purple, numbered by rank) over satellite imagery. On these well-surveyed reefs the exploration areas are outer-edge pockets, reef ends and, at Eyrie, part of the northern back reef that no tow has crossed in five seasons.

3.3 Preliminary larval dispersal patterns vary across the reef cluster

The larval-concentration maps showed uneven exposure to simulated larvae across 2,895 sub-reef sites on 113 reefs during the 2022–23 spawning season. Persistent day-10 hotspots were concentrated on mid-shelf reefs south of the Lizard Island group, including Williamson, Mackay and Osterlund (Figure 4). Reefs toward the northern end of the domain generally ranked lower.

The Lizard Island group was not a persistent concentration hotspot at the later larval ages examined. Some releases produced higher concentrations around Lizard earlier in development, but these peaks generally did not persist (Appendix K). This may partly reflect the domain boundary, which excludes larval sources on reefs outside the model.

Cross-shelf patterns differed along the domain. In its central part, later-age concentrations were

generally higher on mid-shelf reefs than on Ribbon Reefs at similar latitudes; the southern Ribbon Reefs did not show the same deficit. The mid-shelf–Ribbon contrast qualitatively matches the irruption contrasts in the regional monitoring checks (Appendix I), while the one-season concentration result remains sensitive to the domain boundary.

Following this project, network analysis will examine when reefs in the initiation box exchange larvae within tightly connected groups, and when groups become relatively isolated. Combining these results with larval-concentration patterns could identify periods and locations favourable to irruption initiation or escalation and feed into an early-warning system. The planned work will compare these signals with long-term monitoring and reef-level irruption estimates (Stewart et al., 2026). Appendix K describes the network tests needed to assess this use.

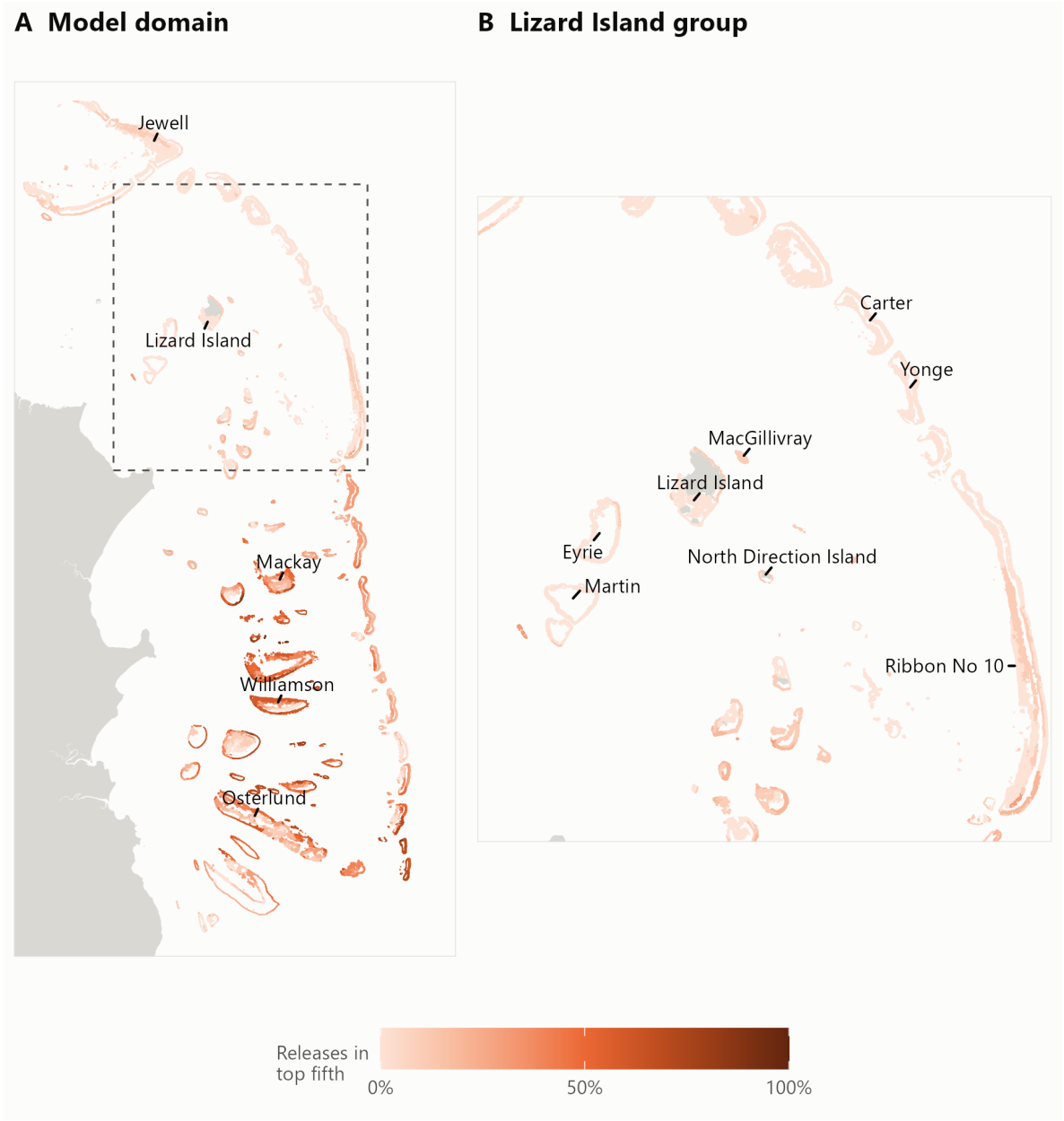


Figure 4: Persistence of simulated larval-concentration hotspots at day 10, across the model domain (A) and around Lizard Island (B). Colour shows the proportion of releases in which a sub-reef site ranked in the highest fifth of concentration within that release. The dashed box marks the enlarged area; grey shows land. This reformats concentration figure lc2 from the project analysis. High concentration does not by itself establish local retention, source strength or settlement.

3.4 Historical COTS and coral dynamics provide a basis for testing control strategies

The updated metacommunity model reproduced the broad sequence of historical irruption waves and coral decline and recovery on the four calibration reefs (Figure 5). The comparison used 5 random-seed pairs, with additional parameter tests giving 65 runs each with and without recorded culling. Lizard Island's first (1990s) COTS peak remains too small, and coral trajectories differed from the surveys in some periods. Manta and photo-transect coral

estimates also differed because they sampled different parts of the reef.

The model represents the build-up of the fifth primary irruption wave, but exceeds observed densities on several reefs. This overshoot could understate the relative effect of contemporary culling, a sensitivity that requires testing. The paired culling trajectories demonstrate the model's response to recorded removals; they do not yet provide an independently validated estimate of the program's impact since 2012.

Exploratory projections compare continued, increased and reallocated diver effort through 2050 to demonstrate the new modelling capability. Increased effort reduced the simulated near-term COTS peaks, but coral gains varied with reef and climate pathway. These scenarios demonstrate the comparisons the model can support; they are not forecasts of likely outcomes. Appendix I presents the scenarios alongside recent-density and additional-reef checks.

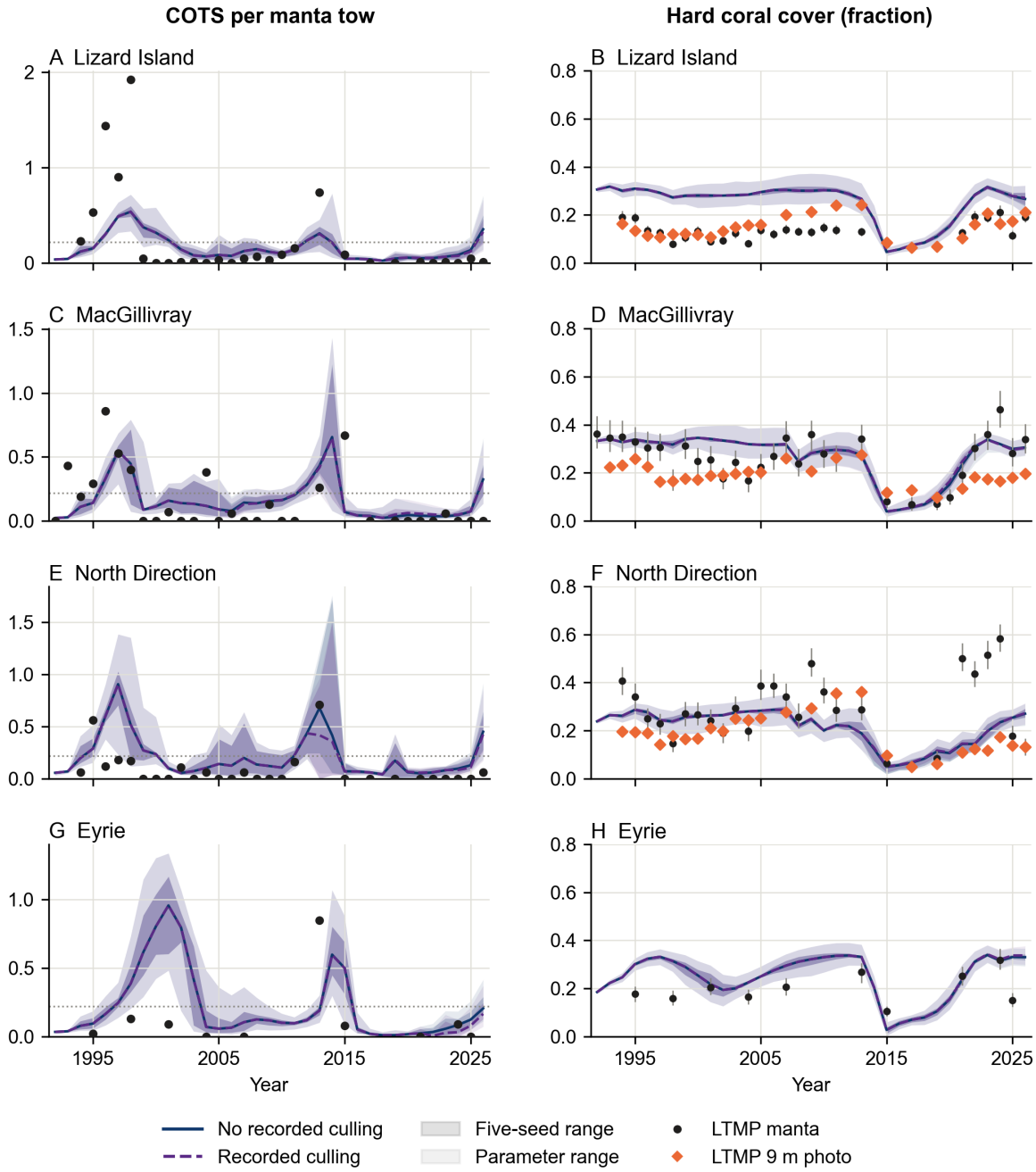


Figure 5: Historical COTS and coral dynamics on the calibration reefs, combining source Figures 1 and 4 from the 21 October snapshot. Left: manta-tow equivalents (dotted line: 0.22 COTS per tow). Right: hard coral cover, modelled relative to habitable area. Blue solid lines omit recorded culling; purple dashed lines apply it. Lines are five-seed means; darker bands span those seeds and lighter bands span all 65 parameter-test runs per culling state. These ranges are not confidence intervals. Black points show LTMP manta observations (coral bars: reported 95% intervals); orange diamonds show 9 m photo-transects. Survey supports differ. These are calibration and diagnostic comparisons.

4 Implications for management

4.1 Combining recent observations with habitat for daily searches

- **Combine habitat with recent observations.** Recent cull and manta evidence improved search priorities and retention of high-density dives. Habitat provides the first search order where observations are missing. Gains depend on the retention required; Appendix H gives the comparisons.
- **Direct effort toward overlooked habitat.** Maps and exploration areas give operators a consistent search order beyond familiar sites. The gains partly formalise crew knowledge; field trials must assess the proposed exploration areas. Appendix G details product design and Appendix H the supporting evidence.
- **Retain follow-up coverage.** Annual full surveys, pre-closure checks and new-evidence checks could find populations deferred by the ranking. Assess missed high-density dives, control delays and all follow-up effort before selecting stopping rules or claiming annual time savings. Appendix L sets out the design.

4.2 Maintaining coverage during manta surveillance

- **Use habitat to guide additional searches.** Habitat-ranked tows did not outperform the evenly spaced thinning benchmark. Thinning recorded tows is not a practical field route; operators need spatial coverage, recent observations and local knowledge. Test added searches in habitats missed by perimeter routes. Appendix H gives the surveillance results and Appendix L explains their operational use.

4.3 Assessing the culling program from 2012 to the present and beyond

- **Measure local and downstream benefits.** The metacommunity model provides a framework for comparing recorded culling since 2012 with the same history without control. Track COTS abundance, larval export and coral loss and recovery, while stating how outside larval supply is treated. Validation and the complete paired assessment remain necessary. Appendices G, I and L describe the design and current model status.

4.4 Projecting suppression of the fifth primary irruption wave

- **Compare control strategies and resources.** Use the model to compare current effort, increased capacity, reallocation among larval-source reefs and starting or stopping rules. Report both local and downstream coral outcomes. The initial projections demonstrate this capability, rather than forecasting management benefits. Appendices I and L give the scenarios and development needs.
- **Develop an early-warning application.** Combine larval concentration with network analysis to identify when initiation-box reefs form connected groups that could favour irruption initiation or escalation. Test the signals against monitoring and reef-level irruption estimates before using them operationally. Appendix K sets out the proposed analysis.

4.5 Monitoring operational performance

- **Trial the ranking with crews in January 2027.** Compare it with normal planning and assess the proposed 10–15% reductions in searches and low-density or empty dives. Agree the allowance for missed >0.04 dives and the maximum delay to checking deferred sites. Record initial and follow-up detections, exploration yield, annual coverage, travel and diver effort. Appendix L retains the field-trial design and author-completion items.

5 Limitations

The search evaluation used outcomes from dives crews chose to make. It cannot establish densities at never-dived sites or the consequences of changing crew behaviour. Reefs without recent observations depend mainly on habitat, where discrimination is weak. Spatially held-out habitat scores reduce bias from scoring a reef with its own training data, but the season-by-season evidence test did not also restrict the habitat fits to preceding years. The evidence regression was tested forward in time; the entire combined modelling process was not a fully prospective forecast.

Spatial coverage also limits interpretation. The primary habitat layer joins models with different predictor coverage, and their rankings differ near that boundary. The explanatory joint habitat model relies heavily on fallback predictors in some regions. Values from the manta-trained model away from tow paths are extrapolations, including at some proposed exploration areas. The retrospective thinning result also depends on an inferred order around the reef, which may be less suitable for surveys that do not follow the perimeter. None of the proposed 2027 products has yet demonstrated its performance in a field trial.

The metacommunity results are calibration, diagnostic comparisons and exploratory projections within one northern reef cluster. Replicated runs and additional-reef checks improve the assessment but do not complete independent regional validation. Historical peak heights and between-irruption densities depend on the assumed relationship between manta observations and true abundance. Recent cull and SALAD observations helped construct corrected irruption inputs and also assess their fit. The current calibration uses five seed pairs and parameter sensitivity tests; projections additionally span five climate models. Those ranges do not capture all structural uncertainty. In particular, excess simulated COTS in 2026 carry into the near-term projections, outside larval supply is only approximately coupled to future coral, and thermal acclimation scenarios have not yet entered the reported ensemble. No estimate of historical program impact or future coral protection should be presented as validated until these assumptions and held-out performance have been assessed.

Larval connectivity has different roles in the two strands. It was not a predictor in the within-reef habitat ranking, whereas it carries population effects among reefs in the metacommunity model. The latter still uses reef-level transport and does not establish fine-scale transport among sites. Uncertainty in that transport, external larval pressure and coral disturbance must accompany any assessment of downstream control benefits.

6 Recommendations and next steps

1. Agree the tolerance for missed dives above 0.04 CPUE, then trial the 2027 ranking with crews. Record site searches, tows, dive effort, intermediate and high-density outcomes, exploration visits and later coverage checks. The revised retrospective trade-off does not establish the proposed adoption targets.
2. Maintain representative surveillance coverage and trial habitat-guided searches beyond familiar routes. The retrospective thinning benchmark is not an operational route; Appendix H documents the comparison.
3. Trial a small number of exploration areas per reef visit and record outcomes, including nil results, so that the evidence base can expand beyond familiar sites.
4. Refresh the ranking each season and repeat its temporal evaluation as new observations arrive. Compare the product with current crew practice as well as a no-information baseline.
5. Complete expert review with GBRMPA, vessel crews and Traditional Owner operators,

and agree ownership of monitoring and integration into Field Maps and the COTS Dashboards. [TO CONFIRM: participants, timing and operational owner.]

6. Obtain survey-effort information for additional observation sources, including ReefScan, to support independent assessment beyond places selected for cull dives.
7. Resolve the recent COTS population mismatch and complete independent validation of the metacommunity model before estimating the program's impact from 2012 to the present.
8. Extend the paired historical culling diagnostics to a complete program-impact assessment, reporting effects on COTS populations, larval export and coral trajectories, with the limits imposed by the external boundary stated explicitly.
9. Refine the exploratory fifth-wave projections under agreed effort budgets, climate conditions and starting and stopping rules. Test the recent population state and outside supply assumptions, and report domain-wide as well as local coral consequences.
10. Test finer-scale hydrodynamic connectivity against a distance-based control, and assess any coupling of search rankings to intervention scenarios before using it for management comparisons. [TO CONFIRM: link to Work Plan output F and the agreed pilot region.]

Acknowledgements

This project was funded by the COTS Control Innovation Program (Reef Trust, AIMS, GBRF), project CCIP-BR-13. [TO CONFIRM: exact funding wording; partner organisations (AIMS, QUT, JCU, GBRMPA); data providers (COTS Control Program vessel crews and contractors, GBRMPA, SALAD, RHIS, ReefGuide); Traditional Owner acknowledgement of Sea Country in the authors' own words.]

References

Bainbridge, S., and Coleman, G. (2024). ReefScan transom. Standard operating procedure number 15 (v1.08). Townsville: Australian Institute of Marine Science. doi: 10.25845/HD2Z-KM55

Beeden, R. J., Turner, M. A., Dryden, J., Merida, F., Goudkamp, K., Malone, C., et al. (2014). Rapid survey protocol that provides dynamic information on reef condition to managers of the great barrier reef. *Environmental Monitoring and Assessment* 186, 8527–8540. doi: 10.1007/s10661-014-4022-0

Bozec, Y.-M., Hock, K., Mason, R. A. B., Baird, M. E., Castro-Sanguino, C., Condie, S. A., et al. (2022). Cumulative impacts across Australia's Great Barrier Reef: A mechanistic evaluation. *Ecological Monographs* 92. doi: 10.1002/ecm.1494

Chandler, J. F., Burn, D., Caballes, C. F., Doll, P. C., Kwong, S. L. T., Lang, B. J., et al. (2023). Increasing densities of pacific crown-of-thorns starfish (acanthaster cf. Solaris) at lizard island, northern great barrier reef, resolved using a novel survey method. *Scientific Reports* 13, 19306. doi: 10.1038/s41598-023-46749-x

Chen, T., and Guestrin, C. (2016). XGBoost: A scalable tree boosting system., in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, (ACM), 785–794. doi: 10.1145/2939672.2939785

Condie, S. A., Anthony, K. R. N., Babcock, R. C., Baird, M. E., Beeden, R., Fletcher, C. S., et

al. (2021). Large-scale interventions may delay decline of the Great Barrier Reef. *Royal Society Open Science* 8. doi: 10.1098/rsos.201296

de Almeida, P. R., Crocker, R., Tan, D., Bairos-Novak, K. R., Ani, C. J., Benthuysen, J. A., et al. (2026). CoralBlox: A computationally efficient coral model for decision support. *bioRxiv [Preprint]*. doi: 10.64898/2026.04.13.718315

De'ath, G., Fabricius, K. E., Sweatman, H., and Puotinen, M. (2012). The 27-year decline of coral cover on the Great Barrier Reef and its causes. *Proceedings of the National Academy of Sciences* 109, 17995–17999. doi: 10.1073/pnas.1208909109

Fletcher, C. S., Bonin, M. C., and Westcott, D. A. (2020). An ecologically-based operational strategy for COTS control: Integrated decision making from the site to the regional scale. Cairns: Reef; Rainforest Research Centre Limited. Available at: https://nesptropical.edu.au/wp-content/uploads/2020/04/NESP-TWQ-Project-3.1.1-Technical-Report-2.pdf

Great Barrier Reef Marine Park Authority (2021a). GBR10 GBRMP Bathymetry. Available at: https://tiles-ap1.arcgis.com/8gXWSCxaJlFIfiTr/arcgis/rest/services/GBR10_GBRMP_Bathymetry/MapServer

Great Barrier Reef Marine Park Authority (2021b). GBR10 GBRMP Geomorphic. Available at: https://reef-geohub-ext-geohubgbrmpa.hub.arcgis.com/maps/2621ac24d749409395d7bb1dfdec5d5c/explore

Great Barrier Reef Marine Park Authority (2026). Crown-of-thorns starfish control program dashboard. Available at: https://www.gbrmpa.gov.au/our-work/programs-and-projects/crown-thorns-starfish-management/crown-thorns-starfish-control-program-dashboard (Accessed October 16, 2026).

Guillaume, A. S., Ferrari, R., Selmoni, O., Mocellin, V. J. L., Denis, H., Naugle, M., et al. (2025). Harnessing multiscale topographic environmental variables for regional coral species distribution models. *Ecology and Evolution* 15. doi: 10.1002/ece3.71292

Matthews, S. A., Beeden, R., Bonin, M. C., Mellin, C., Pratchett, M. S., Ryan, I., et al. (2025). From business intelligence to conservation intelligence: Operationalising adaptive pest control to protect the resilience of the great barrier reef. doi: 10.32942/X2HD3J

Matthews, S. A., Mellin, C., and Pratchett, M. S. (2020a). Larval connectivity and water quality explain spatial distribution of crown-of-thorns starfish outbreaks across the Great Barrier Reef. *Advances in Marine Biology* 87, 223–258. doi: 10.1016/bs.amb.2020.08.007

Matthews, S. A., Shoemaker, K., Pratchett, M. S., and Mellin, C. (2020b). COTSMod: A spatially explicit metacommunity model of outbreaks of crown-of-thorns starfish and coral recovery. *Advances in Marine Biology* 87, 259–290. doi: 10.1016/bs.amb.2020.09.001

Matthews, S. A., Williamson, D. H., Beeden, R., Emslie, M. J., Abom, R. T. M., Beard, D., et al. (2024). Protecting Great Barrier Reef resilience through effective management of crown-of-thorns starfish outbreaks. *PLOS ONE* 19, e0298073. doi: 10.1371/journal.pone.0298073

Morello, E. B., Plagányi, É. E., Babcock, R. C., Sweatman, H., Hillary, R., and Punt, A. E. (2014). Model to manage and reduce crown-of-thorns starfish outbreaks. *Marine Ecology Progress*

Series 512, 167–183. doi: 10.3354/meps10858

open-AIMS (2026a). ADRIA.jl: COTS calibration and exploratory effort-based projections. Available at: https://github.com/open-AIMS/ADRIA.jl

open-AIMS (2026b). ADRIA.jl: COTS calibration and historical culling workflow. Available at: https://github.com/open-AIMS/ADRIA.jl

open-AIMS (2026c). COTSMod.jl: Population dynamics and observed cull removals. Available at: https://github.com/open-AIMS/COTSMod.jl

open-AIMS (n.d.). ReefGuide.jl: Regional and suitability assessment functions. Available at: https://github.com/open-AIMS/ReefGuide.jl

Ovaskainen, O., Tikhonov, G., Norberg, A., Guillaume Blanchet, F., Duan, L., Dunson, D., et al. (2017). How to make more out of community data? A conceptual framework and its implementation as models and software. *Ecology Letters* 20, 561–576. doi: 10.1111/ele.12757

Plagányi, É. E., Babcock, R. C., Rogers, J., Bonin, M., and Morello, E. B. (2020). Ecological analyses to inform management targets for the culling of crown-of-thorns starfish to prevent coral decline. *Coral Reefs* 39, 1483–1499. doi: 10.1007/s00338-020-01981-z

Pratchett, M. S., Caballes, C. F., Burn, D., Doll, P. C., Chandler, J. F., Doyle, J. R., et al. (2022). Scooter-assisted large area diver-based (SALAD) visual surveys to test for renewed outbreaks of crown-of-thorns starfish (acanthaster cf. Solaris) in the northern great barrier reef. COTS Control Innovation Program. Available at: https://www.barrierreef.org/uploads/CCIP-EIP-02-Early-Investment-Project-COTS-Surveys-Final-Report.pdf

Pratchett, M., Caballes, C., Wilmes, J., Matthews, S., Mellin, C., Sweatman, H., et al. (2017). Thirty years of research on Crown-of-Thorns Starfish (1986–2016): Scientific advances and emerging opportunities. *Diversity* 9, 41. doi: 10.3390/d9040041

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 913–929. doi: 10.1111/ecog.02881

Roelfsema, C. M., Lyons, M. B., Castro-Sanguino, C., Kovacs, E. M., Callaghan, D., Wettle, M., et al. (2021). How much shallow coral habitat is there on the great barrier reef? *Remote Sensing* 13, 4343. doi: 10.3390/rs13214343

Stewart, O., Matthews, S., Warne, D., and Bode, M. (2026). Modelling irruptive species with hybrid machine learning: A case study of crown-of-thorns starfish outbreaks on the Great Barrier Reef. *EcoEvoRxiv [Preprint]*. doi: 10.32942/X2J391

Tikhonov, G., Opedal, Ø. H., Abrego, N., Lehikoinen, A., Jonge, M. M. J. de, Oksanen, J., et al. (2020). Joint species distribution modelling with the R-package Hmsc. *Methods in Ecology and Evolution* 11, 442–447. doi: 10.1111/2041-210X.13345

Uthicke, S., Doyle, J. R., Gomez Cabrera, M., Patel, F., McLatchie, M. J., Doll, P. C., et al. (2024). eDNA monitoring detects new outbreak wave of corallivorous seastar (acanthaster cf. Solaris) at lizard island, great barrier reef. *Coral Reefs* 43, 857–866. doi:

10.1007/s00338-024-02506-8

Zurell, D., Franklin, J., König, C., Bouchet, P. J., Dormann, C. F., Elith, J., et al. (2020). A standard protocol for reporting species distribution models. *Ecography* 43, 1261–1277. doi: 10.1111/ecog.04960

Appendices

Appendix A: Delivered layers and products

All layers are in GDA2020 / MGA zone 55, with checksums in `outputs/layers/layer_manifest.csv`.

Table 4: Delivered layers (`outputs/layers/`) and operational products (`outputs/site_ranking/`).

Role	File	Resolution	Contents
Primary search-planning layer	`cots_search_planning_2026_30m.tif` (+ `_priority_`)	30 m	P(problematic site-year, 2026), RG + year with calibrated clean + year fallback; percentile within reef polygon; priority classes (top 10%, 10–20%, 20–30%)
Clean + year	`cots_legacy_clean_year2026_30m.tif`	30 m	P(problematic site-year, 2026) from the 12-layer clean stack + year
RG + year	`cots_legacy_rg_year2026_30m.tif`	30 m	As above with five ReefGuide layers; missing where ReefGuide layers are missing
Manta model	`cots_manta_within_30m.tif`	30 m	Relative log-odds of strong manta evidence against the reef-year average; percentile within reef
Exploration 2027	`cots_exploration_2027_30m.tif`	30 m	Habitat percentile on eligible cells; distances to recent tows and cull sites
HMSC joint (explanatory)	`cots_hmsc_joint_120m.tif` (+ uncertainty, source)	120 m	Posterior mean P(problematic density given an active reef-year); posterior SD and 5th/95th percentiles
Known aggregation sites	`cots_cull_history_kernel_120m.tif`, `cots_hmsc_history_combined_120m.tif`	120 m	Kernel-weighted prior problematic rate and evidence weight; habitat updated by history (forecast for 2027)
2027 cull-site priority	`cots_cull_site_priority_2027.csv`, `.kmz`, `_reefs.csv`, `_classes_2027.csv`	site	Priority class, rank on reef, P(problematic if dived), evidence behind the score

Role	File	Resolution	Contents
Exploration areas 2027	`cots_exploration_areas_2027.csv`, `.gpkg` (also in the KMZ)	as-2027	Rank, area, habitat percentile, depth, distances to cull sites and recent tows

Appendix B: Predictors

Table 5: Predictor codes. The terrain and coral-habitat surfaces follow Guillaume et al. (2025); additional layers were accessed through ReefGuide (open-AIMS, n.d.). GBR10 mapping sources are Great Barrier Reef Marine Park Authority (2021a), Great Barrier Reef Marine Park Authority (2021b) and Roelfsema et al. (2021). [TO CONFIRM: exact predictor product versions and resampling methods.]

Code	Layer	Stack
DEM	Depth	Clean
SLO	Slope	Clean
EAST, NORTH	Eastness, northness (aspect)	Clean
BPI	Bathymetric position index	Clean
VRM	Vector ruggedness measure	Clean
SVF	Sky-view factor	Clean
HCU, VCU	Horizontal and vertical curvature	Clean
AhyaD, Aspat, Aten	Coral habitat suitability for three *Acropora* taxa	Clean
RG_bathy, RG_slope	ReefGuide bathymetry and slope	ReefGuide
RG_turbid	ReefGuide turbidity	ReefGuide
RG_waves_Hs, RG_waves_Tp	ReefGuide significant wave height and wave period	ReefGuide

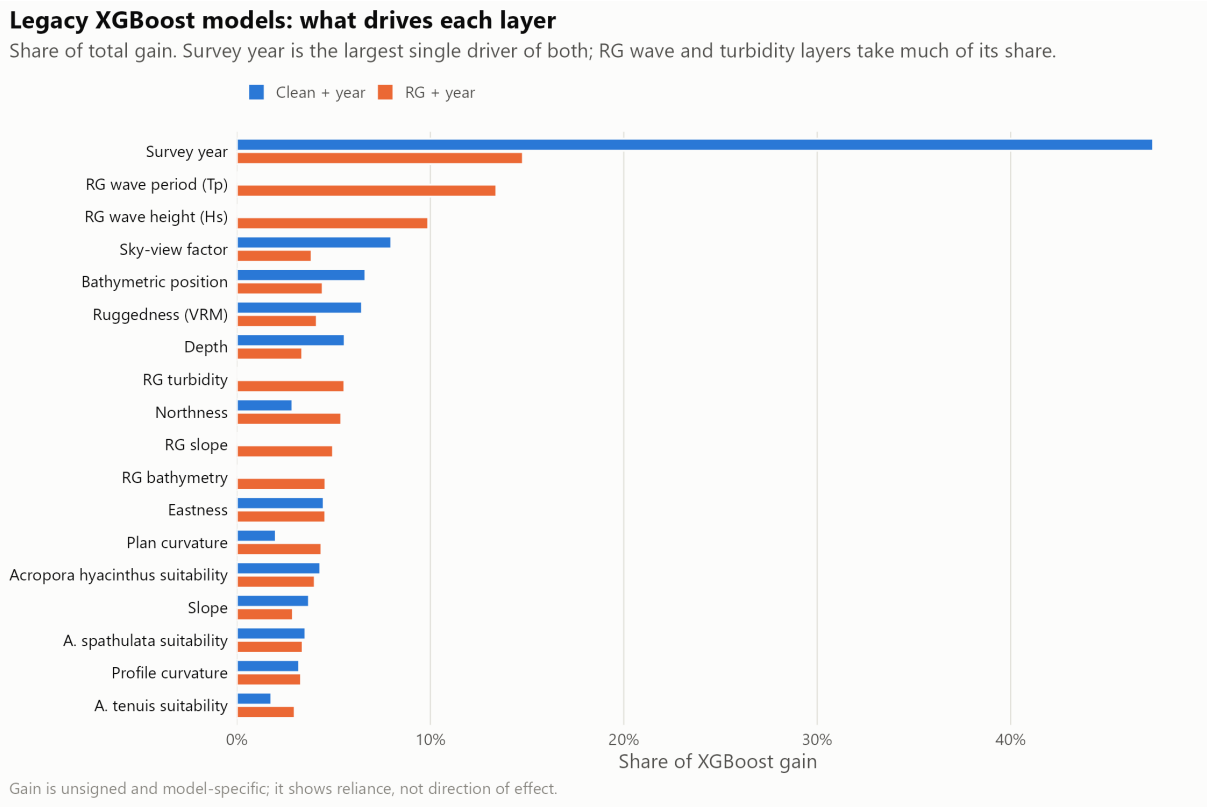


Figure 6: Relative importance (XGBoost gain) of each predictor in the clean + year and RG + year models. Survey year is the largest single driver of both, which is why absolute probabilities mostly reflect prevalence in the prediction year; wave period and height dominate the habitat terms where ReefGuide layers are available.

Appendix C: HMSC joint model

Table 6: HMSC production fits: training rows, convergence diagnostics and apparent (in-sample) AUC for the mapped cull target, and the share of mapped 120 m cells each component fills. All passed the gates (minimum effective sample size ≥ 400, maximum upper PSRF ≤ 1.1).

Component	Training rows	Min ESS	Max upper PSRF	Apparent AUC	Share of mapped cells
enhanced_joint	14,120	705	1.012	0.600	24.6%
clean_joint	15,512	1,645	1.020	0.576	71.8%
clean_native_joint	15,571	409	1.018	0.565	3.6%

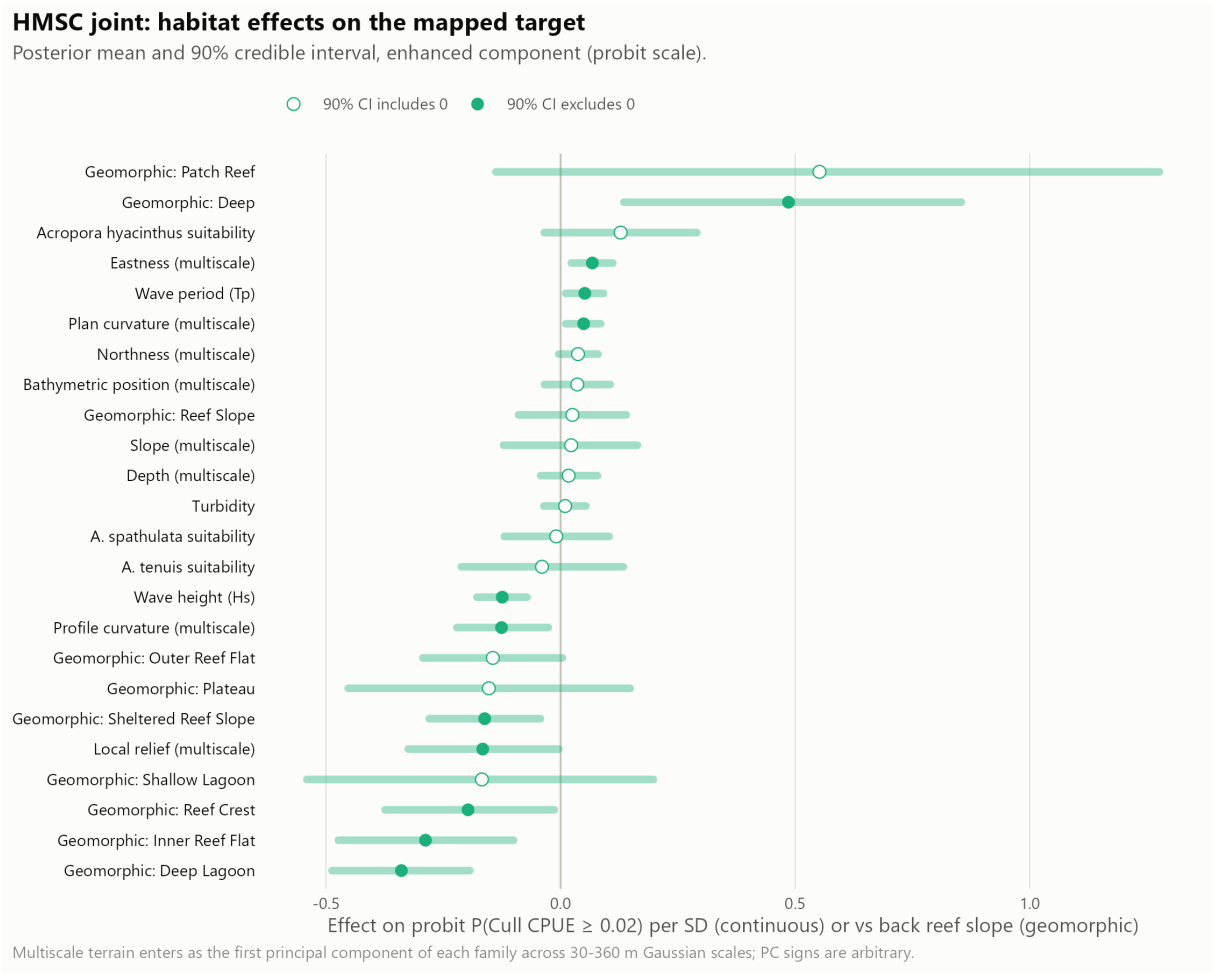


Figure 7: HMSC habitat effects on the mapped cull target, with posterior uncertainty.

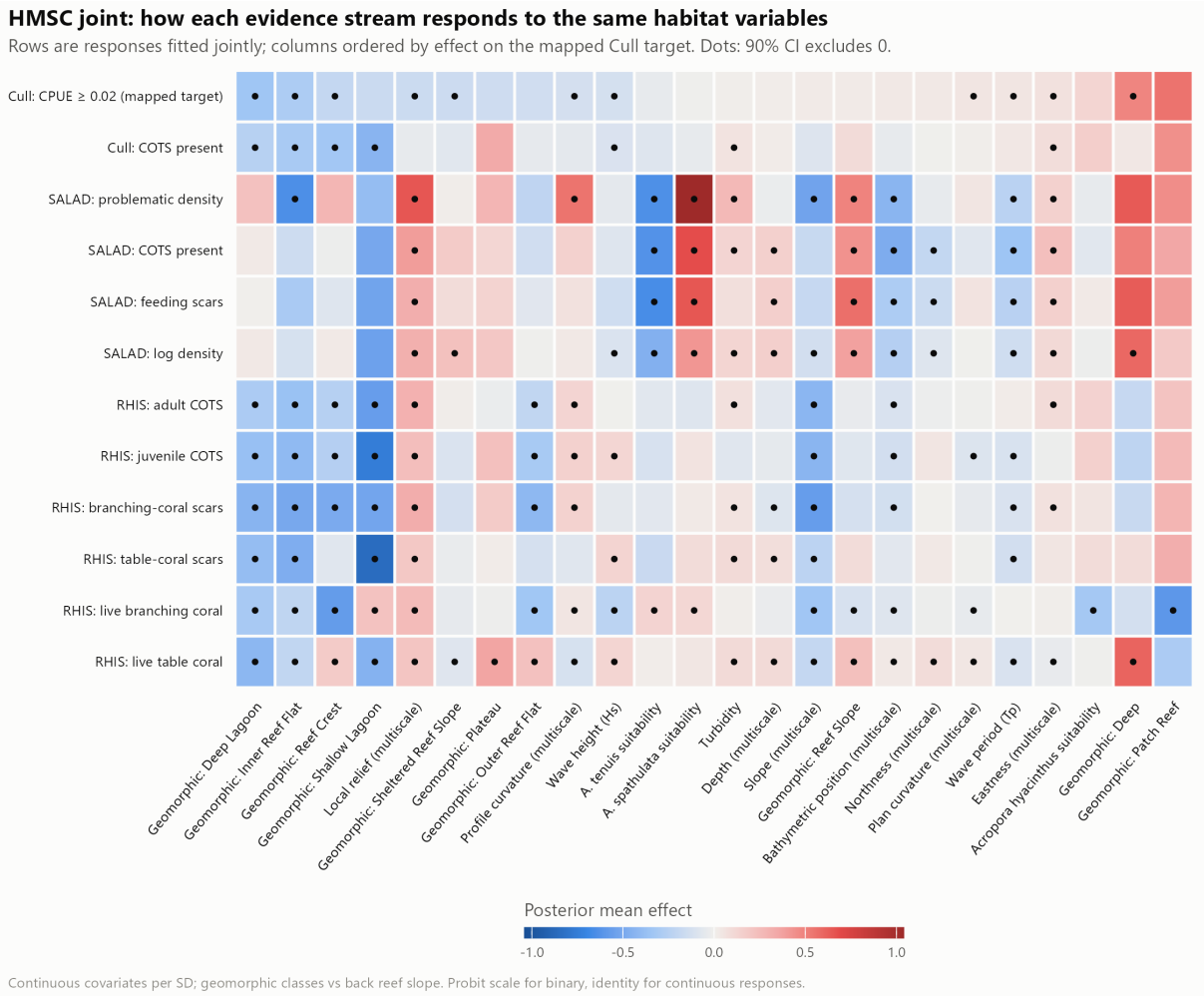


Figure 8: HMSC coefficients across all 12 jointly fitted responses (cull, SALAD and RHIS evidence).

Appendix D: Supplementary validation

Layer agreement and coverage.

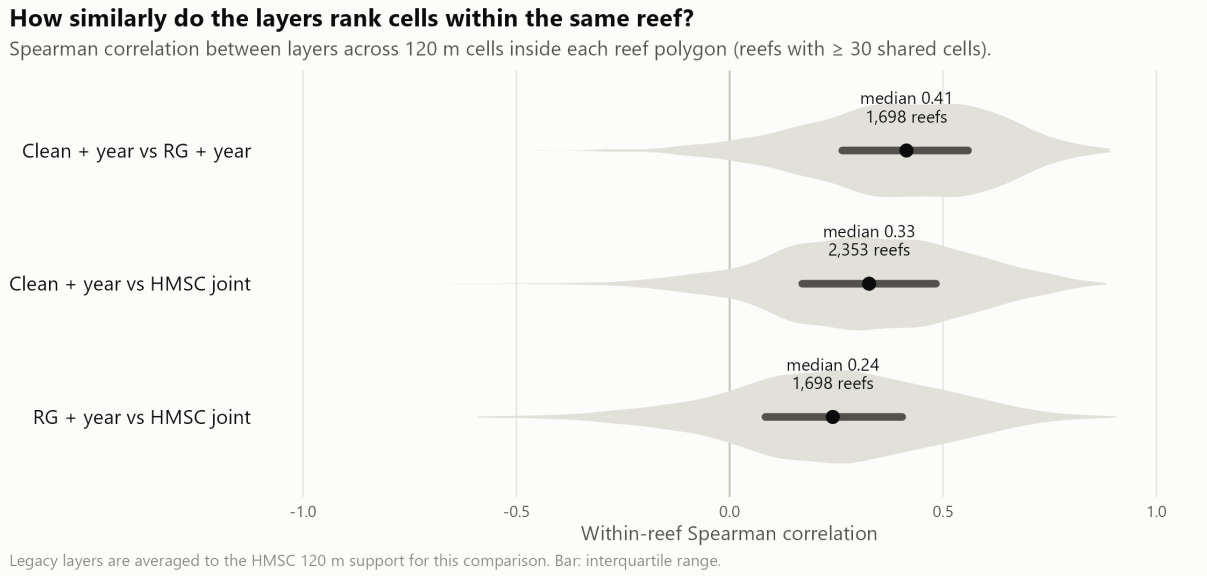


Figure 9: Within-reef agreement (Spearman correlation) between layers across reefs. The layers agree only moderately on which cells rank highest.

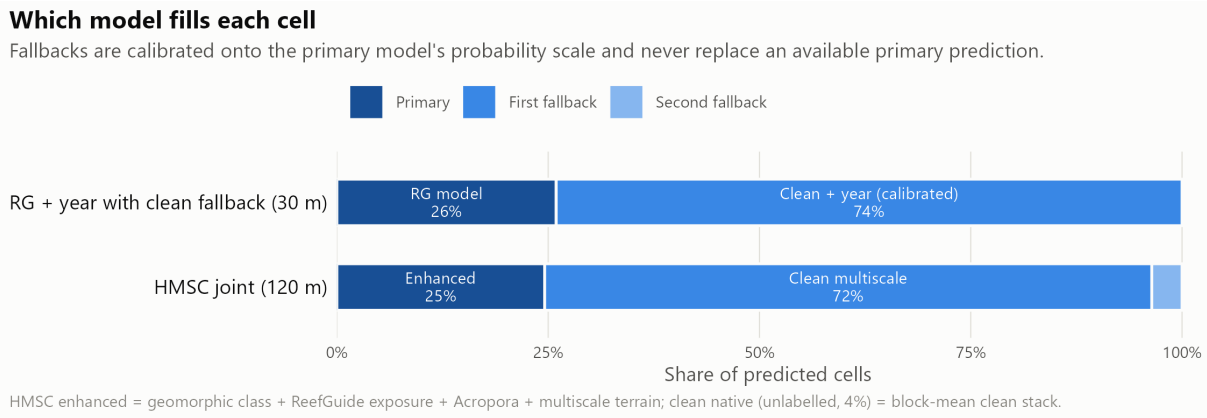


Figure 10: Which model fills each cell of the primary search-planning layer and of the HMSC layer.

Effort cut by the primary layer versus at random. An earlier analysis cut manta and cull effort within each survey or reef-year by the primary layer and compared it with the same cut at random (Table 7, Figure 11). Most of the effort saved is the effect of the cut itself, not of the layer. Systematic thinning and the matched cull-site analysis in the main text supersede this comparison for manta tows and cull sites, respectively.

Table 7: Keeping about 60% of manta tows or cull sites within each survey or reef-year, 2022–2026: primary layer versus the same cut at random. Cull sites were scored by held-out predictions.

Metric	Primary layer	Random
Manta: strong-evidence tows kept	67.6%	60.9%
Manta: outbreak reef-years still detected	88.6%	87.0%
Cull: share of COTS at kept sites	69.9%	62.5%
Cull: share of ghost dives avoided	40.9%	36.5%

Table 7: Keeping about 60% of manta tows or cull sites within each survey or reef-year, 2022–2026: primary layer versus the same cut at random. Cull sites were scored by held-out predictions.

Metric	Primary layer	Random

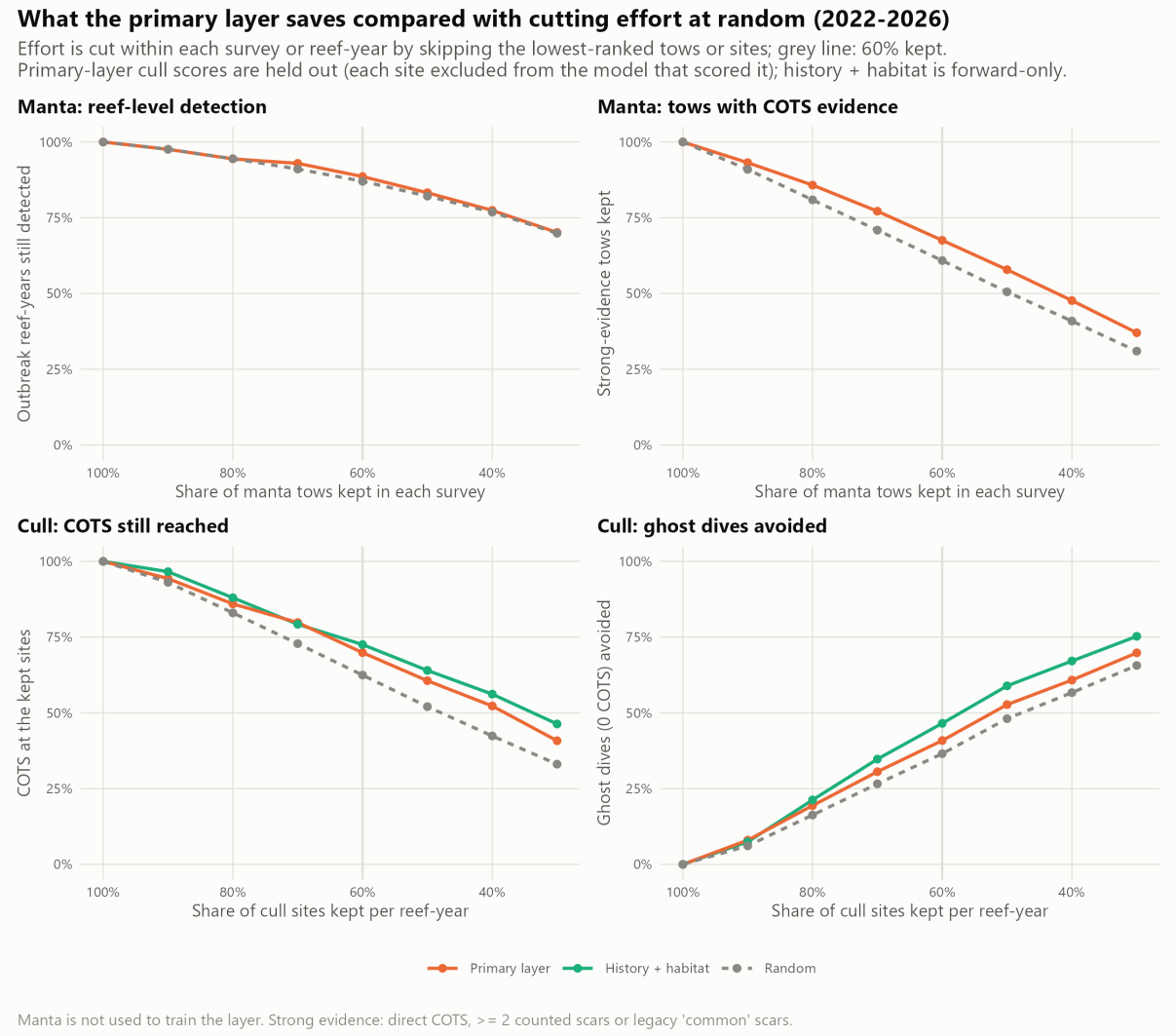


Figure 11: Manta and cull effort cut by the primary layer versus at random (2022–2026).

Example search plans. On Batt Reef (manta survey 2025-08-04), keeping the 79 of 131 tows that crossed the highest-ranked cells would have retained 32 of 42 strong-evidence tows; on Eyrie Reef (2023-10-26), 56 of 92 tows would have retained 9 of 11 (Figure 12). Both reefs are in the layer's training data, so this illustrates the planning logic only; the held-out results in the main text are the evidence.

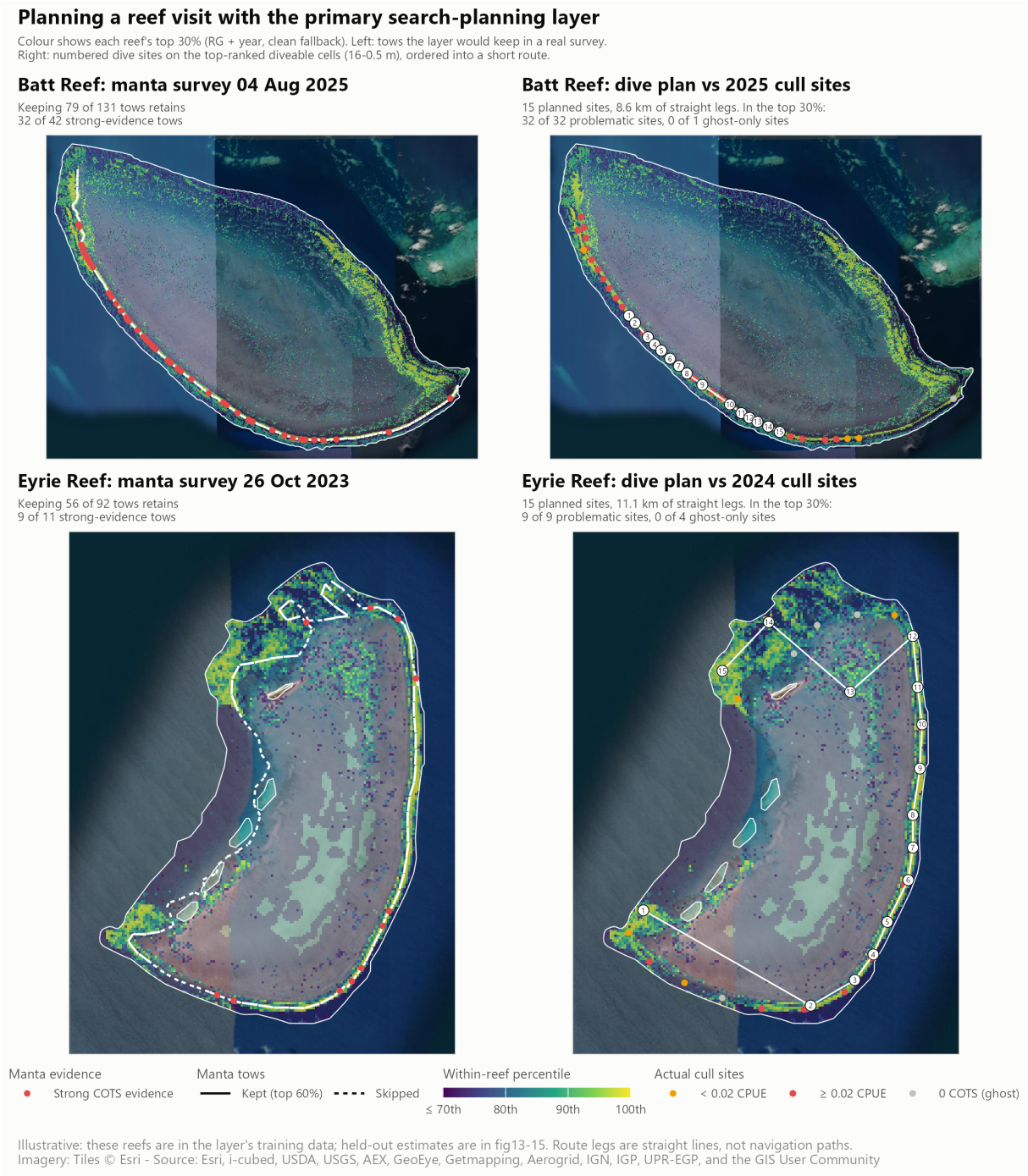


Figure 12: Example use of the primary search-planning layer on Batt Reef (top) and Eyrie Reef (bottom). Left: manta tows the layer would keep (solid) or skip (dashed), with strong-evidence tows in red. Right: fifteen planned dive sites on top-ranked diveable cells (0.5–16 m, at least 400 m apart), ordered into a route, against that season's cull outcomes. Illustrative only: these reefs were in the training data.

Cull history on four reefs.

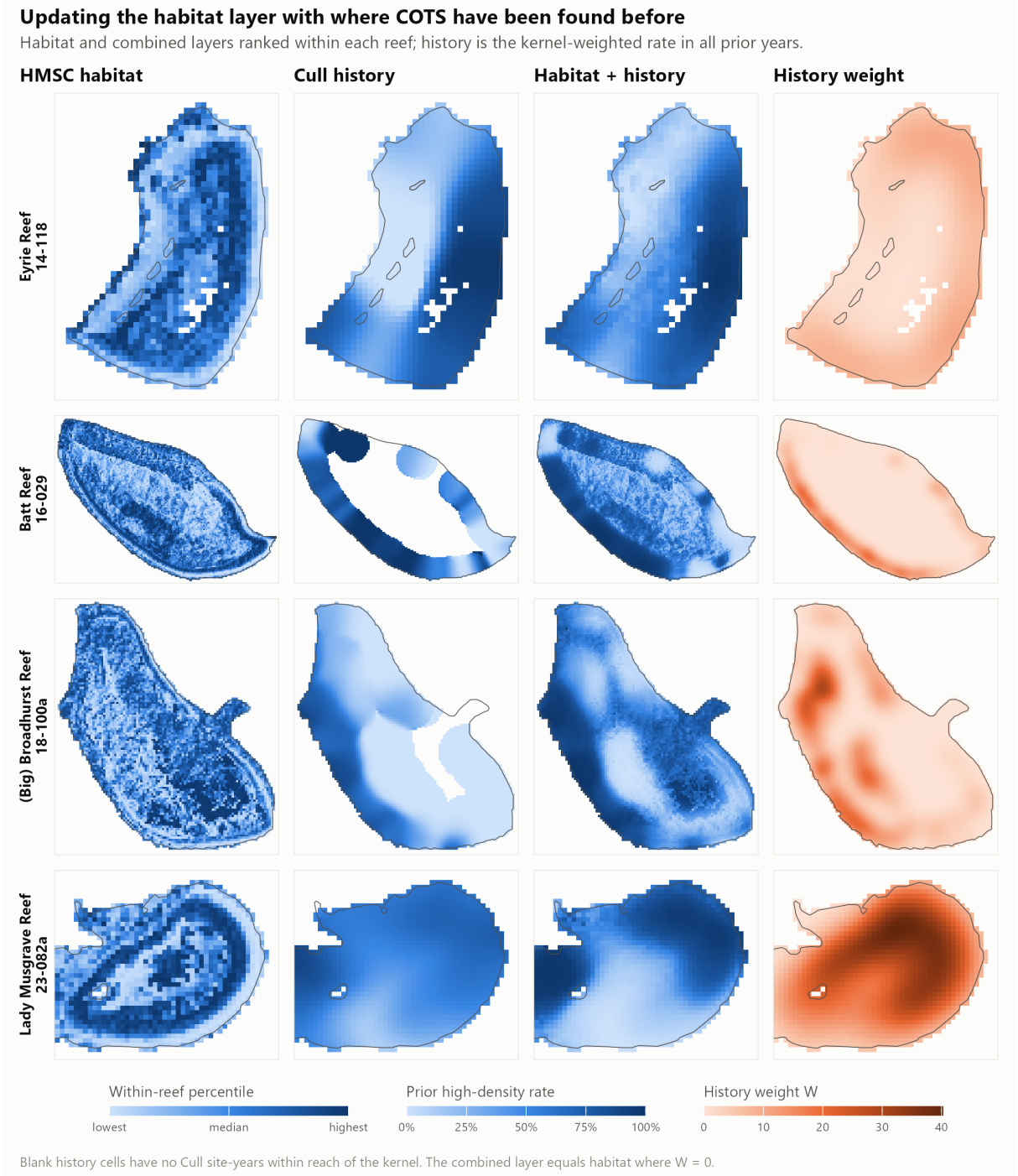


Figure 13: Habitat, cull history and the combined layer on the four reefs, with the history evidence weight. The combined layer is dominated by past finds wherever history exists.

ReefScan check. 237 presence-only ReefScan detections on 29 reefs were compared with each layer's within-reef percentile. Detections sat near or below the median for every layer (mean percentile 0.41–0.53). Without survey tracks that recorded no detections, this mostly reflects where ReefScan was towed, so the check is inconclusive.

Appendix E: Alignment with the CCIP-BR-13 Work Plan

Table 8: Work Plan outputs and where they are reported. The Work Plan proposed validating tow paths at the 50th and 80th percentile thresholds; this report instead evaluates every level of tow retention from 90% to 30% against random and systematic baselines.

Work Plan output	Delivered in this report	Status
E. Finalised GBR-wide habitat suitability model and surveillance paths	30 m habitat layers (Appendix A); held-out evaluation of model-ranked versus systematic manta thinning (Appendix H); example search plans (Appendix D)	[TO CONFIRM]
E (extension). Cull-site selection	2027 cull-site priority ranking and exploration areas (Section 3.1)	[TO CONFIRM]
F. Fine-scale connectivity network at four reef clusters	Hydrodynamic and connectivity Methods (Section 2.5); preliminary larval dispersal Results (Section 3.3)	Concentration patterns described for the Lizard region; expert methods and matrix results pending. [TO CONFIRM: delivery and coverage of all four clusters.]
G. Prototype fine-scale metacommunity model	Main Methods/Results and Appendices G and I	Prototype and exploratory effort projections implemented; independent validation and full program-impact assessment incomplete
H. Final data sets and public report (FAIR)	This report; layers, tables, threshold reassessment and code	[TO CONFIRM: repository, DOI and licence]
I. Peer-reviewed publication	Manuscript in preparation for *Frontiers in Marine Science*	[TO CONFIRM]

Appendix F: Reproducibility

The pipeline is in `DataProcessing/cots_sdm_v2/final/` [TO CONFIRM: public repository URL and release DOI]. Inputs are pinned by SHA-256 checksum in `config.R`; the dated calibration snapshot additionally checks member files against its pinned metadata. Run scripts in the order given in `README.md`, with `07` (figures) last. Script 25 imports the 21 October calibration and projection summaries. Every reported analysis result is read from `outputs/tables/` or `outputs/site_ranking/` when the report is rendered. COTS Control Program cull and manta records are held by GBRMPA and are not public [TO CONFIRM: access statement].

Appendix G: Detailed model specification and evaluation

The main Methods define each model by its purpose. This appendix retains the implementation detail needed to reproduce it. Original software labels are used in archived tables and figure legends; their meanings are mapped below.

Original label	Description used in the main report
clean + year	Annual full-extent model (13 predictors)
RG + year	Annual limited-extent model (18 predictors)
manta / manta with year	Annual habitat model fitted to manta observations
manta within reef-year	Manta habitat model accounting for differences among reef-years
combined site ranking	Recent-evidence site ranking
HMSC joint	Multi-observation habitat model
HMSC + history	Habitat updated by longer cull history; distinct from the operational site ranking

The habitat-model reporting follows the Overview, Data, Model, Assessment and Prediction (ODMAP) protocol for species distribution models (Zurell et al., 2020). This structure specifies the management question and prediction target, the observations and predictors used, the model fitting, how predictions are evaluated, and how the resulting maps should be used. The sections below describe these elements for the habitat and search-ranking analyses; the metacommunity methods describe the population model, calibration and prescribed culling separately.

Data and response definitions

The COTS Control Program's full cull dataset covered 2002–2026 and supplied the site, date, number of COTS culled and bottom time for each dive. Records with missing counts or non-positive bottom time were excluded. Records matched to designated cull sites covered 4,299 sites on 339 reefs; the wider cleaned dataset contained records from 361 reefs. The archived habitat fits used the full-period source dataset, while each validation applied its stated spatial and temporal subsets. [TO CONFIRM: data-access agreement.]

Two CPUE thresholds serve different purposes. The archived habitat fits and original validation defined a problematic dive as CPUE ≥0.02 COTS per minute of bottom time. A ghost dive found no COTS; a low-density dive found COTS at positive CPUE <0.02. This lower threshold identifies dives that might be avoided while retaining some control below the ecological management target. The revised management evaluation instead penalises missed dives above 0.04. Positive dives from 0.02 to 0.04 inclusive form an intermediate group: some can be deferred without the penalty assigned to missed >0.04 dives. Thus exactly 0.02 remains positive under the original response and belongs to the intermediate group in the new evaluation; exactly 0.04 also belongs to the intermediate group. Ecologically sustainable densities vary with coral cover, so 0.04 is the operational criterion for this assessment, rather than a universal coral-growth threshold (Plagányi et al., 2020). Habitat fits used a site-year response, while search selection was assessed against individual dive outcomes. The two thresholds do not change the definition of a manta aggregation.

The habitat model fitted to manta observations used 198,852 labelled tows in 1,015 reef-years, including 16,199 with strong COTS evidence. Strong evidence meant a direct COTS sighting, at

least two counted feeding scars, or scars recorded as “common” in legacy surveys. Strict negatives recorded neither COTS nor scars; weak single-scar records were excluded from model fitting. The observation archives included Control Program crews (2016 onward), the Reef Joint Field Management Program (2012 onward) and AIMS LTMP (1992 onward), as specified by the author. SALAD (Pratchett et al., 2022) and RHIS (Beeden et al., 2014) observations contributed to the joint habitat model. ReefScan imagery (Bainbridge and Coleman, 2024) supplied a separate presence-only check in script 10 and was not a fitted response in the delivered joint model.

Predictors were assembled on a 30 m grid in GDA2020 / MGA zone 55. The clean stack contained depth, slope, eastness, northness, bathymetric position index, vector ruggedness, sky-view factor, horizontal and vertical curvature, and three coral habitat-suitability surfaces for *Acropora* taxa (Guillaume et al., 2025). Five ReefGuide layers added significant wave height, wave period, turbidity, bathymetry and slope. The joint model also used geomorphic classes and reef features. Appendix B lists predictor codes. Sources: Great Barrier Reef Marine Park Authority (2021a), Great Barrier Reef Marine Park Authority (2021b), Roelfsema et al. (2021) and open-AIMS (n.d.). ReefGuide coverage was reduced in some deeper areas, particularly beyond about 15–20 m. [TO CONFIRM: predictor product versions and resampling methods.]

Habitat models and mapped layers

The clean + year model used gradient-boosted trees [XGBoost; Chen and Guestrin (2016)] to predict the problematic-density indicator from the clean stack and survey year. The RG + year model added the five ReefGuide predictors. These cull-trained models learned from locations selected by crews, often following manta evidence, so their scores reflect the program’s sampling footprint as well as habitat. The operational refits used 1,500 boosting rounds, a learning rate of 0.00137382379588326, maximum tree depth 5, minimum child weight 20 and gamma 10^{-10}, retaining the archived hyperparameters. These settings were not retuned on the evaluation reefs.

The primary search-planning layer used RG + year where available and clean + year elsewhere, calibrated onto the RG probability scale before combining them. Each 30 m cell was then ranked within its reef polygon, with map priority classes for the top 10%, 10–20% and 20–30%. RG + year covered 61% of the reef-polygon support cells used in the coverage analysis. Of the mapped reefs, 54 used RG alone, 2,586 mixed the two sources and 920 used the clean fallback alone.

Manta-trained XGBoost models used the RG stack and the same boosting settings. Predictor values were averaged along each tow. A model with survey year was compared with a model designed specifically to rank locations within reef-years. For the latter, the logit of the reef-year’s strong-tow proportion, with a half-count correction, entered training as a fixed offset and survey year was omitted. The offset was omitted when predicting relative habitat scores because it was constant within a reef-year. The resulting map represents relative log-odds, not an annual occurrence probability. The cull-trained and manta-trained ranks were also compared as an average rank and as separate terms in the evidence model.

The hierarchical model of species communities [HMSC; Ovaskainen et al. (2017); Tikhonov et al. (2020)] related cull, SALAD and RHIS responses jointly to habitat at 120 m, conditional on an active reef-year. Three predictor components supplied fallbacks where enhanced predictors were unavailable. Production fits used four chains, each retaining 1,000 samples at thinning 30 after 5,000 transient iterations. Convergence was assessed against a minimum effective sample size of 400 and maximum upper potential scale reduction factor of 1.1 (Appendix C). This model supplied explanatory habitat effects and uncertainty.

An additional layer represented known aggregation sites using a Gaussian kernel of the problematic rate in earlier active cull site-years (σ = 500 m). Its rate H and evidence weight W updated the HMSC habitat probability p as $(\kappa p + WH)/(\kappa + W)$, with $\kappa = 2$ for the fixed-setting comparison. History and dive effort were interpreted as records of past finds and crew targeting, rather than independent habitat evidence.

Evaluation design and search-efficiency analyses

Operational discrimination was measured among locations on the same reef in the same year. Within reef-year AUC compared positive and negative locations only within those groups; an AUC of 0.5 indicated no discrimination. Top-20% lift was the share of positives in the highest-ranked fifth divided by 0.2. These metrics were supplemented by the outcomes retained when effort was reduced.

For operational habitat comparisons, reefs were assigned to five folds and every reef was scored by a model refitted without any reefs in its fold (Roberts et al., 2017). The pipeline calls this leave-reef-out validation; it is grouped five-fold validation, rather than one refit per reef. This prevented a reef's cull or manta observations from contributing to its habitat fit. A separate matched manta check scored the same 117,318 tows in 1,003 active reef-years on 402 reefs where all deployed layers were available. That check used the fitted layers and was not a substitute for the refitted operational evaluation.

Manta aggregations were connected groups of strong-evidence tow midpoints linked within 500 m in a survey (reef and date); successive links could join a group wider than 500 m. The principal aggregation analysis required at least two strong tows, with isolated records examined separately. Surveys required at least five scored tows and some strong evidence. An aggregation was retained if at least one of its tows was kept. This measured retention of observed aggregation evidence under retrospective thinning.

Each thinning strategy kept the same number of tows per survey, over retained shares from 90% to 30%. Model thinning kept the highest-scoring tows. Random thinning used the exact hypergeometric expectation of retaining at least one tow in each aggregation. Systematic thinning kept evenly spaced positions in the tow sequence, averaged over 20 starting offsets. Recorded tow order was unavailable, so sequence was approximated by bearing from the survey centre. The comparison therefore concerned thinning existing surveillance, not the performance of newly designed tow routes.

Cull-site analyses ranked every designated site on a reef, including never-dived sites, by its mean habitat score over the polygon. For each retained share of designated sites, the analysis recorded the share of observed problematic, low-density and ghost dives at retained sites. Random selection supplied a no-information expectation: retaining a fraction of sites gives that fraction's expected retention of each observed outcome. Outcomes at never-dived sites were unknown, so this exercise could not measure COTS missed at those sites or changes in crew behaviour under deployment.

To add recent evidence, each site received indicators for any problematic dive there in the previous season and for having been dived with none problematic. Neighbourhood terms were the problematic rate and dive effort among nearby cull dives, and strong-evidence and total manta tows, using a Gaussian kernel with σ = 300 m. Effort and tow counts entered logistic regressions after $\log(1 + x)$ transformation; missing habitat ranks were assigned the midpoint. Habitat alone, evidence alone, habitat + evidence and a simple flagged-sites-first rule were compared. The regressions were fitted on other reef folds, and evidence for a test year came only from preceding seasons. This spatial test did not also restrict regression training outcomes to years before the test year.

Rankings were compared at matched retention of problematic dives because evidence could move both problematic and low-density dives upwards. Retention curves were evaluated over 5% steps in designated sites searched and interpolated at 80%, 85%, 90% and 95% problematic-dive retention. The separate season-by-season test used 80% and 90%. For each test year from 2019 to 2026, its evidence regression was fitted only on earlier dive outcomes, comparing all earlier seasons with the last five or three, and cull-only, averaged cull–manta or separate habitat terms. Habitat covariates in this test were the reef-held-out ranks from the spatial analysis; the habitat fits themselves were not restricted to preceding years. Thus the evidence regression was tested forward in time, while the habitat component was held out spatially.

The >0.04 reassessment kept the saved rankings from the habitat-only, recent-evidence and manta-habitat comparisons and the season-by-season evidence test; none was refitted to the higher threshold. Dive counts and CPUE were rejoined to the saved scores using reef, year and designated site. For the forward test, the recovered dive counts and original outcome classes were checked against the saved evaluation sample, preventing repeated dives at one site from generating duplicated records. The candidate set remained every designated site on each evaluated reef, including never-dived sites. At each rank cutoff, effort was the actual number of designated site-years selected, summed across evaluated reef-years, divided by all candidate site-years. Consequently the proportion searched could differ slightly from the nominal cutoff on reefs with few sites or tied habitat ranks.

The reassessment reported retention and omission of >0.04 dives, low-density and ghost dives avoided, and intermediate dives retained. Random selection used the exact expected outcomes for the same number of sites on each reef; it matched the combined ranking's effort in the spatial comparison and all rankings' effort in the forward comparison. Curves were also interpolated at 80%, 90%, 95% and 99% retention of >0.04 dives, using the first crossing of each target over 5% rank-cutoff steps. This interpolation describes a trade-off between adjacent effort levels, rather than a validated stopping rule for an individual reef. Confidence intervals used 500 paired reef-year bootstrap draws for this reassessment, including the forward scores. Earlier ≥0.02 results retain their original calculations and bootstrap counts. The high-density reassessment retains the original spatial and temporal holdout limits of each ranking.

Site search counts describe candidate locations selected for a visit; tow effort, dive effort and annual coverage are separate quantities. A field trial will record tows and their duration, dives and bottom time, COTS outcomes, added exploration searches, departures from the proposed search order, and later checks of deferred sites. It will compare the ranking with crews' normal planning. Sites deferred on a visit will still be checked annually or before the reef closes to further control visits. Analyses of the full season must count those later checks and added exploration, so that shifting effort between sites or visits is distinguished from reducing total vessel and diver time. Existing records cannot estimate outcomes at unvisited sites or the yield of the proposed new routes.

The HMSC–history forecast used a separate temporal design: for each test year from 2022 to 2026, habitat models were refitted on earlier years and history used earlier active site-years only. First-visit cull site-years were compared within reef-years containing both outcomes, using habitat, prior dive effort, history and their blend. This distinguished habitat transfer from the persistence of previous finds.

Uncertainty intervals used paired bootstrap resamples of reef-years for tow and cull metrics, or surveys for aggregation retention. Operational spatial comparisons used 500 resamples; the season-by-season site-ranking test used 200. The same resampled units were used across scores to preserve paired comparisons.

Seasonal products

For 2027, the selected cull-habitat + evidence model was fitted on 28,431 dives from 2019-2026. Recent evidence covered 1 June 2025 to 31 May 2026, the latest 12 months available, because calendar 2026 contained only January–May. Sites were ranked within reefs and assigned classes A (top 20%), B (20–50%), C (50–80%) and D (bottom 20%). The deployed habitat layer supplied scores for this product; the operational validation used refitted habitat scores.

Exploration areas were drawn from 30 m cells 0.5–16 m deep in the reef's top 30% of habitat, more than 100 m from designated cull sites and more than 150 m from manta tows in the preceding five seasons. Habitat was the mean of the primary-layer and manta-model percentiles, re-ranked within the reef, with the primary layer alone where manta scores were missing. Nearby pockets were joined, large patches split into compact areas of about 9 ha, and outlines smoothed and clipped to the reef outside cull sites. Areas smaller than 0.5 ha were removed; up to ten areas per reef polygon, at least 350 m apart, were retained. These areas identified places to explore beyond the recent search footprint.

Metacommunity model and historical culling

The detailed calibration and spatial-kernel removal specification below records the 15 October setup. Appendix I describes the subsequent changes to settlement, coral settings, reef-wide removal allocation and effort-based projections used in the current Results.

The strategic modelling strand extended COTSMod (Matthews et al., 2020b) through the Julia package COTSMod.jl, coupled to ADRIA and the CoralBlox coral model (de Almeida et al., 2026; open-AIMS, 2026c, 2026b). COTSMod.jl supplied population transitions, coral predation and larval processes; ADRIA supplied the spatial domain, disturbance forcing and scenario execution. The prototype used the Lizard Island cluster, with 2,895 sub-reef sites on 113 reefs. Historical simulations beginning in 1991 were assessed against AIMS Long-Term Monitoring Program (LTMP) manta COTS and coral records on Lizard Island, Eyrie, MacGillivray and North Direction reefs. The model state was expressed as recruit, juvenile and adult densities per hectare of site area, with coral cover represented relative to habitable area.

Site-level habitable fractions came from benthic maps in domain v0.3 and its v0.4 extension. Historical heat stress was supplied through ADRIA, with cyclone exposure examined as an option. Reef-to-reef larval flows were divided evenly among sites within a receiving reef. Habitable fractions and prescribed removals varied among sites, but transport did not yet resolve within-reef flows. The 30 m habitat layers and evidence-based search rankings were candidate inputs for future coupling, rather than validated components of the metacommunity model. [TO CONFIRM: source and attribution of the annual 2018–2022 connectivity matrices, status and delivery date of Work Plan output F, and the planned coupling to the habitat layers.]

The working COTS formulation combined preferred-first per-capita grazing, survival linked to remembered preferred coral cover, adult senescence and preferred-food effects on reproductive condition. Larval production included a low-density Allee effect, juvenile survival included a saturating predator-pit term, and larvae were transported using reef-level connectivity and supplied from outside the domain. External supply was derived from the reef-level irruption-probability hindcast (Stewart et al., 2026); its production multiplier was a calibration assumption, not measured recruitment. Mechanisms suggested by MICE, CoCoNet and ReefMod-GBR were examined as optional alternatives (Morello et al., 2014; Plagányi et al., 2020; Condie et al., 2021; Bozec et al., 2022). Testing a mechanism did not make it part of the working configuration. [TO CONFIRM: attribution and final choice of the annual connectivity matrices; the sub-reef hydrodynamic network had not replaced these reef-level inputs.]

Calibration used bounded comparisons of candidate mechanisms and parameter settings. Each screen was preregistered, included an exact-replay reference and recorded settings, seeds, input checksums and repository revisions. Candidate mechanisms had to pass a qualitative peak-and-trough gate before wider optimisation. Irruption peak presence, timing and height, inter-irruption troughs and coral trajectories were assessed together. Coral dynamics were also assessed with COTS grazing forced from observed records to separate coral-model errors from population-dynamics errors. Simulated adult density was converted to manta CPUE through an explicit observation model. The working linear conversion used $CPUE = qA$, with A in adults ha^{-1} and q = 0.015 COTS tow^{-1} per adult ha^{-1}; alternative scales and density-dependent detection, $CPUE = qA^2/(A + A½)$, were sensitivity analyses. These alternatives were kept distinct because they change both initial-density estimates and the comparison with observations. In-sample calibration and diagnostic screens were not treated as held-out validation or evidence of management benefit.

The replicated historical comparison used score v2, which scored North Direction against its raw series and omitted Eyrie's unobserved first wave from the peak count. Its three retained candidates were compared over five recorded seed pairs. The later v0.4 diagnostic extended forcing to 2026 and used score v3: the manta-based score on 1992–2021 plus 1.5 times a diver-anchored recent-density log error. Original and cull/SALAD-corrected irruption boundaries were crossed with earlier and revised coral thermal settings; exploratory chlorophyll-derived larval-survival modifiers were also screened. These later runs used one seed pair and prescribed culling was off. Their corrected boundary and recent-density score shared evidence, so the comparison was a diagnostic fit rather than a validation. The two score versions were reported separately.

Observed historical culling was included as prescribed removals. Cull records from the licensed CSIRO compilation were retained for reefs in the domain, grouped by site and year, and mapped to model sites on the same reef. Counts were distributed with weights proportional to site area × exp(−$d^2/(2\sigma^2)$) within 3σ of the cull-site centroid, normalised to preserve total counts. The primary width was σ = 210 m, with nearest-site allocation (σ = 0) and σ = 420 m as sensitivities; the nearest site was used if no centroid lay within the kernel. Allocated counts were divided by model-site area to obtain removals in COTS ha^{-1}. COTS below 15 cm were assigned to the juvenile stage and those at least 15 cm to adults, an explicit size-to-stage approximation.

September–December removals were assigned to the following LTMP reporting year and applied before spawning; January–August removals were applied after spawning in their reporting year. The annual implementation placed these removals immediately before or after the COTS predation and population-transition call. Requested removals were capped at the simulated juvenile or adult population present, recruits were left unchanged, and adult age and size substates were scaled by the surviving adult fraction. The model logged actual removals separately from requested removals, allowing a shortage of simulated COTS to be diagnosed rather than removing animals that were absent.

This culling implementation prescribed observed catches. It did not yet predict diver search, catchability, effort costs or adaptive starting and stopping decisions. The intended scenarios—primary irruption suppression, starting and stopping rules, and downstream consequences—require further calibration and independent validation before their management performance can be assessed.

Design of the historical and future control assessments

The intended historical assessment pairs simulations with recorded removals from 2012 onward and simulations without program removals, starting from the same pre-control population and coral state and using matching environmental sequences and random seeds. Outcomes will

include actual removals, infeasible requested removals, COTS abundance, larval export, cumulative coral consumption and coral cover. Holding external larval supply identical estimates the effect of control within the model domain conditional on unchanged outside reefs. A regional no-culling assessment instead requires an external supply history consistent with no control outside the domain. This experiment is a planned extension of the prescribed-removal diagnostics reported here; no completed paired program-impact estimate is asserted.

Future comparisons will examine stopping when local coral protection is no longer achievable versus continuing to reduce larval export, and when to begin or intensify control on reefs whose COTS populations cannot be reduced below a local target. Reallocation within existing vessel capacity and additional capacity will be compared separately, using current control and no control as references. Feasible removals must follow population availability and an effort–catchability relationship, rather than prescribed catches alone. Each paired comparison will share climate, disturbance and random-seed sequences, and repeat across plausible population, coral, connectivity and observation assumptions. Local and downstream coral outcomes and the resources required for fifth-wave suppression will be reported together. [TO CONFIRM: final effort–catchability model, projection horizon, vessel budgets, intervention rules and climate scenarios agreed with partners.]

Software and reproducibility

The SDM analyses used R 4.5.0 with xgboost, Hmsc and terra. Inputs were pinned by SHA-256 checksum in the final pipeline configuration, and reported results were read from output tables when the documents were rendered. [TO CONFIRM: package versions for the archived fits and a release identifier for the final analysis.] Metacommunity methods were checked against local COTSMod.jl and ADRIA code on 15 October 2026. The citations record both repository revisions and declare the uncommitted ADRIA changes inspected for this pass. [TO CONFIRM: Julia and package versions and the frozen code release to accompany submission.]

Appendix H: Detailed tactical model comparisons

These comparisons retain the original ≥0.02 response and the separate >0.04 management assessment. They support the highlights in the main Results and should be read with the model definitions in Appendix G. Original labels in figure legends and source tables refer to the same frozen models.

Recent evidence combined with habitat reduces the cull-site search

In the original ≥0.02 evaluation, ranking designated cull sites by last season's cull dives and manta tows, with the habitat layer ordering sites that lack recent evidence, reduced the search needed for the same problematic dives (Table 10, Figure 14). The ranking is a logistic model fitted on other reefs (leave-reef-out) that uses evidence from the previous season only, never from the season being searched. Held out over 2018–2026 (30,037 dives in 926 reef-years), with 90% of problematic dives retained, the combined ranking:

- reduced the designated sites searched from 88% to 82%;
- raised low-density dives avoided from 10% to 16%;
- raised ghost dives avoided from 13% to 21%.

All paired bootstrap intervals excluded zero. Gains were larger in recent seasons. In 2022–2026, keeping 80% of problematic dives needed 61% of designated sites instead of 74%. Low-density dives avoided rose from 20% to 32%.

Table 10: Habitat-only ranking → habitat plus previous-season evidence, compared at matched retention of problematic dives (CPUE ≥0.02; held out). Brackets: 95% paired bootstrap interval of the difference, in percentage points.

Period	Problematic dives kept	Designated sites searched	Low-density dives avoided	Ghost dives avoided
2018–2026	90%	88% → 82% (-11 to -1)	10% → 16% (+2 to +10)	13% → 21% (+4 to +13)
2018–2026	80%	73% → 67% (-11 to -3)	22% → 27% (+2 to +10)	26% → 36% (+5 to +14)
2022–2026	90%	88% → 78% (-14 to -4)	10% → 19% (+6 to +13)	12% → 25% (+8 to +17)
2022–2026	80%	74% → 61% (-22 to -4)	20% → 32% (+6 to +20)	25% → 41% (+8 to +24)

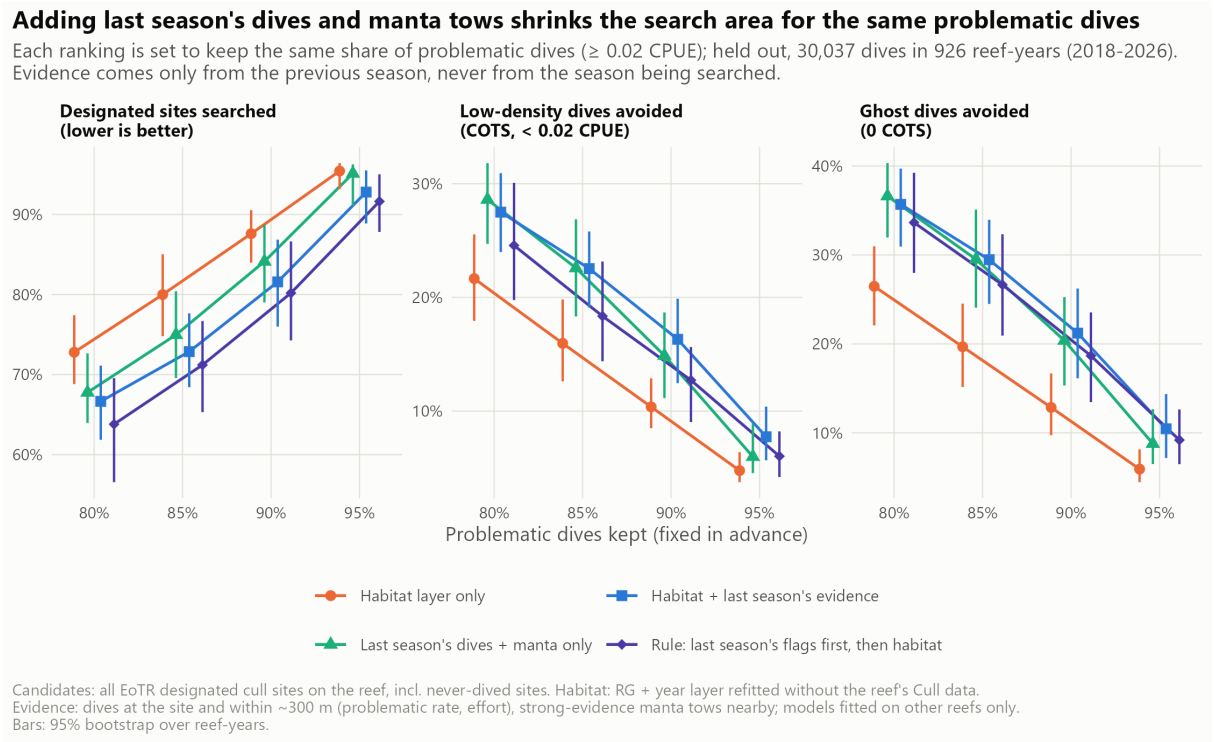


Figure 14: Habitat-only and evidence-informed rankings of designated cull sites compared at the same share of problematic dives retained (80–95%). Panels: share of designated sites searched (lower is better), and shares of low-density and ghost dives avoided. Evidence is from the previous season only. Bars are 95% bootstrap intervals over reef-years (2018–2026). Adding last season's evidence shifts every curve in the useful direction.

The fitted model weighs the evidence in ways that match field experience:

- strong manta tows or problematic dives nearby raised a site's rank;
- a site dived the previous season with nothing problematic dropped sharply;
- a site that had itself been problematic ranked slightly lower than its neighbourhood suggested, consistent with culling having reduced it.

Most of the reduction came from ranking quiet and never-dived sites lower. Among dives actually made, discrimination between problematic and low-density dives rose only from 0.57 to 0.59.

A simple rule ("last season's flagged sites first, then habitat") searched the fewest sites but avoided fewer low-density dives than the fitted model. A three-season evidence window gave the same picture.

The gain holds season by season

The evidence model was also tested using earlier seasons to predict later ones. Its habitat models were withheld by reef but were not restricted to earlier years. It avoided more low-density dives than habitat alone in every season from 2022, and searched fewer sites in every season from 2023 (Table 11). It did not help in 2019–2021, when fewer earlier seasons were available to learn from.

Pooled over 2019–2026 (28,431 dives in 861 reef-years), with 90% of problematic dives retained, sites searched fell from 86% to 81%. Low-density dives avoided rose from 11% to 17%. Training on all earlier seasons matched the last five (81% of sites) and outperformed the last

three (86%). Adding the manta model to the habitat term made no difference (80%), so the simpler cull-trained habitat term was kept.

Table 11: Year-by-year forward test of the cull-site ranking with 90% of problematic dives (CPUE ≥0.02) retained. Each season uses an evidence model fitted only on earlier seasons, with habitat models withheld by reef. Values are habitat alone → cull-trained habitat + last season's evidence (all earlier seasons).

Test season	Dives	Sites searched	Low-density dives avoided	AUC, problematic vs low-density
2019	4,380	88% → 85%	11% → 14%	0.59 → 0.50
2020	3,311	75% → 73%	23% → 21%	0.61 → 0.59
2021	3,769	78% → 80%	19% → 17%	0.54 → 0.56
2022	4,109	80% → 81%	13% → 15%	0.55 → 0.61
2023	4,119	89% → 74%	10% → 21%	0.59 → 0.61
2024	3,910	88% → 84%	7% → 15%	0.54 → 0.61
2025	3,676	88% → 84%	11% → 18%	0.52 → 0.60
2026 (Jan–May)	1,157	96% → 76%	5% → 25%	0.53 → 0.71
2019–2026	28,431	86% → 81%	11% → 17%	0.57 → 0.58

A caveat applies to all cull-site results. A ranking is credited when it agrees with where crews chose to dive. Crews already use the previous season's knowledge, so part of the gain formalises current practice. COTS densities at designated sites that were never dived cannot be verified from these data.

Retaining dives above 0.04 CPUE

Re-evaluating the unchanged rankings against dives above 0.04 CPUE showed how the acceptable omission rate constrains search reductions (Table 12, Figure 2). In the forward evidence test over 2019–2026, retaining 95% of these dives required searching 90.2% (95% CI 85.4%–93.4%) of designated site-years with the combined ranking, compared with 94.0% (95% CI 89.7%–95.5%) using habitat alone and 95.6% (95% CI 95.3%–95.9%) under random selection. The combined ranking avoided 8.9% (95% CI 6.2%–12.9%) of low-density dives and 11.2% (95% CI 8.0%–16.8%) of ghost dives at this retention. These are retained outcomes from observed dives; they do not establish the number of aggregations missed or the consequences of deferring unvisited sites.

The acceptable loss makes a large difference. Retaining 99% of >0.04 dives required 98.2% (95% CI 97.2%–98.7%) of site-years with the combined ranking, leaving much less scope to avoid low-density and ghost dives. At the nominal top-85% rank cutoff, the forward ranking searched 84.2% of candidate site-years and missed 8.0% (95% CI 5.4%–10.7%) of >0.04 dives. The proposed 10–15% search reduction therefore cannot be assumed compatible with negligible omissions; the allowed omission rate must be agreed and tested in the field.

Table 12: Unchanged rankings evaluated at >0.04 CPUE. Percentage of designated site-years searched and low-density or ghost dives avoided at matched high-density retention; brackets are 95% paired reef-year bootstrap intervals. Low-density means positive CPUE <0.02. Spatial: 2018–2026; forward evidence test: 2019–2026.

Test	Retention (%)	Ranking	Sites (%)	Low avoided (%)	Ghost avoided (%)
forward	80	Habitat + evidence	66.6 [62.0–71.1]	28.7 [24.5–32.8]	34.5 [30.3–39.2]
forward	80	Habitat	72.6 [67.1–77.2]	21.6 [17.7–26.5]	25.3 [20.9–30.6]
forward	90	Habitat + evidence	80.1 [73.9–85.2]	17.5 [13.5–22.3]	21.8 [17.0–27.9]
forward	90	Habitat	85.8 [81.4–88.7]	11.1 [9.0–14.4]	13.4 [10.9–17.2]
forward	95	Habitat + evidence	90.2 [85.4–93.4]	8.9 [6.2–12.9]	11.2 [8.0–16.8]
forward	95	Habitat	94.0 [89.7–95.5]	4.9 [3.7–8.2]	6.8 [4.9–10.5]
forward	99	Habitat + evidence	98.2 [97.2–98.7]	1.8 [1.2–2.7]	2.2 [1.6–3.6]
forward	99	Habitat	98.8 [98.3–99.1]	1.0 [0.7–1.4]	1.4 [1.0–2.0]
spatial	80	Habitat + evidence	66.8 [62.1–71.6]	27.0 [23.2–30.6]	35.1 [30.2–39.4]
spatial	80	Habitat	72.6 [68.1–78.0]	21.6 [17.3–26.4]	26.4 [21.1–31.4]
spatial	90	Habitat + evidence	81.6 [75.0–86.9]	15.8 [11.6–20.4]	20.6 [15.3–27.1]
spatial	90	Habitat	87.1 [83.1–90.2]	10.1 [7.9–12.9]	12.5 [9.2–16.8]
spatial	95	Habitat + evidence	92.5 [87.9–94.9]	7.2 [5.1–10.5]	9.6 [6.5–14.4]
spatial	95	Habitat	94.6 [92.1–95.9]	4.6 [3.4–6.3]	5.7 [4.2–8.1]
spatial	99	Habitat + evidence	98.6 [97.9–99.0]	1.4 [1.0–2.1]	1.9 [1.3–2.8]
spatial	99	Habitat	98.9 [98.6–99.2]	0.9 [0.7–1.2]	1.1 [0.8–1.6]

See Figure 2 in the main Results.

Intermediate dives were tracked separately rather than treated as failed high-density detections. At 95% >0.04 retention, the forward combined ranking retained 95.2% (95% CI 93.2%–96.7%) of dives from 0.02 to 0.04 inclusive. Allowing their omission in the management criterion did not itself change the search order: the saved model still learned the original ≥0.02 response and recent crew observations. The spatial comparison and manta-habitat alternatives were likewise rescored without refitting. Full fixed-effort, matched-retention, paired-difference and per-season results are provided in `cull_high_density_*.csv`.

Each source covers the other's gaps

Habitat layers: a consistent first ordering on every reef

The habitat layers cover every reef. They give a consistent ordering within each reef, which is useful wherever crews have little recent evidence. Across the GBR, each layer's absolute values mostly reflect survey year and region (Figure 15). Within reefs, the cull-trained layers placed their highest ranks along reef margins and outer slopes (Figure 1).

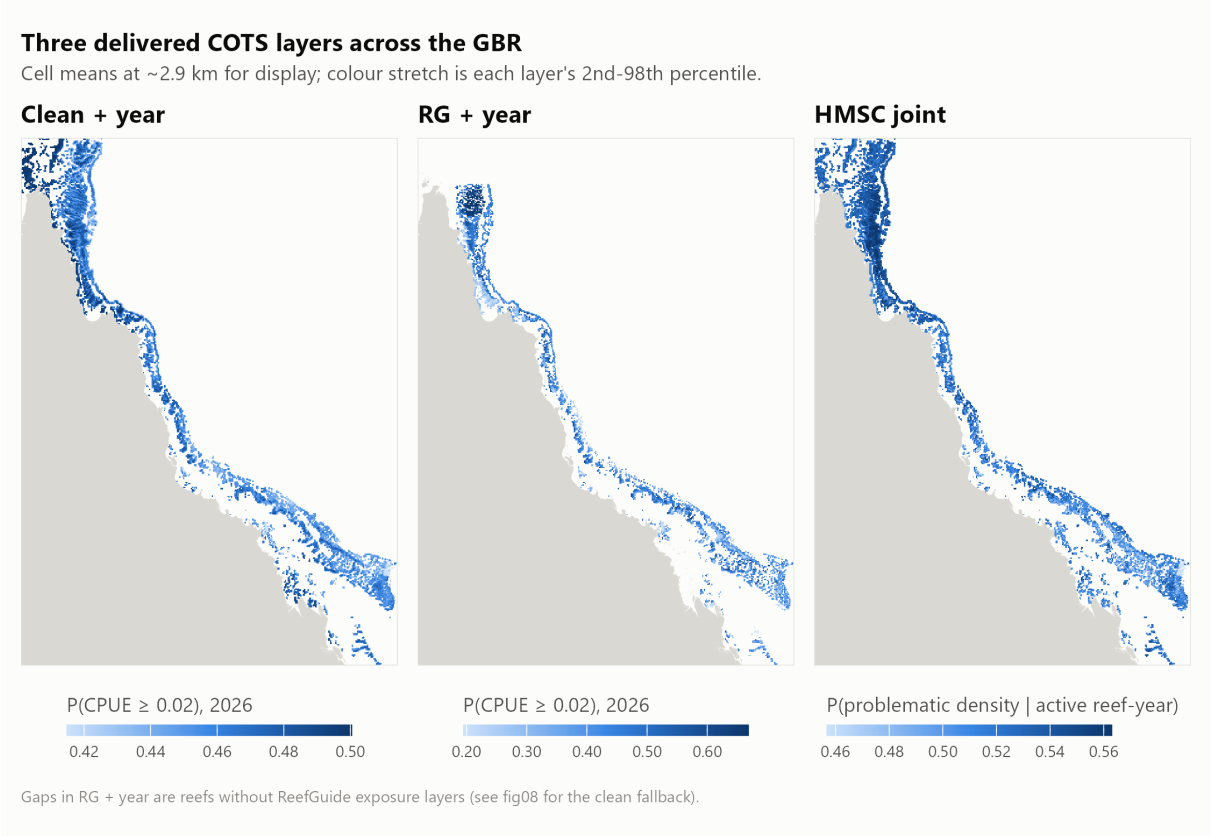


Figure 15: The three main habitat layers across the Great Barrier Reef, shown as cell means at about 2.9 km. The two cull-trained layers (left, centre) are annual probabilities of a problematic cull site-year in 2026; the HMSC layer (right) is conditional on an active reef-year. Gaps in RG + year are reefs without ReefGuide layers, filled by the clean + year fallback in the primary search-planning layer.

See Figure 1 in the main habitat Results. Imagery attribution: Tiles © Esri — Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community.

On their own, the layers discriminate modestly between places on the same reef. On independent manta tows, RG + year reached a within reef-year AUC of 0.576 (95% CI 0.548–0.602) (Table 13, Figure 16). It placed 23% more strong-evidence tows in each reef's top-ranked 20% than chance. The manta model, trained on tows but scored leave-reef-out, reached 0.612 (95% CI 0.578–0.636) (Appendix H).

Table 13: Discrimination between strong-evidence and strictly negative manta tows within reef-years. Manta tows were not used to fit any of these layers. Lift is the share of strong-evidence tows in a reef-year's top 20% of tows, divided by 20%. Intervals are 95% bootstrap intervals over reef-years.

Layer	Within reef-year AUC (95% CI)	Top-20% lift (95% CI)
Clean + year	0.516 (0.500–0.532)	1.11 (1.06–1.15)
RG + year	0.576 (0.548–0.602)	1.23 (1.18–1.29)
HMSC joint	0.567 (0.549–0.585)	1.15 (1.11–1.19)

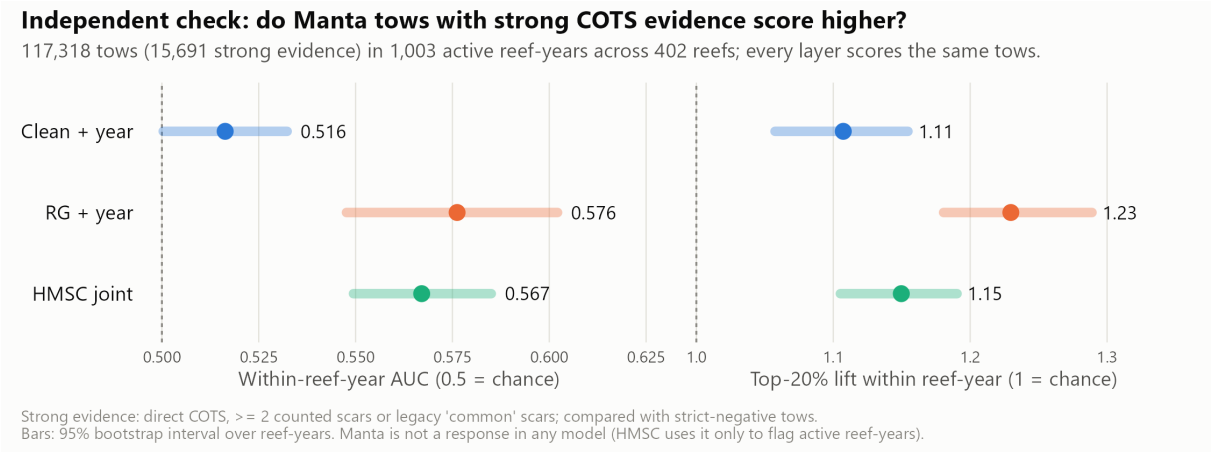


Figure 16: Independent manta tow check of the three habitat layers. Left: within reef-year AUC; right: top-20% lift. Bars are 95% bootstrap intervals over reef-years; dotted lines mark chance.

The layers agree only in part on which cells rank highest. Median within-reef Spearman correlations were 0.41 between clean + year and RG + year, 0.33 between clean + year and HMSC, and 0.24 between RG + year and HMSC (Appendix D). Where both cull-trained layers exist, the fallback and RG rankings correlate at 0.34, so ranks near the RG coverage boundary within a reef are less reliable.

Habitat alone as a cull-site ranking

Used alone to rank all designated sites on a reef, the held-out habitat layer retained problematic dives only slightly faster than low-density or ghost dives (Table 14, Figure 17). Searching the top 70% of sites retained 77.2% (95% CI 72.9%–80.8%) of problematic dives. A ranking with no information would have retained 75.6%, the share of all dives at those sites. Within reef-years, the AUC was 0.57 for problematic versus low-density dives and 0.60 versus ghost dives. The in-sample layer looked stronger (0.61 and 0.66) only because it had been trained on those dives.

Whether a site holding COTS is above or below the action threshold on a given visit depends mainly on irruption stage, earlier culling and the movement of aggregations. Recent evidence captures these; static habitat cannot. The habitat ranking does reach places crews have not tested: 43% of site-years ranked in a reef's top 30% were at sites that had never been dived.

Table 14: Diving only the top-ranked designated cull sites on each reef, held-out habitat ranking alone, 2018–2026. 'Dives kept' is what a ranking with no information would retain of every outcome. Brackets: 95% bootstrap interval over reef-years.

Designated sites searched	Dives kept (no-information baseline)	Problematic dives kept	Low-density dives avoided	Ghost dives avoided
top 80%	83.8%	84.9% (81.9%–87.7%)	16.1%	19.8%
top 70%	75.6%	77.2% (72.9%–80.8%)	24.6%	29.5%
top 60%	66.5%	68.7% (64.1%–72.8%)	35.2%	38.9%
top 50%	57.4%	59.8% (55.5%–64.5%)	44.9%	48.2%

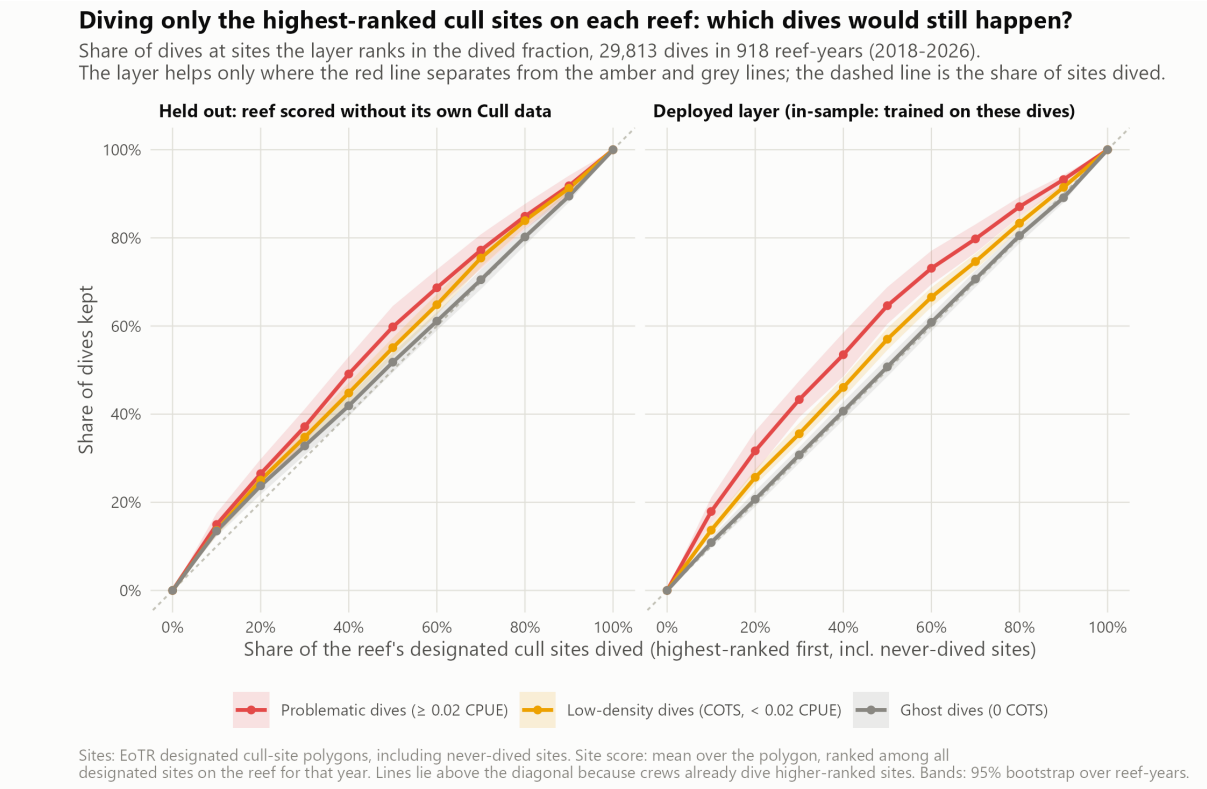


Figure 17: Original ≥0.02 evaluation: share of cull dives that would still occur if only the highest-ranked designated cull sites on each reef were dived, ranked by habitat alone, by dive outcome. Left: held-out ranking; right: deployed layer (in-sample). The dashed diagonal is the share of sites dived. Bands are 95% bootstrap intervals over reef-years. Without recent evidence, problematic, low-density and ghost dives fall away at almost the same rate.

Cull history: strong where it exists

Problematic sites persist between seasons at a scale of about 0.5–1 km. In the strictly forward test, the blend of habitat and cull history reached a within reef-year AUC of 0.681 (95% CI 0.640–0.718), against 0.569 for habitat alone (Table 15, Figure 18).

The forecast is strongest where history exists. It reached 0.716 with history nearby, 0.540 where history was sparse and 0.553 at sites never culled before. History lies within reach of only 8.6% of 120 m reef cells, so across most of the reef the blend falls back to habitat.

Part of the skill reflects operator behaviour: a score based only on how often crews had dived a site reached 0.636. This layer is therefore delivered as a map of **known aggregation sites** for briefing crews, not as habitat. Reef-context predictors at 0.5–1 km (windward position, distance to reef edge, channels, eReefs currents) did not improve habitat forecasts robustly (0.582 vs 0.569, overlapping intervals).

Table 15: Forward validation: within reef-year AUC for problematic cull site-years, test years 2022–2026 pooled. Each score uses only data from before the test year. The deployed blend's 95% bootstrap interval over reef-years is given for all site-years.

Score	All site-years (n = 4,170)	With nearby history (n = 3,212)	Little or none (n = 958)	New sites (n = 1,664)
Habitat (HMSC refitted before test year)	0.569	0.584	0.554	0.524
Habitat + reef context	0.582	0.600	0.520	0.514
Dive effort only	0.636	0.656	0.491	0.513
History kernel	0.654	0.705	0.465	0.476
Habitat + history (deployed)	0.681 (95% CI 0.640–0.718)	0.716	0.540	0.553

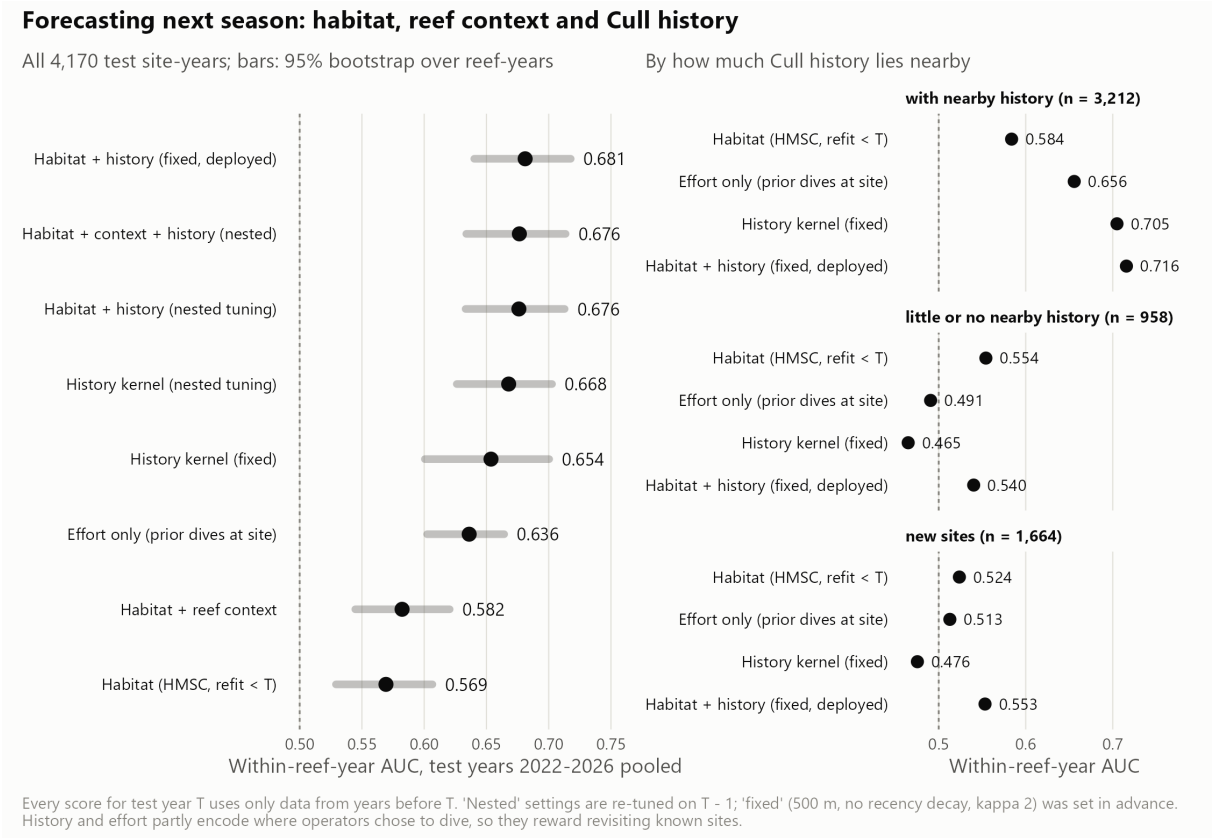


Figure 18: Forward validation of habitat, reef context and cull-history scores. Left: all test site-years with 95% bootstrap intervals over reef-years. Right: the same scores split by how much cull history lies within reach of the site. History carries the forecast where it exists; habitat carries it where history is absent.

Where recent evidence is missing

Recent evidence is unevenly spread. 30% of reef-years were on reefs not visited the season before. On those reefs (3,221 dives) every ranking was close to chance (AUC about 0.53), and habitat was the only information available.

Habitat also reaches ground that recent search has not covered. A tow was counted as *new ground* if no tow lay within 300 m in the previous five seasons. On new ground, the manta model ranked tows with a within reef-year AUC of 0.619 (95% CI 0.587–0.651), and the cull-trained layer at 0.56. Strong evidence was more common on new ground than on repeat ground (13.6% vs 5.8% of tows), partly because new ground is often surveyed early in an irruption. This is the basis for the exploration areas in the 2027 product.

Manta surveillance: spread tows evenly, add where habitat ranks highest

Systematic thinning found the most aggregations at every level of effort (Table 16, Figure 19). With 60% of tows kept, it found 96.6% of the 1,900 aggregations of two or more strong-evidence tows in 1,082 surveys. Random thinning found 94.0%, and thinning by the held-out cull-trained layer found 88.1% (95% CI 86.6%–89.5%).

The largest cut that still found at least 95% of aggregations was:

- 40% of tows for systematic thinning (40% with bootstrap assurance);
- 30% for random thinning;

- 20% for thinning by habitat rank.

Results for 2022–2026 alone were the same: 40%, 30% and 20%.

Ranking concentrates the kept tows in a few high-scoring stretches of reef, so aggregations in low-scoring stretches are missed entirely. Habitat rank is therefore not a good basis for dropping tows. It did find slightly more isolated single strong tows than random (64% vs 61% at 60% kept). Scoring with the deployed (in-sample) layer instead of the held-out one overstated aggregations found by 2 percentage points at 60% kept.

Reef-level detection is robust to thinning whichever tows are kept. Irruption reef-years have many positive tows, so keeping about 60% of tows in 2022–2026 still flagged 89% of irruption reef-years with the layer and 87% at random (Appendix D).

Table 16: Share of COTS aggregations (two or more strong-evidence tows within 500 m) still found when manta tows are cut within each survey, all years. The layer is the cull-trained RG + year model scored leave-reef-out; the deployed column is the same layer scored in-sample. Brackets: 95% bootstrap interval over surveys.

Tows kept	Layer, held out	Layer, deployed	Random	Systematic
80%	96.2% (95.3%–97.2%)	96.9%	98.8%	99.7%
70%	93.0% (91.8%–94.2%)	93.7%	97.0%	98.7%
60%	88.1% (86.6%–89.5%)	90.4%	94.0%	96.6%
50%	81.8% (80.0%–83.4%)	85.2%	89.7%	93.0%

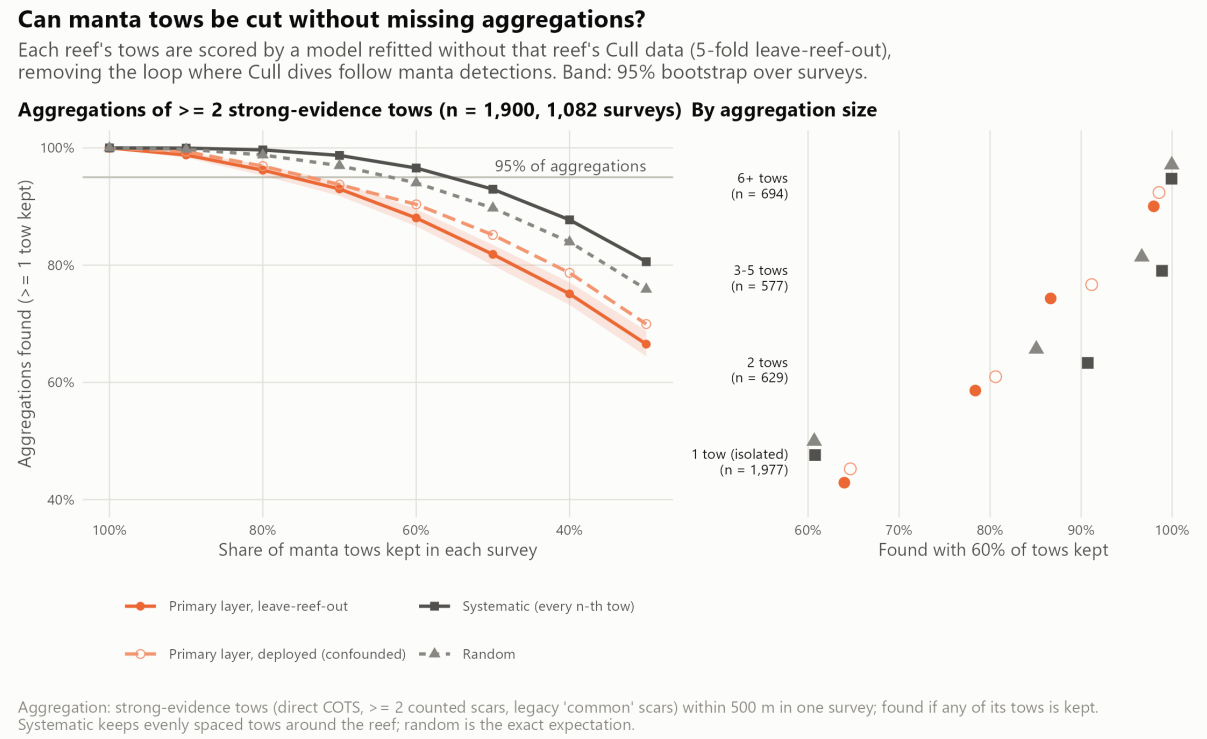


Figure 19: Manta tows kept versus COTS aggregations found. Left: aggregations of two or more strong-evidence tows found as tows are cut, for the cull-trained layer (held out and deployed), random selection and systematic thinning; bands are 95% bootstrap intervals over surveys. Right: share found at 60% of tows kept, by aggregation size. Evenly spaced thinning loses the fewest aggregations.

A habitat model trained on manta tows is the best guide to where to add tows. Manta tows provide about 20 times more training rows than cull site-years. The within reef-year manta model ranked held-out tows best (Table 17, Figure 20). It found 3 percentage points more aggregations than the cull-trained model at 60% kept (95% CI +1 to +5). That is still below random (94.0%) and systematic thinning (96.5%), so its role is to direct extra tows rather than to remove them. Its top predictors were wave period and height, bathymetry and slope.

Training on manta did not improve cull-site selection. With survey year as a predictor, the manta model ranked cull dives at chance (AUC 0.48), because survey year took 70% of its fitted gain. The within reef-year version matched the cull-trained model over 2018–2026 (0.566 vs 0.567). Once last season's evidence was added, every habitat model gave the same result. Manta tows see COTS and scars from the surface along the reef perimeter, whereas a problematic dive measures how many COTS a diver finds in a site; the two are related but not the same.

Table 17: Cull-trained and manta-trained habitat models on held-out manta tows (all three fitted on the same leave-reef-out folds). AUC: strong-evidence vs strictly negative tows within reef-years, with 95% bootstrap intervals. Aggregations: share found at 60% of tows kept.

Model	Tow AUC (95% CI)	Top-20% lift	Aggregations found, 60% kept
Cull-trained	0.547 (0.516–0.574)	1.15	88.2%
Manta-trained (with year)	0.575 (0.547–0.599)	1.27	88.4%
Manta-trained, within reef-year	0.612 (0.578–0.636)	1.33	91.0%

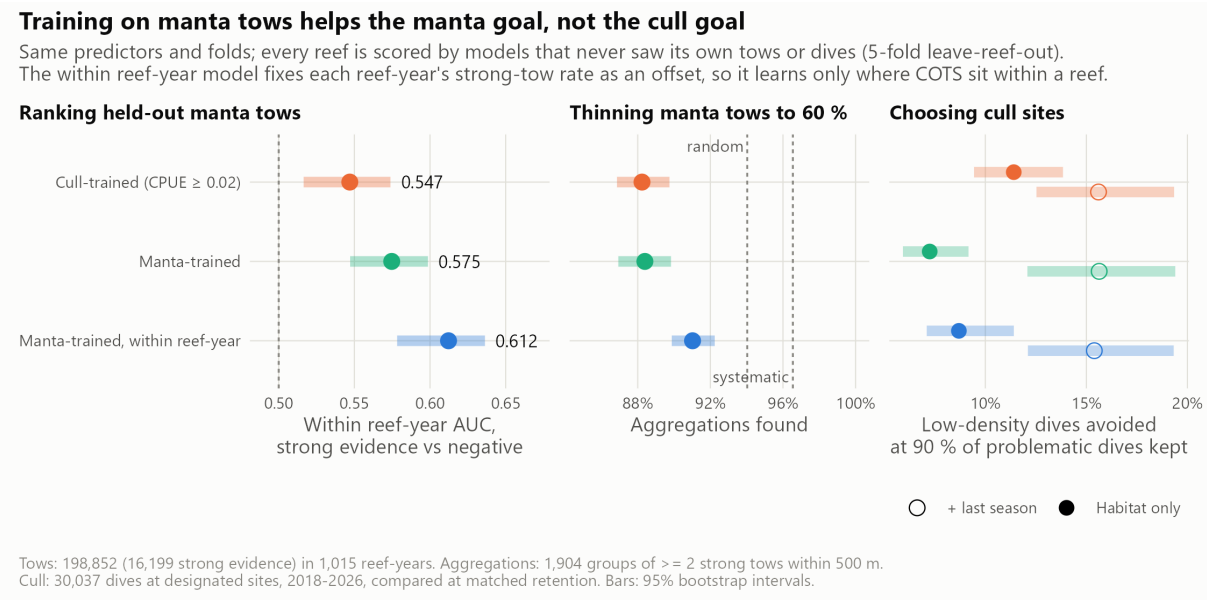


Figure 20: Cull-trained and manta-trained habitat models on both goals. Left: within reef-year AUC on held-out manta tows. Centre: aggregations found with 60% of tows kept, with random and systematic thinning for reference. Right: low-density cull dives avoided at 90% of problematic dives (CPUE ≥0.02) retained, for the habitat model alone and with last season's evidence. Bars are 95% bootstrap intervals. The manta model ranks tows best; for cull sites, last season's evidence brings every habitat model to the same result.

The 2027 cull-site priority ranking and exploration areas

Every designated cull site (15,884 sites on 576 reefs) was ranked within its reef for the 2027 season, using the combined model with the most recent 12 months of evidence. The ranking is delivered as a CSV and as the EoTR cull-site KMZ, with each polygon coloured by priority class. Each polygon also carries its rank, its modelled chance of a problematic dive if dived, and last season's evidence. Table 18 shows what each class kept in the forward test.

Table 18: Original ≥0.02 evaluation: 2027 priority classes and what searching through each class retained in the forward test (dives 2019–2026).

Searching through	Sites on the reef	Problematic dives kept	Low-density dives avoided	Ghost dives avoided
A	top 20%	33%	75%	79%
A + B	top 50%	64%	42%	49%
A + B + C	top 80%	89%	18%	22%
All (incl. D)	100%	100%	0%	0%

The separate >0.04 class summary (Table 19) makes the cost of stopping at a fixed class explicit. Searching through A + B + C retained 89.7% of observed >0.04 dives in the forward test; the remainder would have been omitted on those visits. These classes describe search order, not an established safe stopping point.

Table 19: High-density outcomes for the unchanged class cutoffs, using the forward test of 2019–2026. Retained shares are percentages with 95% reef-year bootstrap intervals. Intermediate means 0.02–0.04 inclusive; high-density means >0.04. The fractions are rank cutoffs, not measured reductions in annual tow or dive effort.

Searching through	Rank cutoff (%)	High-density retained (%)	Intermediate retained (%)
A	20	33.6 [28.0–40.0]	33.1 [29.3–36.5]
A + B	50	63.8 [56.3–71.4]	65.8 [62.0–69.3]
A + B + C	80	89.7 [86.4–92.7]	88.7 [87.0–90.3]
All	100	100.0 [100.0–100.0]	100.0 [100.0–100.0]

The two sources share the ranking:

- **Evidence ranks the top.** 678 sites had a problematic dive last season and 809 have strong manta evidence nearby.
- **Evidence also lowers sites.** 293 sites were dived last season with nothing problematic, which strongly lowers their rank.
- **Habitat ranks the rest.** 10,654 sites (67%) are on reefs with no dives or tows in the last 12 months. Habitat alone orders them, as a first search order to update once manta tows come in.

The exploration layer adds high-ranked habitat outside the current search footprint (Figure 3), so that the ranking does not keep consolidating search on the same places. It contains 3,807 areas on 488 reefs, with a median size of 8 ha, a median habitat percentile of 0.84, and a median distance from a cull site of 170 m. On well-surveyed reefs such as Batt and Eyrie, cull sites and five seasons of tows already cover most high-ranked habitat. The larger opportunities are on reefs with little recent towing.

See Figure 3 in the main Results.

Appendix I: Updated metacommunity calibration and exploratory projections

This appendix reports the completed 21 October 2026 figure set for the current working model, identified in the source files as W-wave-lf04. The COTS and coral comparison combines source Figures 1 and 4 in Figure 5. The remaining figures appear below, with panels rearranged for portrait pages and the source summaries and uncertainty bands retained. Appendix J preserves the 15 October snapshot as a development record.

Calibration updates and uncertainty

Since the earlier snapshot, the working model has incorporated revised coral thermal responses, a chlorophyll-based larval-survival modifier, wave-exposure effects on settlement, and revised initial population and coral settings. Initial fast-growing coral at Lizard Island was increased by 0.04 in absolute cover. Larger increases were rejected because they left too many COTS between irruptions. The linear observation conversion remains 0.015 COTS per tow per adult per hectare; nonlinear alternatives failed regional checks. These decisions improved the working calibration without identifying a unique population mechanism (open-AIMS, 2026a).

The comparison comprises 5 base seed pairs plus twelve one-at-a-time parameter perturbations of ±25%, giving 65 runs in each culling state. Perturbations cover outside larval supply, background coral mortality, bleaching under low heat stress, the chlorophyll response, juvenile survival and initial fast-growing coral shares. Inner bands span the base seeds; outer bands span all runs. Neither is a probability interval, and joint parameter and structural uncertainty are not fully represented.

Recorded cull totals are now distributed among sites on the same reef in proportion to simulated population counts, retaining juvenile/adult stages and pre-/post-spawning timing. Removals remain limited by the animals available. This replaces the spatial-kernel allocation in the earlier diagnostic. Calibration scoring keeps culling off; paired culling-on runs provide the overlay. Earlier unmet-removal percentages therefore do not describe the current model.

Recent densities and additional reefs

The recent-density comparison (Figure 21) shows partial agreement with diver estimates and differences among reefs. Corrected external irruption probabilities and density targets share cull and SALAD evidence, so this is a diagnostic comparison. The additional reefs (Figure 22) show the contrast between historical irruption reefs and low-density ribbon reefs, while individual peak timing and size remain imperfect. These checks do not complete regional validation; Ribbon No 10 has no LTMP series here.

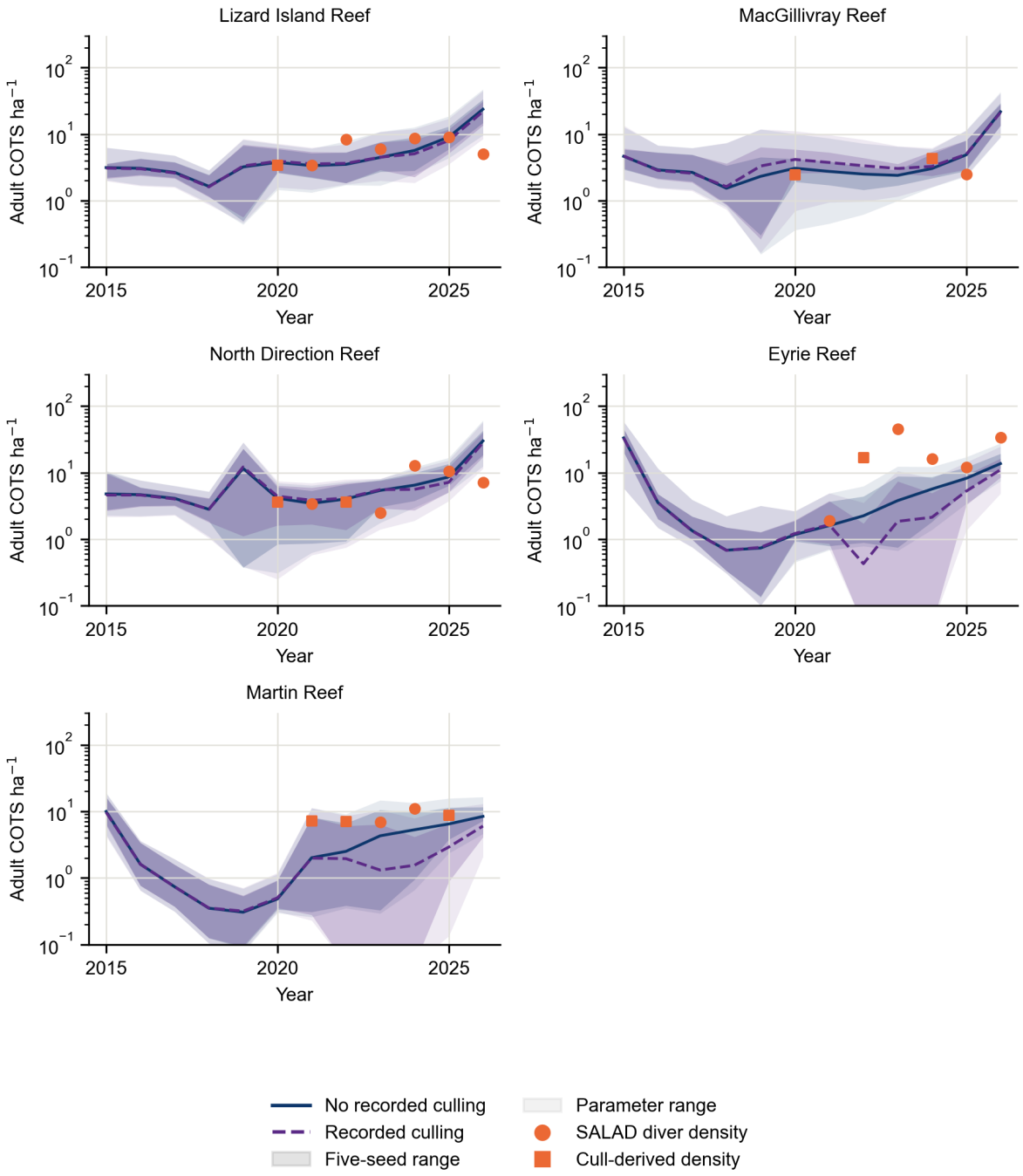


Figure 21: Recent adult COTS density with and without recorded culling, against SALAD diver estimates and catches converted to density (source Figure 2). The vertical axis is logarithmic. Lines are five-seed means; bands span seeds and parameter tests as in the main figure. Larval-input corrections and density targets share observations, so agreement is not independent validation.

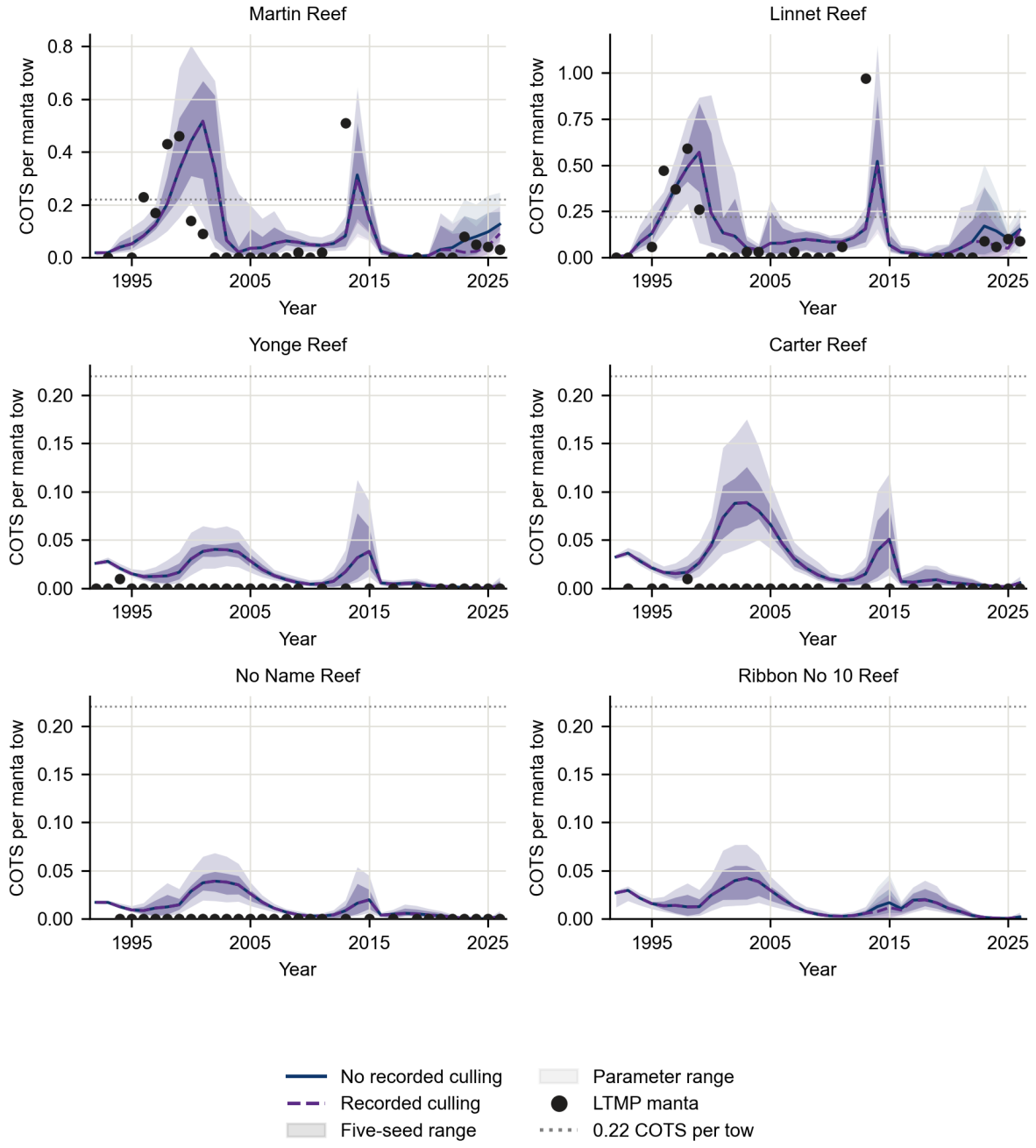


Figure 22: COTS trajectories on additional reefs outside the four-reef calibration fit (source Figure 3). Points show LTMP manta observations; Ribbon No 10 has no LTMP series. Blue and purple indicate culling-off and culling-on trajectories with the main figure's range definitions.

Initial control projections to 2050

The completed ensemble spans 5 climate models, five seed pairs, two climate pathways, nine culling levels and two treatments of the 2026 population state. The figures use the 450 continuous-history runs without a population reset. Historical catches apply through 2026; effort-based culling begins in 2027. "Recent effort" uses 2023–25 diver-hours on each reef. Increased effort multiplies those hours; targeted effort reallocates the same total each half-year among reefs with at least 3 adults per hectare. Density-dependent catch rates determine

adult removals, capped by the available population. Vessel travel and search costs are not represented explicitly.

Outside larval supply resamples historical irruption patterns and scales them by projected coral in the domain. These conditional scenarios include neither thermal acclimation nor external coral-larval supplementation. Excess simulated COTS in 2026 carry into the near-term projections. Proposed changes to the population reset, thermal acclimation scenarios and source-reef culling corrections had not yet entered the reported ensemble.

Increased effort lowered near-term COTS peaks, with climate-dependent coral outcomes (Figure 23, Figure 24). Reallocation could increase domain-wide removals without providing the largest benefit at the five displayed reefs (Figure 25). Population and coral summaries cover those reefs, whereas removals cover the whole domain. These summaries cannot establish domain-wide coral protection or an optimal allocation; local and downstream outcomes and operational constraints need further assessment.

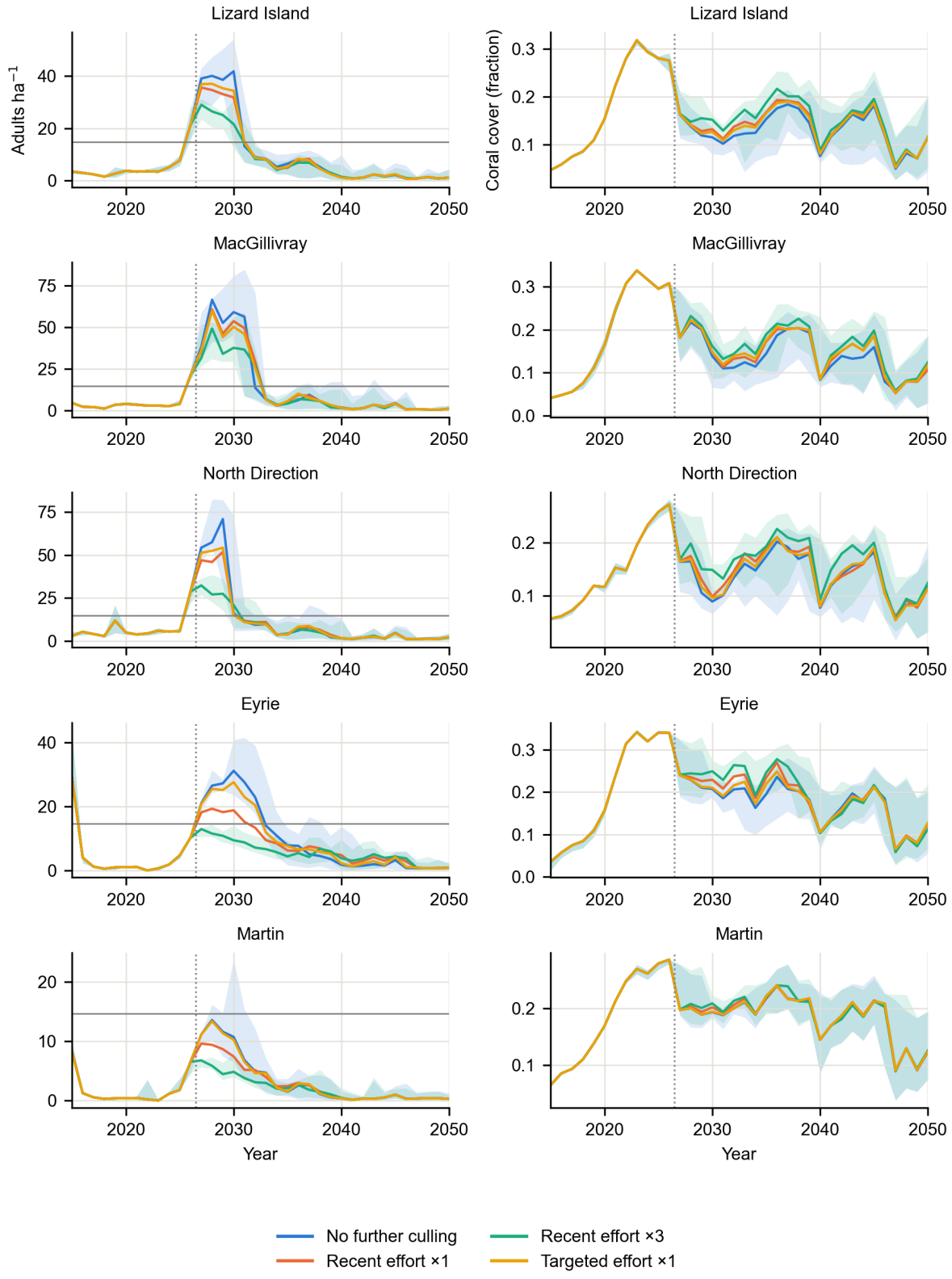


Figure 23: Exploratory COTS and coral projections under SSP2-4.5. Lines are medians across five climate models and five seed pairs. Bands show the 25th–75th percentiles for no further culling and three times recent effort, rather than confidence intervals. “Targeted ×1” reallocates the recent total effort. The dotted line separates history from projections. Recent population and outside-supply uncertainty carry forward.

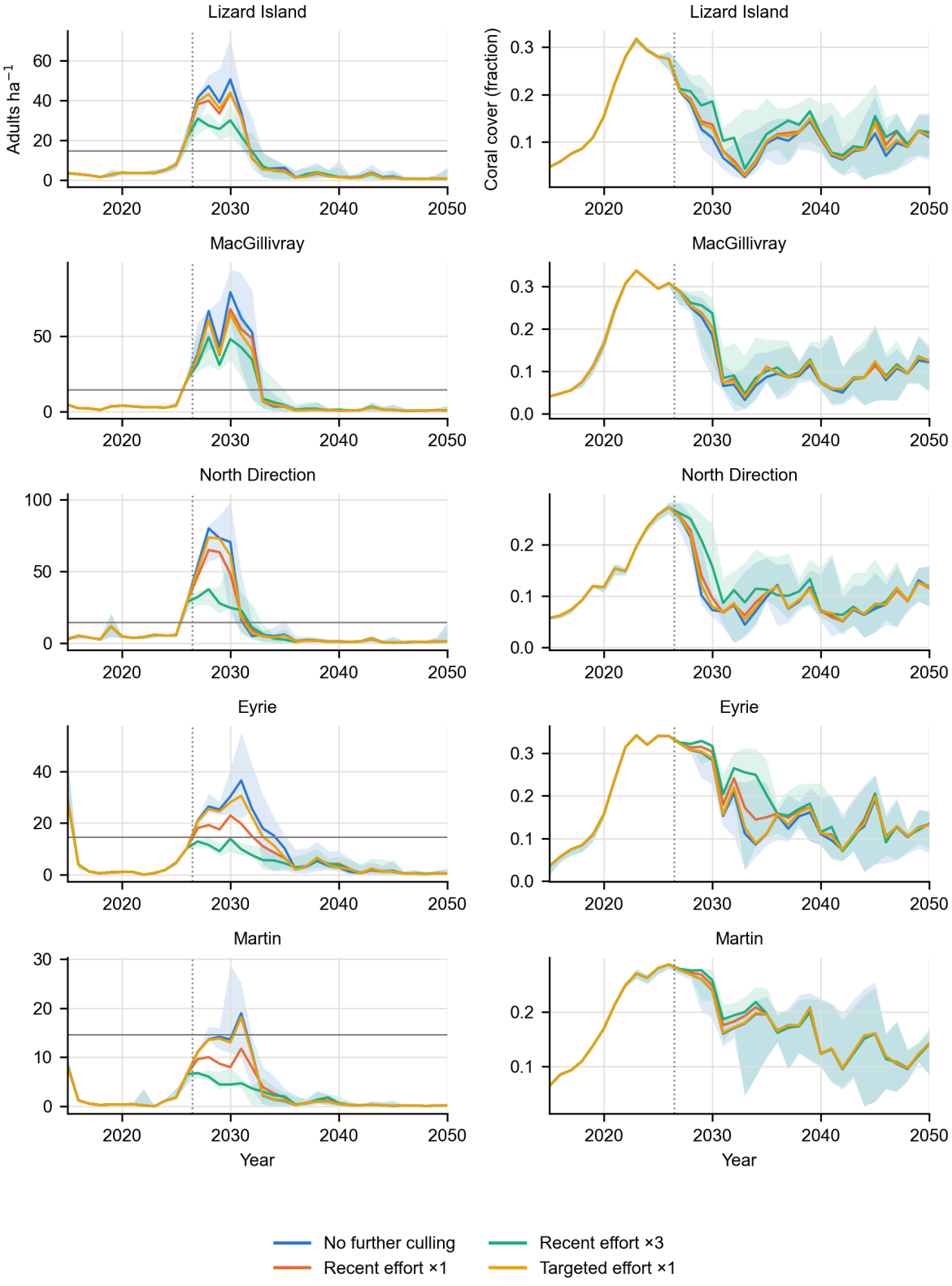


Figure 24: The same exploratory effort comparisons under SSP3-7.0, with definitions matching the preceding figure. Lower COTS densities later in the period can accompany reduced coral food and should not be read as a control benefit.

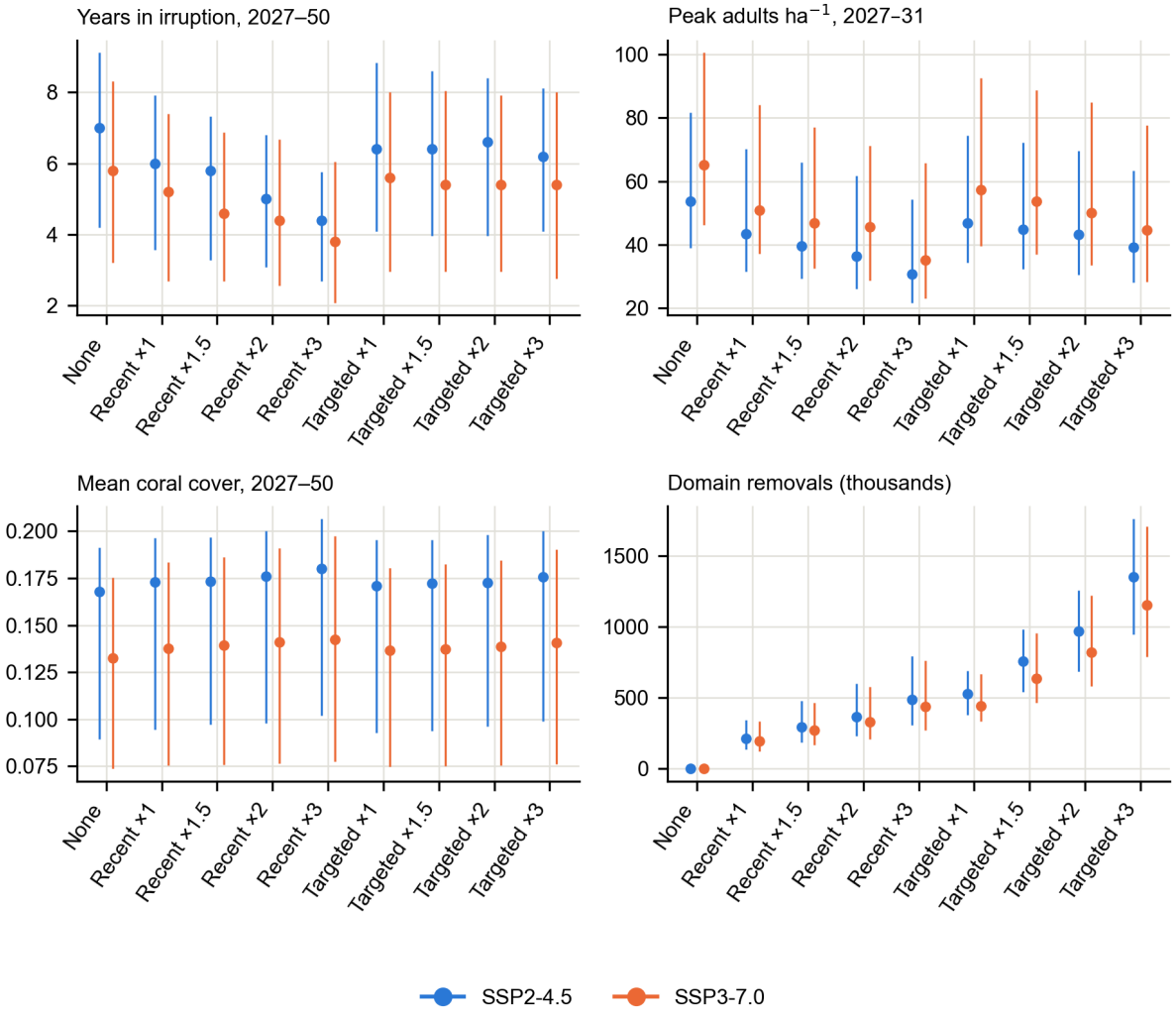


Figure 25: Exploratory effort comparisons for 2027–2050. Points are medians; bars span the 10th–90th percentiles across climate models and seeds. The first three panels average Lizard Island, MacGillivray, North Direction, Eyrie and Martin; the last counts adult removals across the whole domain. “Recent” retains the recent allocation, “targeted” reallocates it, and multipliers scale total hours. These are model-conditional scenarios.

Appendix J: Archived 15 October metacommunity comparisons

The candidate codes below are technical identifiers. W denotes the working configuration; the version with more initial preferred coral at Lizard Island is C3a; the version with lower background coral mortality is C6. They remain calibration alternatives, not validated control policies. Methods and scenario design are documented in Appendix G; the following results retain the frozen 15 October 2026 reporting snapshot.

Metacommunity model: calibration and current limitations

Replicated candidates

The metacommunity model now reproduced both historical irruption waves on the reefs with two observed waves, although irruption heights and intervening densities remained imperfect (Figure 26). Three configurations were retained for further testing: the working configuration W, C3a with a higher initial preferred-coral share at Lizard Island, and C6 with lower chronic background coral mortality. Each was run with 5 recorded pairs of ecological and connectivity seeds. Under the revised score v2, each candidate matched 7 of 7 observed historical peaks in every seed. Matching allowed a timing window; it did not establish accurate peak heights or performance on independent data.

The candidates retained different compromises (Table 20). Mean COTS losses overlapped in their variation across seeds. C3a produced a larger first Lizard irruption, 56.1 ± 6.5 adults ha^{-1}, compared with 29.0 ± 2.3 for W, and had a lower coral error. However, it left more COTS between irruptions. C6 reduced the troughs at the cost of a higher coral error, 0.279 against 0.250 for W.

The qualitative gate was sensitive to the observation model. At $A½ = 30$ adults ha^{-1}, W passed in 5 of 5 seeds, C6 in 3 and C3a in 0. At $A½ = 10$, W passed in 0 and C6 in 3. These are conditional calibration checks: the detection curve remains uncertain, and a passing qualitative gate does not replace held-out or regional validation.

Table 20: Replicated metacommunity calibration on the four Lizard Island reefs (run 20261014T_48). Means ± SD across five recorded seed pairs; these are measures of stochastic variation, not confidence intervals. Lower COTS loss (score v2) and coral J indicate better calibration fit. Gate columns count seeds passing G2′ under the indicated density-dependent manta detection assumption, with A½ in adults per hectare. All candidates matched seven historical peaks in every seed at the linear observation mapping. No entry represents held-out validation.

Candidate	COTS loss v2	Coral J	Gate A½ = 10	Gate A½ = 30
W	6.79 ± 0.79	0.250 ± 0.002	0/5	5/5
C3a	6.72 ± 0.70	0.228 ± 0.002	0/5	0/5
C6	7.13 ± 0.85	0.279 ± 0.002	3/5	3/5

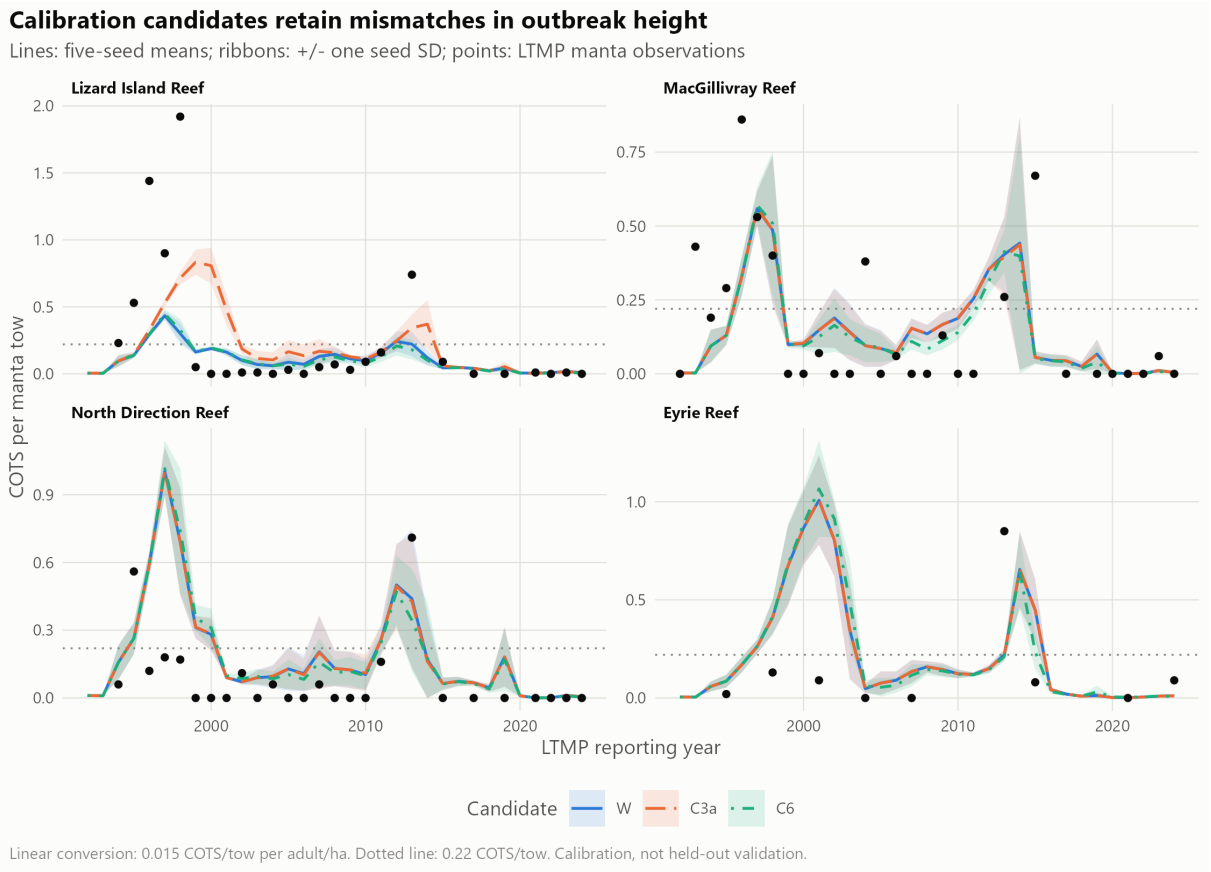


Figure 26: Historical COTS trajectories for W, C3a and C6. Lines are means across five seed pairs and ribbons are one standard deviation across seeds; points are AIMS LTMP manta observations. Adult density is shown through the fixed linear conversion of 0.015 COTS per tow per adult ha^{-1}, and the dotted line marks 0.22 COTS per tow. Peak-height and trough mismatches remain despite successful historical peak matching.

Historical culling exposed missing populations

Prescribing the recorded historical removals exposed populations that were too small to support the observed catches. In the working configuration with the primary 210 m allocation kernel, 88% of requested removals at Eyrie in 2021–24 could not be applied because the simulated juvenile or adult population was absent. Across the model domain, the corresponding share was 70%. This discrepancy identified a failure to represent the recent build-up, rather than evidence that actual culling was ineffective.

Where simulated COTS were available, removals changed the historical trajectory. North Direction's simulated second-wave maximum fell from 0.82 to 0.55 COTS per tow. The summed calibration loss v2 increased from 6.44 to 6.57. These single-seed comparisons tested the removal implementation and consistency with recorded catches. They did not estimate the benefit of a control policy, and culling was left off in the subsequent build-up diagnostics.

Extension to 2026

The latest completed diagnostic runs extended the domain to 2026 and examined the recent build-up separately from the historical candidate comparison (Figure 27). They crossed original and cull/SALAD-corrected irruption-boundary inputs with changes to coral thermal settings. With the earlier thermal settings and no chlorophyll modifier, correcting the boundary

reduced score v3 from 15.61 to 12.63. With revised thermal settings, losses were 11.34 and 11.34, respectively. The revised settings removed the extra bleaching multiplier and raised the threshold below which bleaching mortality was reduced. The trajectories showed that both the food trajectory and boundary supply affected the timing and magnitude of the new irruption.

Exploratory chlorophyll-based larval-survival modifiers gave smaller additional changes: the lowest score v3 in the corrected-boundary screen was 10.99, against 11.34 with those modifiers off. Score v3 includes a diver-anchored recent-density error and is not numerically comparable with the earlier score v2. All of these extended runs used one seed pair. Cull and SALAD evidence contributed to both the corrected boundary and its assessment, so their apparent improvement was not independent validation.

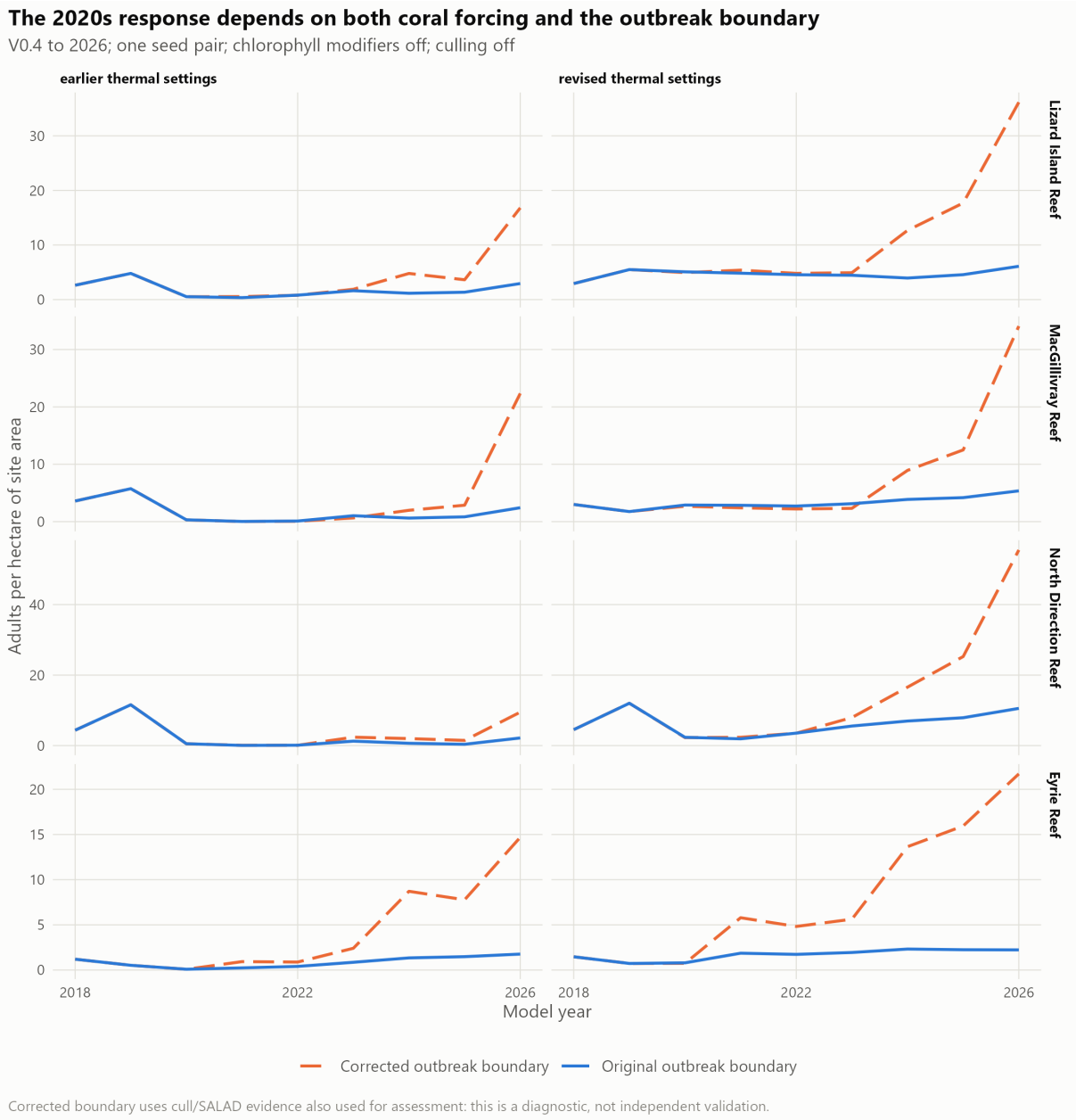


Figure 27: Single-seed extension to 2026 with original or corrected irruption-boundary inputs, under the earlier or revised coral thermal settings. Lines show adult density, with chlorophyll modifiers and culling off. Corrected boundary supply produces a larger late build-up; its response also depends on coral forcing. The same cull/SALAD evidence helped construct and assess the corrected boundary. These are diagnostic trajectories, without an uncertainty interval or an independent validation claim.

No configuration has completed the full validation sequence. The model remains a platform for testing irruption mechanisms and developing control scenarios. Held-out reefs and years, regional transfer and sensitivity tests are still required before reporting benefits of primary-irruption suppression or rules for starting and stopping control. The fine-scale habitat rankings have not been validated as inputs to the metacommunity model.

Appendix K: Preliminary larval-concentration comparisons

These comparisons support Section 3.3 and reformat project figures lc4 and lc6. They describe concentrations, rather than source-to-destination exchange, at larval ages 5, 9 and 10 days across 10 releases. The full source set, including concentration ranks, hotspot persistence, zone comparisons and the north–south profile, remains in `DataProcessing/connectivity_lizard/outputs/figures/`.

Variation among releases

Concentrations around Lizard varied among release dates. Elevated early-age concentrations in some releases did not consistently persist to the later ages (Figure 28). Values compare the unweighted mean reef concentration within a zone with the mean across all reefs in the domain. They are relative to that release and larval age, rather than a fraction of larvae produced locally. The Lizard group comprised reefs within 20 km of Lizard Island after the outer-barrier and Ribbon classifications were applied. Full zone rules and the component tables are retained with the source analysis.

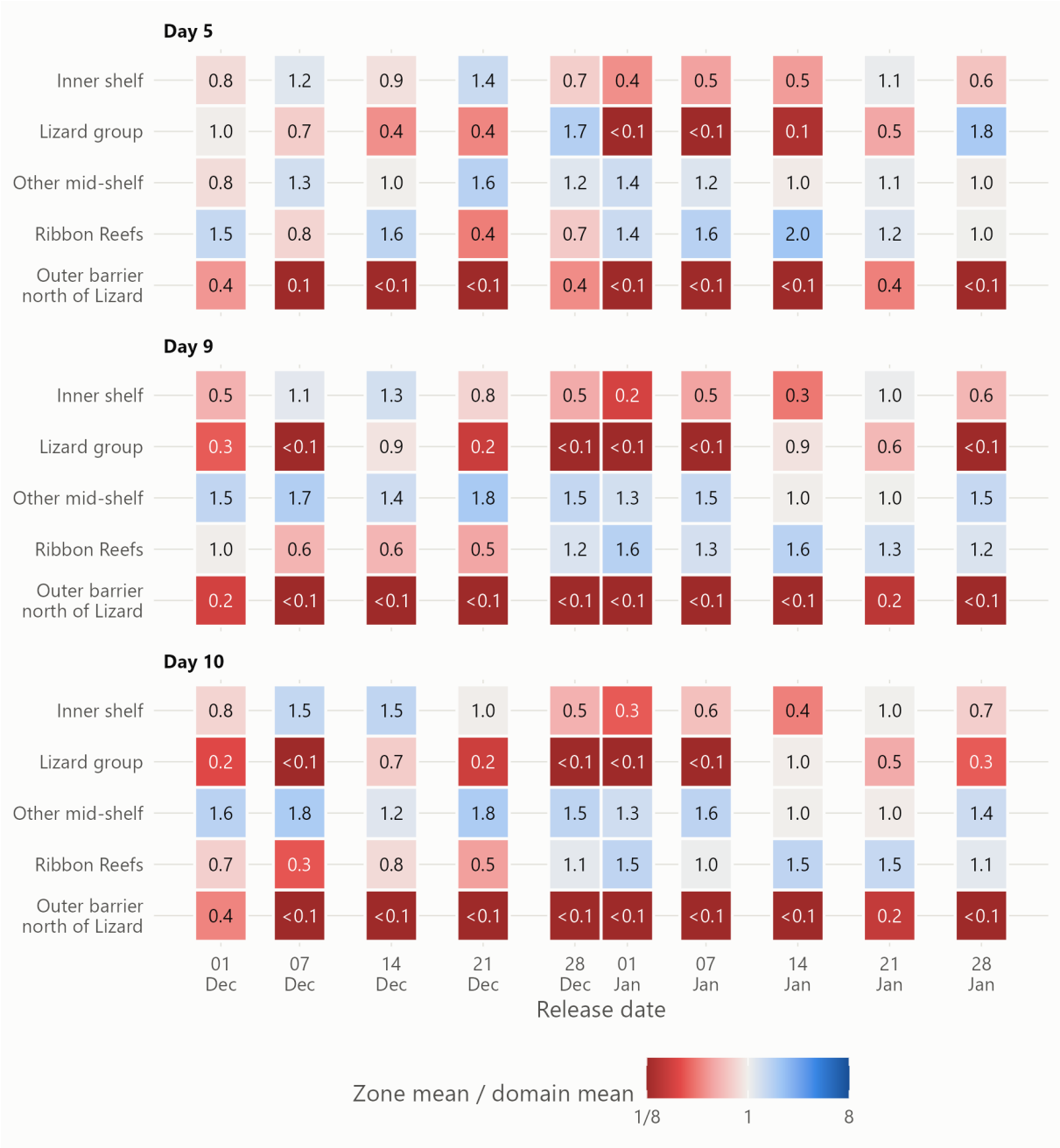


Figure 28: Mean reef concentration in each shelf zone relative to the domain mean, by release and larval age. A value of one means the zone matched the domain mean for that release and age. Colours are clipped at one-eighth and eight times the domain mean; labels retain the source values, with values below 0.1 grouped. This reformats source figure lc4. The ratios do not identify larval origins or measure local retention.

Cross-shelf patterns and the domain boundary

The broad north–south gradient makes an overall Lizard-versus-other-reefs contrast difficult to interpret. Comparing cross-shelf positions within latitude bands gives a more useful view (Figure 29). The central Ribbon Reefs generally ranked below nearby mid-shelf reefs at the later ages, whereas the southern Ribbon Reefs did not show the same pattern. Latitude bands reduce the influence of the broad gradient but do not remove all spatial dependence or possible boundary effects.

These results describe one spawning season. The local analysis plan calls for source-to-destination matrices for individual releases, explicit checks of sources and losses at the model boundary, and comparisons against connections based on distance alone. Those tests are needed before attributing concentration patterns to retention within reef groups, interpreting Ribbon Reef isolation, or using the patterns to identify reefs for source control. Exploratory reef-label permutations in the source analysis did not account for spatial autocorrelation; their p-values are not used here as confirmatory evidence. [TO CONFIRM: concentration units, larval biology and source-buffer coverage with Sev and Owen, alongside the Methods placeholders.]

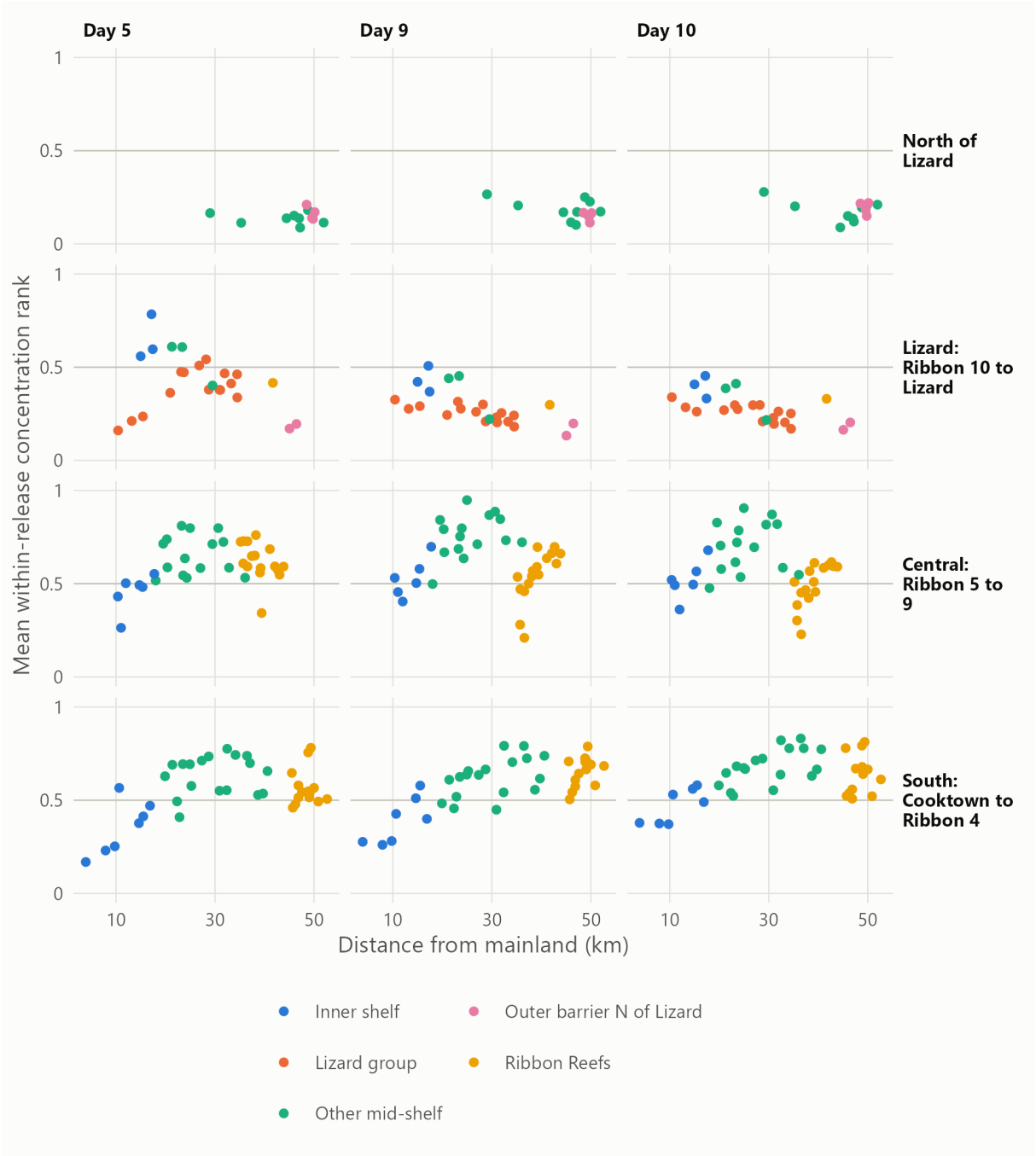


Figure 29: Cross-shelf concentration ranks within latitude bands, with the northern band at the top. Each point is a reef, showing its mean within-release rank over the simulated releases; the horizontal line marks rank 0.5. Columns show larval age and colours identify shelf zones. This reformats source figure lc6. Broad latitude bands aid comparison but do not eliminate domain-boundary effects or spatial dependence.

Appendix L: Management assessment and field-trial design

The main management implications summarise the five uses below. This appendix retains the detailed interpretation, scenario design and field-assessment requirements that support them.

Combining recent observations with habitat for daily searches

The practical contribution of the search models is a consistent way to combine what crews have recently observed with habitat information covering places they have not visited. Recent cull dives and manta tows carry most of the information about the current irruption, while habitat provides a first search order where that evidence is absent. The resulting ranking reduced the designated sites searched for a matched share of problematic dives in the retrospective comparisons. Its value is therefore in helping crews decide where to look first and releasing effort to search habitat outside the established footprint. Whether this improves overall detection depends on what is found in the added searches, as well as which familiar sites can be deferred.

This use follows from the different information each source contains. Habitat describes the setting in which COTS occur, but it cannot identify the current population after recruitment, movement and earlier control. Conversely, recent observations are informative about the places crews have searched but leave large gaps. Combining them is useful even when habitat alone separates high- and low-priority locations only weakly. The combined score should be read as a relative search priority within a reef, with the evidence behind it available to planners.

The >0.04 evaluation shows a benefit from ranking sites, with its size depending on the retention required. At 80% and 90% retention, the combined ranking searched 66.6% and 80.1% of sites, respectively, compared with 80.4% and 90.5% at random (Table 3). Requiring 95% retention still reduced searches, whereas 99% required almost all sites. The smaller reductions at high retention should not be attributed to the change in CPUE threshold alone: earlier ≥0.02 examples used lower retention targets. Any operational choice must weigh the sites deferred against the high-density dives potentially missed.

One option for field assessment is to use the ranking for repeated targeted visits while retaining an annual full-coverage survey that includes all deferred designated sites. Positive manta observations, feeding scars or nearby high-density detections could trigger earlier checks, and a full coverage check before reef closure could provide another opportunity to find missed populations. These measures could reduce the time that false negatives remain undetected, but their recovery rate has not been measured. Their effort, detection limits and the ecological consequences of delayed control must be included when judging the overall benefit (Section 4.5).

The retrospective gains also include knowledge already held by experienced crews. Sites were selected for dives using previous observations and local experience, so the ranking partly formalises existing practice. A field trial is needed to establish how much it adds to crew planning, particularly at never-dived sites. Exploration areas provide a way to extend the evidence base: visiting a small number of them and recording nil results as well as detections would allow future rankings to learn from a broader sample of reef habitat. Historical tows on previously unsurveyed ground found strong evidence more often than repeat tows, but this does not establish the yield of the proposed exploration areas; irruption stage may contribute to that difference.

Maintaining coverage during manta surveillance

The retrospective manta comparison provides a spatial-coverage benchmark for assessing search strategies; evenly spaced omissions are not a proposed field route. Model-based selection concentrated tows in a few highly ranked areas and could miss whole aggregations

elsewhere. Even spacing retained better coverage of the observed aggregations. This finding concerns retrospective thinning of existing surveys: tow order was inferred from position around the reef, and new routes have not been trialled. Habitat remains useful for proposing where to add or repeat observations, subject to field assessment, especially beyond the current footprint. The surveillance objective therefore has two parts: preserve coverage when thinning familiar routes, and test additional searches in lagoon bommie fields, shallow crests and isolated patches that perimeter routes can overlook. The trial should account for effort used on both, so that shifting effort to new habitat is distinguished from reducing total surveillance.

Assessing the culling program from 2012 to the present and beyond

The fine-scale metacommunity model was developed to reconstruct the third and fourth irruption waves and the emerging fifth wave in the initiation zone, assess how culling has changed COTS populations and coral trajectories since 2012, and guide future prioritisation. The historical assessment compares the observed program with a counterfactual in which the same reefs receive no control. Counts of COTS culled document the program's removals, but do not show how many animals would otherwise have survived, reproduced and dispersed larvae, or how much coral those animals and their descendants would have consumed. Linking these processes allows the model to assess effects extending beyond the treated sites.

We compared paired simulations beginning from the same population and coral state before control. One applied recorded removals from 2012 onward, focusing on the periods of control in 2014–15 and 2020–2026; the other omitted program removals. Both received the same heat stress, cyclone and other environmental forcing and used matched random seeds. Reproduction, larval dispersal and coral consumption evolved within each simulation. Differences in COTS abundance, larval export, cumulative coral consumption and coral cover provided the comparison of program effects within the model's assumptions. Requested removals exceeding the simulated population were recorded explicitly. [TO CONFIRM BEFORE FINALISATION: completion of the full paired historical assessment and removal-record coverage through 2026; this paragraph is drafted in past tense as requested.]

Matching historical peak timing alone is insufficient: the model must also represent irruption sizes, populations between irruptions and the recent build-up documented by divers. Initial population estimates, the relationship between manta observations and actual density, and coral recovery all contribute to uncertainty in the estimated effects of culling.

Larvae arriving from reefs outside the Lizard Island model domain also affect this comparison. Their supply is currently informed by the GBR-wide irruption predictions of Stewart et al. (2026). For a regional history without culling, we would ideally need corresponding predictions of what happened on those outside reefs without culling, so that their larval supply reflects the same counterfactual. The existing prediction layer does not by itself provide that causal comparison. Using the same outside supply in both simulations instead estimates the effect of culling inside the Lizard Island domain while holding conditions on outside reefs unchanged. Any estimate of coral protected must state which of these questions it answers.

Projecting suppression of the fifth primary irruption wave

The forward-looking extension will assess how attempts to suppress the fifth primary irruption wave in the initiation box during 2020–2026, including increased resources during 2026–2028, could change subsequent irruptions and coral protection outcomes. Within the current domain, the first test is whether suppression reduces COTS densities and larval supply at its southern extent, including Irene Reef and Startle Reefs. Those results would help assess potential consequences farther south toward the Cairns region; a direct estimate of coral protection there

would require a model domain or linked model that includes those reefs. This extension is partly developed and builds on the strategic questions addressed by earlier COTS population and reef ecosystem models (Morello et al., 2014; Matthews et al., 2020b; Condie et al., 2021; Bozec et al., 2022).

The scenario comparisons will focus on what increased effort from now can achieve in suppressing the primary irruption. This question differs from protecting coral at individual reefs after secondary irruptions are established, as discussed by Matthews et al. (2024). One set of comparisons will reallocate existing capacity among reefs, for example by targeting dense populations on reefs that supply larvae to others. A second set will assess the additional benefit of one or two vessels. Comparisons within each capacity level will show the value of changing allocation, while comparisons across capacity levels will show what extra resources could achieve. Continuation of current control and no control will provide reference trajectories. Starting and stopping rules will be examined within this primary-suppression objective. A stopping rule based only on the inability to save local coral could abandon a reef that remains an important larval source. Scenarios should therefore compare stopping when local protection is no longer achievable with continued control to reduce downstream recruitment. They should also test when to begin or intensify control on source reefs whose populations cannot feasibly be reduced below the local ecological target. These comparisons will show whether downstream benefits justify effort that produces little immediate local coral protection.

Each paired comparison will use the same climate and disturbance sequences. Repeating it across plausible population states, biological processes, larval transport and future environmental conditions will help identify strategies that are robust to deep uncertainty in the system. The aim is to find strategies that continue to meet management objectives across a wide range of plausible futures, rather than selecting a strategy that performs best under one assumed future.

The outcomes will connect suppression to the duration and extent of high COTS densities, larvae exported to connected reefs, coral loss through feeding, and subsequent coral recovery. Fewer COTS at a treated reef need not produce a proportional local coral benefit, particularly if coral spared from feeding is later affected by heat stress or cyclones. Conversely, culling a large aggregation might not prevent substantial coral loss at that reef but could still reduce COTS recruitment and protect coral on connected reefs. The assessment therefore needs to follow both local and downstream consequences.

Future scenarios must also represent what crews can achieve. The initial projections now link diver effort and COTS density to achievable removals, alongside prescribed historical catches. That relationship still requires assessment against operational records, including search and travel costs, before starting and stopping rules can describe operational choices credibly. Site rankings from the species distribution models could help allocate searches, although that coupling has not been validated. Scenario design will depend on future funding and direct consultation with GBRMPA, program partners and contractors to reflect operational realities. The exploratory horizon is 2050 under SSP2-4.5 and SSP3-7.0. [TO CONFIRM: vessel budgets, intervention rules and climate scenarios agreed with program partners.]

Monitoring operational performance

A field trial of the 2027 site rankings from the species distribution models is planned for January 2027, while the strategic metacommunity model continues to be developed. Before each visit, the trial will record the recommended priority of each site. It will then record which sites crews searched by manta tow or dive, the effort used, the COTS found and any reasons for departing from the suggested order. Outcomes will be compared with the search order crews would

normally use and with the modelled expectations in Table 18. The original class expectations use ≥0.02 COTS per minute of bottom time. The separate >0.04 reassessment (Section 3.1.1) supplies the more relevant high-density trade-offs, but neither retrospective test establishes what will be found in the trial or at unvisited sites.

The trial will aim for a 10–15% reduction in the number of designated sites requiring manta-tow searches on individual visits, with a negligible share of dives above 0.04 COTS per minute of bottom time missed. It will also aim for a 10–15% reduction in low-density or empty dives under the same allowance for missed dives above 0.04. Here, empty dives find no COTS, and low-density dives retain the report's definition of positive CPUE below 0.02. Dives from 0.02 to 0.04 will be reported separately: some may be deferred without the penalty assigned to missed dives above 0.04. These remain proposed adoption criteria: the retrospective reassessment does not establish both reductions at an agreed negligible omission rate. [TO CONFIRM: the numerical tolerance for missed dives above 0.04 and whether reductions are measured per visit or over the full season.]

A fallback option is an annual full-coverage surveillance round that includes every designated site, including those deferred by the ranking, using manta tows and follow-up dives as appropriate. A further coverage check before reef closure could catch populations that develop or remain undetected between visits. New COTS sightings, feeding scars or nearby high-density dives could prompt earlier checks of low-ranked sites. Crews should record coverage and reasons for revisiting, rather than treating a low rank as evidence that COTS are absent.

The trial should record >0.04 dives found during follow-up at deferred sites, the time since the initial visit, and the extra tow and dive effort required. Independent checks of a sample of deferred sites during the initial visit would help estimate the false-negative rate; annual detections alone could also include newly developed populations. A negative manta observation alone cannot establish that a site has no high-density population. Annual surveys and pre-closure checks offer opportunities to detect false negatives, without guaranteeing complete detection or coral protection during the delay. The 80% and 90% options therefore need assessment with an agreed maximum delay and omission allowance, rather than assuming later coverage makes their initial 20% or 10% omissions negligible.

Full-season effort must include annual full surveys, pre-closure and triggered checks, repeat dives and exploration visits. Sites deferred on one visit may be searched later, so the initial site reductions in Table 3 cannot be read directly as annual time savings. Visits to exploration areas should be reported separately, with dates and irruption conditions recorded. If the trial meets the 10–15% improvement targets while remaining within the agreed missed-dive allowance and completing the coverage checks, the approach can be considered for adoption; otherwise, the ranking or search rules should be refined. [TO CONFIRM: trial design, monitoring owner and reporting route agreed with GBRMPA and vessel operators.]

Savings should be attributed to the ranking only to the extent that it improves retention of high-density dives at the same effort, or reduces effort at the same retention. Translating those gains into vessel time or money requires an agreed account of travel, search, diving and culling time. [TO CONFIRM: whether to estimate financial savings and which operational cost data to use.]

Developing an early-warning system

The proposed application combines concentration patterns with release-specific network analysis to identify periods when reefs in the initiation box exchange larvae within tightly connected groups. Comparisons with distance-based connections and explicit checks of sources beyond the domain boundary are needed to distinguish hydrodynamic clustering from reef layout and boundary truncation. Repeating the analysis across seasons and comparing its signals with

subsequent COTS observations and reef-level irruption estimates would assess warning lead time and false alarms. Appendix K describes the planned connectivity tests; no early-warning performance has yet been evaluated.